MARKETSANDMARKETS™

RNA THERAPEUTICS MARKET

GLOBAL FORECAST TO 2028

BY PRODUCT (VACCINES, DRUGS), TYPE (MRNA THERAPEUTICS, RNA INTERFERENCE, ANTISENSE OLIGONUCLEOTIDES), INDICATION (INFECTIOUS DISEASES, RARE GENETIC DISEASES), END USER (HOSPITALS & CLINICS)

www.marketsandmarkets.com

MarketsandMarkets™ is the world's largest revenue impact company, serving over 7500 customers. 80% of the top 2000 companies globally rely on us for identifying new high-growth and niche revenue opportunities. In the face of constant technology innovation and market disruption, we help organizations plan and operationalize their future revenue mix decisions by identifying over 30,000 high-growth opportunities ranging from $1B to $500B across 90+ industry trends and markets. Organizations choose MarketsandMarkets™ to stay ahead of the curve and accelerate their revenue decisions and implementations by 6 – 12 months, giving them a unique, first-mover advantage. Our revenue impact methodology provides quantified and actionable insights on converged, granular, and connected market eco-systems that result from disruptive technologies and high-growth markets. We provide an extended lens on not only what will impact our client's revenue but also what will impact their clients' revenues, continually uncovering latent opportunities.

We work across all major B2B industries with C-level executives in functions such as Strategy, Marketing, Sales, R&D, Product, and M&A. MarketsandMarkets™ brings exclusive high-growth markets intelligence generated by over 850 SMEs and analysts along with its proprietary Revenue Impact platform (Knowledge Store).

MARKETSANDMARKETS™

TABLE OF CONTENTS

LIST OF TABLES

LIST OF FIGURES

LIST OF ABBREVIATIONS

ABBREVIATION	FULL FORM
ACS	American Cancer Society
APEC	Asia-Pacific Economic Cooperation
ARM	Alliance for Regenerative Medicine
ASGCT	American Society of Gene & Cell Therapy
ASO	Antisense Oligonucleotide Therapeutics
CAR T-cell	Chimeric Antigen Receptor (CAR) T-cell Therapy
cGMP	Current Good Manufacturing Practice
circRNA	Circular RNA
CRD	Chronic Respiratory Diseases
CVD	Cardiovascular Diseases
dsRNA	Double-stranded RNA
EMA	European Medicines Agency
ESGCT	European Society of Gene and Cell Therapy
GHIT Fund	Global Health Innovative Technology Fund
HGP	Human Genome Project
HIV	Human Immunodeficiency Virus
HVTN	HIV Vaccine Trials Network
LSIF	Life Sciences Innovation Forum
MFDS (Republic of Korea)	Ministry of Food and Drug Safety
miRNA	microRNA
MOU	Memorandum of Understanding
mRNA	Messenger RNA
ncRNA	Non-coding RNA
NGS	Next-Generation Sequencing
NIH	National Institutes of Health
PH	Primary Hyperoxaluria
RHSC	Regulatory Harmonization Steering Committee
RNAi	RNA Interference Therapeutics
RNAi	RNA Interference
saRNA	Small Activating RNA
siRNA	Small Interfering RNA

US FDA/FDA	US Food and Drug Administration
WHO	World Health Organization

1 INTRODUCTION

1.1 STUDY OBJECTIVES

- To define, describe, and forecast the global RNA therapeutics market based on product, type, indication, end user, and region.
- To provide detailed information regarding the major factors influencing the market growth (such as drivers, restraints, challenges, opportunities, and trends)
- To strategically analyze micromarkets[1] with respect to individual growth trends, prospects, and contributions to the overall RNA therapeutics market
- To analyze the opportunities in the market for stakeholders and provide details of the competitive landscape for market leaders operating in the RNA therapeutics market
- To forecast the revenue of the market with respect to four regional segments, namely, North America, Europe, the Asia Pacific, and Rest of the World (ROW)
- To profile key market players and comprehensively analyze their product (vaccines and drugs) portfolios, market positions, and core competencies[3]
- To track and analyze competitive developments such as product launches, regulatory approvals; expansions; agreements, collaborations, partnerships; and acquisitions in the global RNA therapeutics market

1.2 MARKET DEFINITION

RNA therapeutics uses both coding RNA (such as mRNA) and non-coding RNAs, which include small interfering RNAs (siRNA), aptamers, antisense oligonucleotides (ASO), ribozymes, and clustered regularly interspaced short palindromic repeats-CRISPR-associated (CRISPR/Cas) endonuclease to target proteins and DNA. RNA therapeutics have diverse abilities to target the desired protein/gene, and the ongoing research in RNA modification and delivery systems has developed unique RNA-based formulations. As of 2022, the use of RNA therapeutics is approved for indications such as COVID-19, hemophilia A and B, cholesterol reduction, spinal muscular atrophy (SMA), polyneuropathy of hereditary transthyretin-mediated amyloidosis-(ATTR), and familial chylomicronaemia syndrome (FCS).

1.2.1 INCLUSIONS AND EXCLUSIONS

PARTICULAR	INCLUSION	EXCLUSION
Product	Qualitative and quantitative (in terms of USD million) analysis of RNA-based vaccines and drugs. The pipeline candidates (only Phase 3 and 4) for both products are not considered in the forecast model.	Technologies licensed to develop vaccines/drugs.
Type	Qualitative and quantitative (in terms of USD million) analysis of RNA therapy types such as mRNA therapeutics, RNA interference (RNAi) therapeutics, antisense oligonucleotide (ASO) therapeutics, and other therapeutics such as RNA aptamers, small-activating RNA.	NA
Indication	Qualitative and quantitative (in terms of USD million) analysis of applications, including infectious diseases, rare genetic diseases/hereditary diseases, and other indications such as ophthalmology, oncology, and cardiovascular diseases.	NA

End User	Qualitative and quantitative (in terms of USD million) analysis of RNA therapeutics end users such as hospitals and clinics and research settings	NA

1.3 MARKET SCOPE

RNA THERAPEUTICS MARKET

PRODUCT
- Vaccines
- Drugs

INDICATION
- Infectious Diseases
- Rare Genetic/Hereditary Diseases
- Other Indications

END USER
- Hospitals and Clinics
- Research Settings

TYPE
- mRNA Therapeutics
- RNA Interference (RNAi) Therapeutics
- Antisense Oligonucleotide (ASO) Therapeutics
- Other Therapeutics

REGION
- North America *(US and Canada)*
- Europe *(Germany, UK, France, and Rest of Europe)*
- Asia Pacific *(Japan, China, South Korea, and Rest of Asia Pacific)*
- Rest of the World

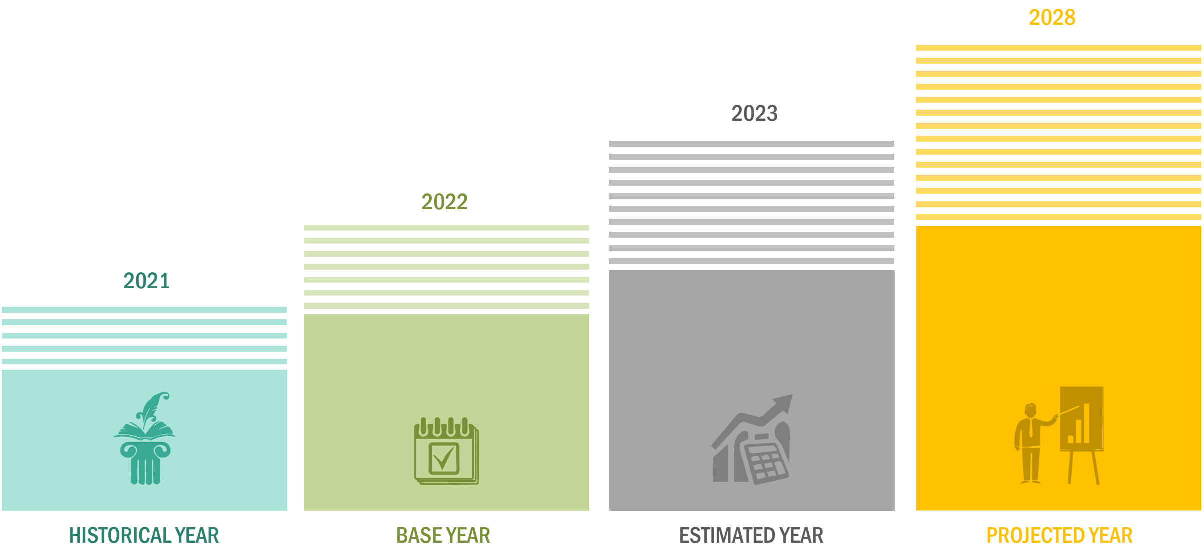
Note 1: Other types include RNA aptamers and small activating RNA.

Note 2: Other indications include oncology, ophthalmology, and CVD.

Note 3: Rest of Europe includes Italy, Spain, Russia, Poland, and other countries.

Note 4: Rest of Asia Pacific includes Australia, India, Singapore, and other countries.

1.3.1 YEARS CONSIDERED



1.4 CURRENCY CONSIDERED

- The currency used in this report is the United States Dollar (USD), with market sizes indicated in terms of USD million/billion.
- For companies reporting their revenues in USD, they were taken from their respective annual reports/SEC filings. For companies reporting their revenues in other currencies, the average annual currency conversion rate was used for that particular year to convert the value to USD.

1.5 RESEARCH LIMITATIONS

- The market study does not cover the volumetric analysis for the product, type, indication, and end users.
- A limited number of industry experts participated in primary research from Africa, the Middle East, and Latin America due to specific language barriers. In such cases, the geographical market size was derived based on weightages assigned to these markets based on the qualitative insights from global industry experts, the analysis of demographic parameters, and the adoption trends for RNA-based vaccines and drugs observed in these markets.
- Financial information for privately held and unlisted companies was not available in the public domain. Hence, their financial details are not provided in the company profiles section.
- Company developments not reported in the public domain are not included in this report.
- Company profiles showcase pipeline candidates that are in Phases 3 and 4; the remaining pipeline candidates (Phase 1 and Phase 2) are showcased in pipeline analysis.
- Market shares are estimated on a year-on-year basis using historical data of companies and secondary sources.
- Several strategic approaches and recommendations from industry experts were utilized to collate the research study to overcome the limitations and maintain the effectiveness of the study.
- The list of products included in the products offered' tables is only an indicative list of the product portfolio of a company.

1.6 STAKEHOLDERS

- Research Institutes
- Pharmaceutical and Biotechnology Companies
- Drug Discovery Companies
- Venture Capitalists and Investors
- Contract Development and Manufacturing Organizations
- Government Associations
- Healthcare Associations/Institutes
- Business Research & Consulting Service Providers

1.6.1 RECESSION IMPACT: RNA THERAPEUTICS MARKET

The impact of the recession is analyzed in the current report. The factors considered while analyzing the growth of the market in the forecast period are economic and financial recession, the Russia-Ukraine war, and supply chain disruptions, which are anticipated to affect the market directly or indirectly in the future.

- The Russia-Ukraine war will have a significant impact on different regional markets with a direct impact on the European market.
- The decreasing demand for COVID-related drugs and vaccines is also expected to reduce the number of drug discovery and development initiatives for the same, which, in turn, is expected to affect the revenue of key players in the coming years.
- Declining GDPs will affect R&D expenditures, which, in turn, will restrain the overall market to a certain extent.

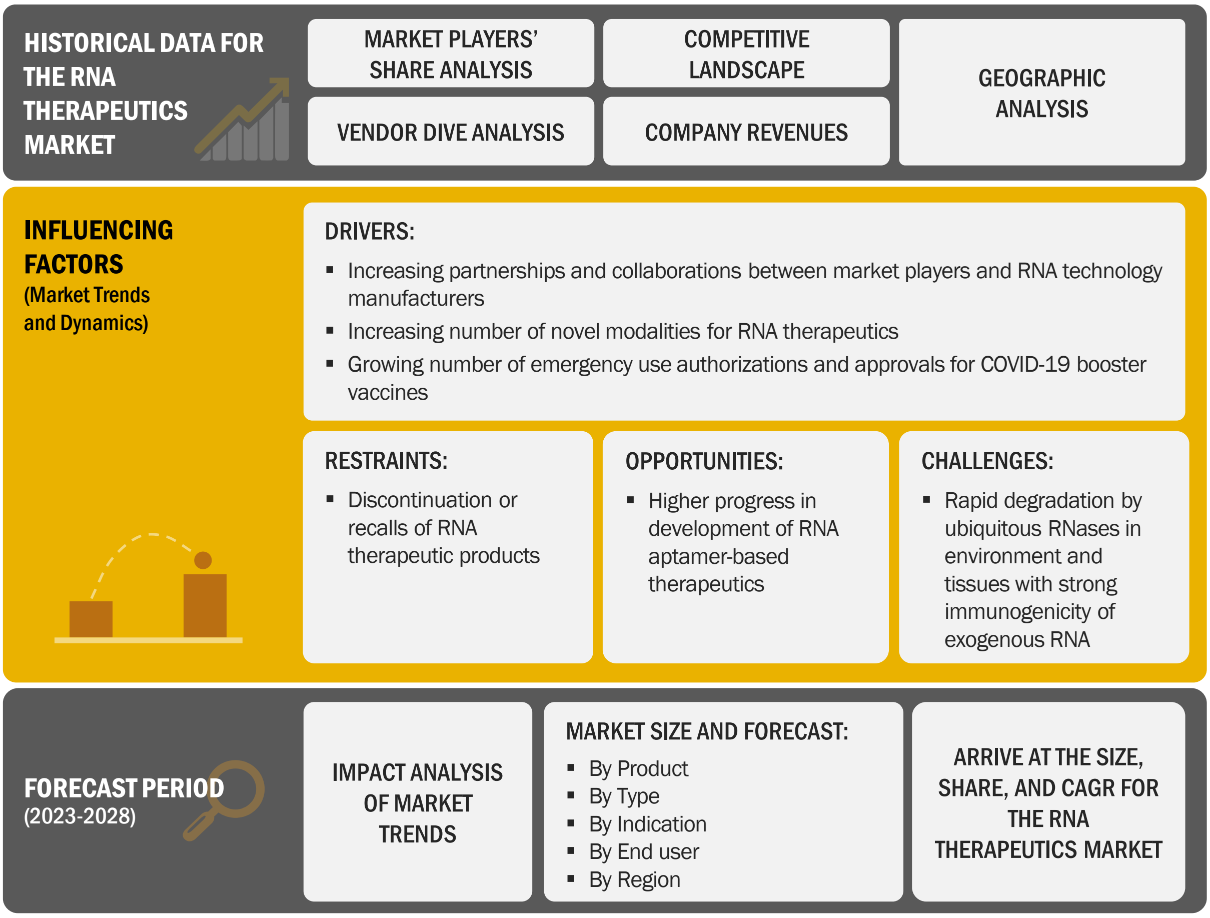
The direct and indirect impact of these factors on the growth of the RNA therapeutics market is further analyzed in the current edition of the report.

2 RESEARCH METHODOLOGY

2.1 RESEARCH DATA

This research study involved the extensive use of secondary sources, directories, and databases to identify and collect information useful for this technical, market-oriented, and financial study of the global RNA therapeutics market. In-depth interviews were conducted with various primary respondents, including key industry participants, subject-matter experts (SMEs), C-level executives of key market players, and industry consultants, to obtain and verify critical qualitative and quantitative information and assess the growth prospects of the market. The global market size estimated through secondary research was then triangulated with inputs from primary research to arrive at the final market size.

FIGURE 1 RESEARCH DESIGN



Sources: Secondary Research and MarketsandMarkets Analysis

2.1.1 SECONDARY DATA

The secondary sources referred to for this research study include publications from government sources, such as National Institutes of Health (NIH), US Food and Drug Administration (US FDA), the World Health Organization (WHO), the European Pharmaceutical Review, American Society of Gene & Cell Therapy (ASGCT), European Society of Gene and Cell Therapy (ESGCT), RNA Society (US), European Medicines Agency (EMA), Alliance for Regenerative Medicine (ARM), Vaccines Europe, and Central Drugs Standard Control Organization (India). Secondary sources also include corporate and regulatory filings, such as annual reports, SEC filings, investor presentations, and financial statements; business magazines and research journals; press releases; and trade, business, and professional associations. Secondary data was collected and analyzed to arrive at the overall size of the global RNA therapeutics market, which was validated through primary research.

SECONDARY RESEARCH INVOLVED THREE MAJOR ACTIVITIES:

BACKGROUND STUDY

- Building a basic understanding of the RNA therapeutics market
- Analyzing the MarketsandMarkets data repository followed by developing and updating data tables using current information
- Identifying data gaps and key global players in each segment

MARKET UNDERSTANDING

- Identifying stakeholders and key decision-makers in different regions
- Identifying the selection criteria of key decision-makers
- Analyzing the competitive landscape
- Analyzing major global players with their respective product portfolios
- Analyzing the strategies adopted by global players to position their products
- Analyzing markets at the regional/country level

TRENDS

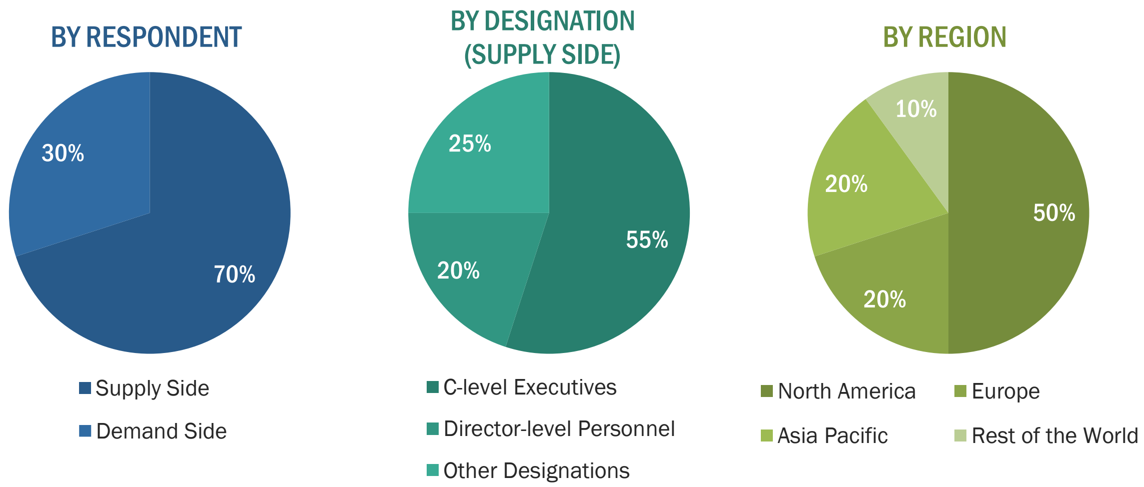
- Identifying and analyzing key products and industry trends to draw notes and include considerations in the final forecasts
- Identifying major drivers, restraints, challenges, and opportunities in the market
- Analyzing the regulatory landscape

2.1.2 PRIMARY DATA

Extensive primary research was conducted after acquiring basic knowledge about the global RNA therapeutics market scenario through secondary research. Several primary interviews were conducted with market experts from the demand side, such as personnel from the pharmaceutical and biopharmaceutical industries and experts from the supply side, such as C-level and D-level executives, product managers, marketing and sales managers of key manufacturers, distributors, and channel partners. These interviews were conducted across four major regions, including North America, Europe, the Asia Pacific, Latin America, and the Middle East & Africa. Approximately 70% and 30% of primary interviews were conducted with supply side and demand side participants, respectively. This primary data was collected through questionnaires, e-mails, online surveys, personal interviews, and telephonic interviews.

FIGURE 2 BREAKDOWN OF PRIMARIES: RNA THERAPEUTICS MARKET

SUPPLY SIDE PARTICIPANTS		DEMAND SIDE PARTICIPANTS
Arrowhead Pharmaceuticals, Inc. (US)	Arcturus Therapeutics, Inc. (US)	UMass Chan Medical School (US)
CRISPR Therapeutics (Switzerland)	Arbutus Biopharma (US)	University at Albany (US)



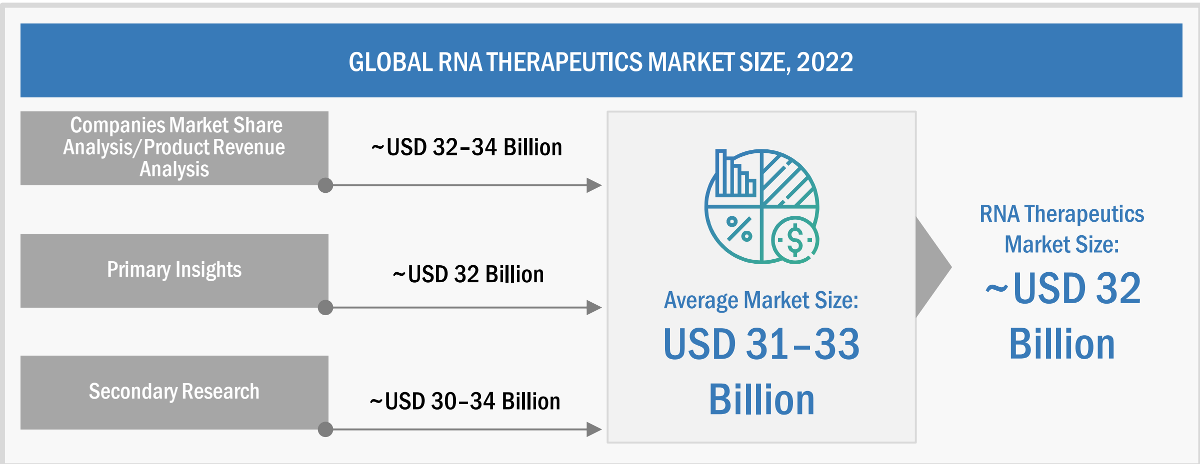
Note 1: Others include sales managers, marketing managers, and product managers.

Note 2: Tiers are defined based on a company's total revenue. As of 2020, Tier 1 = >USD 1 billion, Tier 2 = USD 500 million to USD 1 billion, and Tier 3 <USD 500 million.

2.2 MARKET ESTIMATION METHODOLOGY

The global size of the RNA therapeutics market was estimated through multiple approaches. A detailed market estimation approach was followed to estimate and validate the value of the RNA therapeutics market and other dependent submarkets, as mentioned below:

FIGURE 3 RNA THERAPEUTICS MARKET SIZE ESTIMATION (SUPPLY SIDE ANALYSIS), 2022

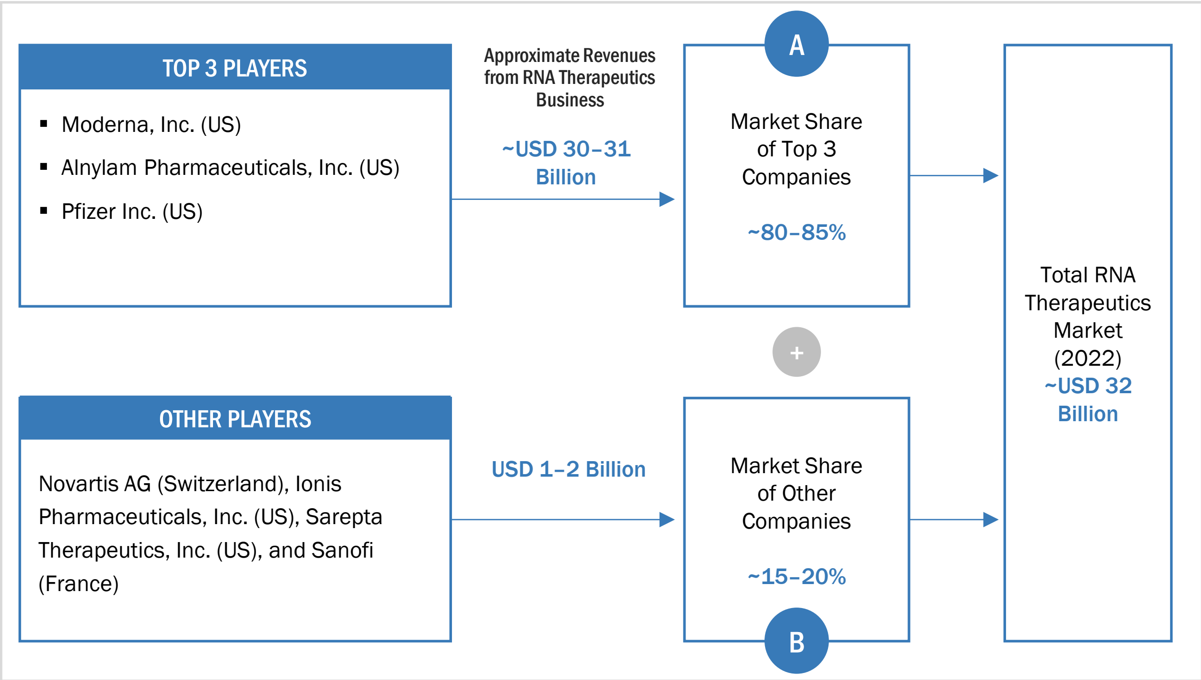


Source: MarketsandMarkets Analysis and Primary Insights

APPROACH 1: COMPANY REVENUE ANALYSIS (BOTTOM-UP APPROACH)

- Revenues for individual companies were gathered from public sources and databases.
- The RNA therapeutics-related business shares of leading players were gathered from secondary sources to the extent available. In certain cases, the shares of RNA therapeutics product (vaccines + drugs) businesses were ascertained through a detailed analysis of various parameters, including product portfolios and market positioning.
- Individual shares or revenue estimates were validated through Interviews with Experts.

FIGURE 4 MARKET SIZE ESTIMATION: COMPANY REVENUE ANALYSIS-BASED ESTIMATION, 2022



Source: MarketsandMarkets Analysis and Primary Insights

For the calculation of the global market value, the segmental revenue was arrived at based on the revenue mapping of major players active in the RNA therapeutics market. This process involved the following steps:

- A list of the major global players operating in the RNA therapeutics market was generated.
- The annual revenues generated by major global players from the RNA therapeutics market, or the nearest reported business unit/product category, were mapped.
- The revenues of the top five players (approved products) in the RNA therapeutics market were identified.
- The revenue of the top three players was assumed to be approximately 80–85% in 2022 and added up to 100% to derive the global value of the RNA therapeutics market.

APPROACH 2: SECONDARY RESEARCH

- The size of the RNA therapeutics and subsegments was obtained from secondary sources.
- The market size of RNA therapeutics in the overall market was then derived and validated through primary participants.

APPROACH 3: PRODUCT REVENUE ANALYSIS/COMPANY MARKET SHARE ANALYSIS

- For estimation of the RNA therapeutics market, product sales were mapped for the top 5 players [Moderna, Inc. (US), Alnylam Pharmaceuticals, Inc. (US), Pfizer Inc. (US), Novartis AG (Switzerland), Ionis Pharmaceuticals, Inc. (US)] from 2019 to 2022. Predicted sales were also gathered for 2023 for individual products (vaccines/drugs) from secondary resources and validated by industry experts. The final global market size was triangulated through the average of all approaches and validated through primary interviews with industry experts.

APPROACH 3: PRIMARY INTERVIEWS

- As part of the primary research process, individual respondents were interviewed for insights on the market size and market growth.
- All responses were collated, and a weighted average was taken to derive a probabilistic estimate of the market size and growth rate.

2.2.1 INSIGHTS FROM PRIMARY EXPERTS

FIGURE 5 MARKET SIZE VALIDATION FROM PRIMARY SOURCES

FACTORS AFFECTING MARKET GROWTH

R&D/Development Head, *RNA Vaccines Manufacturer*	A higher number of vaccine approvals against COVID-19 by Moderna and Pfizer majorly contribute to the revenue in the RNA therapeutics market.
Regional Marketing Manager, *RNA Therapeutics Manufacturer*	Demand for RNA therapeutics is presently targeted toward the treatment of patients with hereditary/genetic disorders. Non-oncology applications have also expanded the potential of RNA therapeutics, propelling industry growth.

MARKET TRENDS

- Partnerships and collaborations among market players with RNA technology manufacturers
- Emergency Use Authorizations (EUA) and approvals for COVID-19 booster vaccines

– *RNA Therapeutics Market Experts*

TOP THREE MARKET PLAYERS RANKING

Moderna, Inc. (US)	1
Alnylam Pharmaceuticals, Inc. (US)	2
Pfizer Inc. (US)	3

Source: Primary Insights

The final market size was estimated based on the weighted average of values arrived at through various approaches, such as company revenue analysis/product revenue analysis, secondary research, and primary insights. The research methodology and market numbers were further validated from primaries.

SEGMENT ASSESSMENT (BY PRODUCT, TYPE, INDICATION, AND END USER)

The segmental assessment was conducted using the following approaches:

- Approach for product: Revenue contributions/splits of different segments to the total revenues of leading players in the market (wherever available) and respective growth trends were derived by studying product revenues disclosed in the annual reports, SEC filings, and investor presentations published by key market players from 2019-2022.
- Approach for type: Splits for segments were derived by studying the type of RNA vaccines and drugs approved till 2022, including mRNA, RNAi, and antisense RNA. This information was validated through primary interviews. Similarly, the adoption of various products for each type segment was assessed to arrive at the percentage splits.
- Approach for indication: Splits for segments were derived by studying the type of RNA therapeutics for approved indications till 2022. This information was validated through primary interviews. Similarly, the adoption of various products for each indication segment was assessed to arrive at the percentage splits.
- Approach for end users: Splits for segments were derived by studying the adoption pattern of RNA therapeutics products by different end users in the RNA therapeutics market.

GEOGRAPHIC MARKET ASSESSMENT (BY REGION AND COUNTRY)

- The geographic assessment was conducted using the following approaches:
- The geographic assessment was conducted by studying geographic revenue contributions/ splits of leading players in the market (wherever available) and respective growth trends.

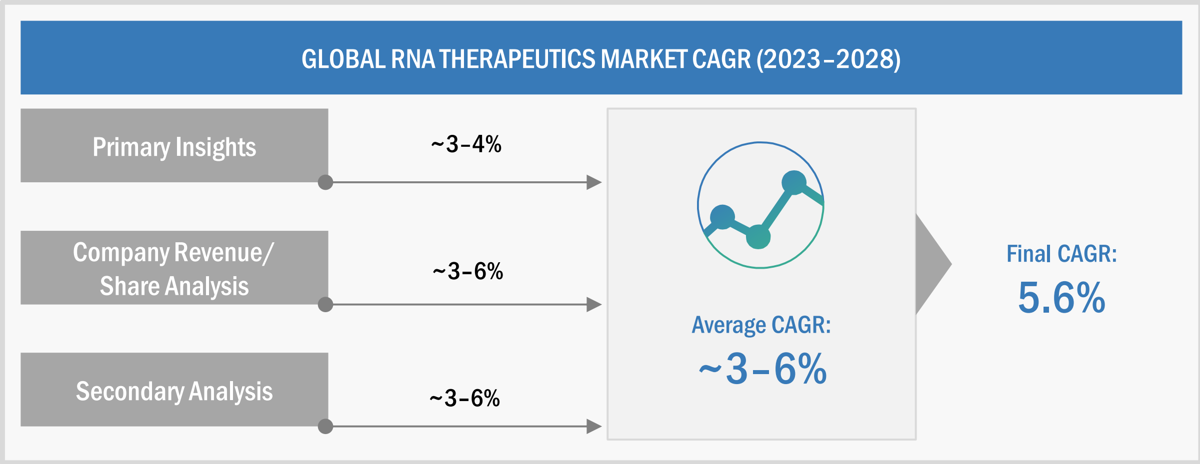
At each point, the assumptions and approaches were validated through industry experts contacted during primary research. Considering the limitations of data available from secondary research, revenue estimates for individual companies (for the overall RNA therapeutics market and geographic market assessment) were ascertained based on a detailed analysis of their respective product offerings, geographic reach/strength (direct or through distributors or suppliers), and shares of the leading players in a particular region or country.

2.3 MARKET GROWTH RATE PROJECTION

The growth forecast model was defined based on an assessment of the qualitative and quantitative factors affecting the market growth. These mainly included:

- Historic revenue growth trends of the leading players in the market
- Demand-side factor assessment, including developments in novel medicine manufacturing in various regions, approvals and recalls for RNA therapies (2019–2022), clinical trials conducted for various therapeutic areas, and pharmaceutical R&D spending
- Impact assessment of the key factors, such as drivers, restraints, challenges, and opportunities influencing market growth
- Impact assessment of the historic competitive trends driving market growth—product launches, strategic growth initiatives, innovation trends, and R&D initiatives
- Impact assessment of regulatory guidelines/mandates on market growth over the forecast period. For this assessment, only the defined regulatory guidelines related to the study period were considered. The impact of any future regulatory uncertainties on market growth is not considered

FIGURE 6 RNA THERAPEUTICS MARKET (SUPPLY SIDE): CAGR PROJECTIONS



Source: MarketsandMarkets Analysis and Primary Insights

Taking into consideration the average of all the growth rates from the above-mentioned approaches, the final CAGR for 2023–2028 was estimated to be 5.6% for this report.

FIGURE 7 GROWTH ANALYSIS OF DEMAND SIDE DRIVERS: RNA THERAPEUTICS MARKET

DRIVERS	IMPACT LEVEL
Increasing partnerships and collaborations between market players and RNA technology manufacturers	●
Increasing number of novel modalities for RNA therapeutics	●
Growing number of emergency use authorizations and approvals for COVID-19 booster vaccines	●
RESTRAINTS	**RATING**
Discontinuation or recalls of RNA therapeutic products	●
OPPORTUNITIES	**RATING**
Higher progress in development of RNA aptamer-based therapeutics	●
CHALLENGES	**RATING**
Rapid degradation by ubiquitous RNases in environment and tissues with strong immunogenicity of exogenous RNA	●

IMPACT LEVEL: ● LOW ● MEDIUM ● HIGH

Source: Secondary Research, MarketsandMarkets Analysis, and Primary Insights

2.4 MARKET ESTIMATION AND DATA TRIANGULATION

After arriving at the market size from the market size estimation process explained above, the total market was divided into several segments and subsegments. Data triangulation and market breakdown procedures were employed, wherever applicable, to complete the overall market engineering process and arrive at the exact statistics for all segments and subsegments.

The following figure shows the market validation sources structure and the data triangulation methodology implemented in the market engineering process for this report.

FIGURE 8 DATA TRIANGULATION METHODOLOGY

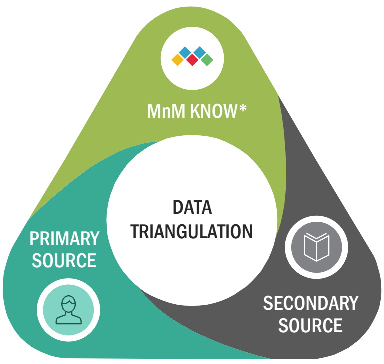
PRIMARY SOURCES

DEMAND SIDE

INTERVIEWS WITH:
- Personnel from Pharmaceutical Companies, Biotechnology Companies, Government, and Academic Research Institutes

SUPPLY SIDE

INTERVIEWS WITH:
- C-level and D-level Executives
- Product Managers
- Marketing and Sales Managers of Key Manufacturing Companies
- Distributors
- Channel Partners



INFORMATION SOURCED

SECONDARY SOURCES

MARKET RANKING ANALYSIS:
- Major players were mapped based on revenues and product portfolios
- Annual sales and geographical penetration of leading players were mapped

MARKET SIZE ANALYSIS:
- Sales data was mapped from company websites and industry whitepapers
- Based on primary and secondary research, market shares of the product segments were identified to calculate the global market size and splits for subsegments

STEP 1: Identifying the key stakeholders in the RNA therapeutics market

STEP 2: Gathering information from major suppliers and customers through secondary research and primary interviews

STEP 3: Providing detailed data on the market, key segments, key stakeholders, and the competitive scenario

*MnM KNOW stands for MarketsandMarkets' "Knowledge Asset Management" framework. In this context, it stands for the existing market research knowledge repository that acts as an independent source to help us validate information gathered from primary and secondary sources. In this report, reference was taken from MnM reports such as previous version of Genome Editing/Genome Engineering Market report, CRISPR Technology Market report.

2.5 STUDY ASSUMPTIONS

- Market values have been rounded off to the nearest decimal place. Due to this, there may be slight differences in actual totals and the totals mentioned in market tables.
- Region-wise inflation/deflation for product pricing during 2023–2028 has not been considered.
- It has been assumed that dollar fluctuations will not affect the growth forecast to a significant extent.
- The revenues of private companies acquired from paid databases were assumed to be in line with actual company financials.

- A positive economic climate will continue in emerging countries until 2028
- There will be no major changes in government regulations that will have a significant market
- All the market segments and subsegments listed in this report are considered to be mutually exclusive
- Wherever the exact segmental market splits were not available, the historical trend and inputs from primary respondents have been considered.

2.6 RISK ANALYSIS

DESCRIPTION	RISK ANALYSIS
The growth rate for each segment is considered based on growth ranks.	Moderate
The percentage splits for the product segment are given based on the matrix mapping of product analysis.	Low to Moderate

2.7 IMPACT OF ECONOMIC RECESSION ON RNA THERAPEUTICS MARKET

On the basis of the overall analysis of the pharmaceutical and biotechnology industry and inputs from our primary experts, the pharmaceutical and biotechnology industry is projected to withstand the downturn due to the recession. The demand for products and services is expected to remain stable, and the recession will have a low to minimal impact on healthcare-oriented businesses. Healthcare, inclusive of pharmaceutical and biotechnology companies, is a defensive industry (less sensitive to recession as compared to other industries) that is likely to hold up relatively well to recession as compared to other industries, such as automotive, telecom, and energy, as drugs for treating health conditions are a necessity for patients. Besides, they are backed by insurance and health plans, which makes this sector recession-resilient. Primary experts mentioned that investors are now looking forward to invest in defensive businesses (as a hedge against recessionary views), such as the pharmaceutical and biotechnology industry, and would avoid recession-sensitive businesses like consumer durables.

On the other hand, small to medium-sized pharmaceutical and biotechnology companies might face uncertainty due to changes in macroeconomic conditions, such as a decrease in global healthcare expenditure, an increasing cost of raw materials, reduced margin due to rising inflation, changes to the government funding environment, and limited or significantly reduced points of access of products. Such factors might have an adverse effect on the growth of the market in the forecast period.

TABLE 1 GLOBAL INFLATION RATE PROJECTIONS, 2021–2027 (% GROWTH)

	2021	2022	2023	2024	2025	2026	2027
Inflation Rate	4.7%	8.8%	6.5%	4.1%	3.2%	3.3%	3.2%

Source: IMF, World Economic Outlook Update, July 2022

TABLE 2 US HEALTH EXPENDITURE, 2019–2022 (USD MILLION)

Year	2019	2020	2021	2022
Total National Health Expenditure	37,57,382.0	41,44,077.0	42,55,127.0	44,42,352.6

Source: Centers for Medicare & Medicaid Services, 2022 (estimated on the basis of CMC data)

TABLE 3 US HEALTH EXPENDITURE, 2023–2027 (USD MILLION)

Year	2023	2024	2025	2026	2027
Total National Health Expenditure	46,57,806.7	49,04,670.4	51,69,522.6	54,49,969.3	57,46,992.6

Source: Centers for Medicare & Medicaid Services, 2022–2027 (estimated on the basis of CMC data).

As per the Centers for Medicare & Medicaid Services data, US healthcare spending will grow at an average of 4.9% per year from 2022 to 2024 and 5.3% per year beyond that, hitting USD 6.8 trillion in 2030.

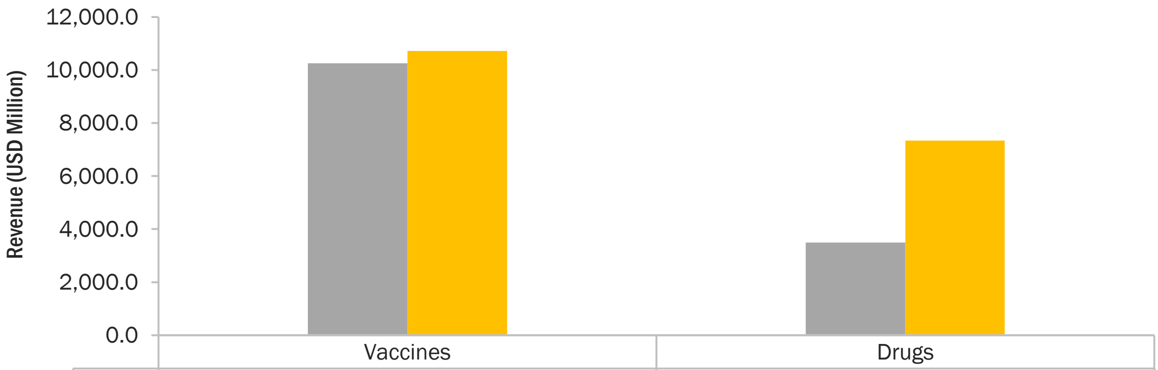
As per the Healthcare Outlook 2023 report published by The Economist Intelligence Unit Limited (EIU), total healthcare spending (public and private) was expected to rise by 4.9% in 2023, propelled by higher costs and wages. However, due to high inflation and slow economic growth, EIU estimated that the healthcare spending would remain similar to 2022 throughout 2023, meaning that 2023 will be the second successive year of real terms funding declines. OECD data suggests that after the global financial crisis of 2008–09, healthcare spending on preventive care and pharmaceuticals was reduced, and the same pattern is expected to be repeated.

The impact of the global recession on the RNA therapeutics market has been anticipated in this report with respect to changes in buying criteria, decline in GDP, decrease in healthcare spending, increase in inflation, decline in government funding for R&D, and supply chain constraints. The recession is anticipated to impact the top market players. Companies experienced slightly lower revenue growth during 2021–22 compared to previous years due to reduced demand for COVID-19 testing products, fewer shipments due to political and economic disturbances, high inflation, supply chain constraints, and unfavorable foreign exchange rate fluctuations. This impact will be partially offset by the growth of the pharmaceutical industry, international companies' expansion in developing geographies, and investments made in 2021 and 2022. For instance, Illumina, the market leader, has funded numerous start-ups in 2022. In June 2022, Illumina (US) announced its participation in Time Boost Capital I LP, a USD 37 million genomics venture fund dedicated to providing match funding to start-ups graduating from Illumina Accelerator Cambridge. Since its inception in July 2020, Illumina Accelerator Cambridge has launched 13 start-ups focused on utilizing genomics applications in diagnostics, synthetic biology, novel therapeutics, and agriculture. Also, during the pandemic, technologies such as genomics and bioinformatics became essential tools for research as they facilitated the development of advanced testing methods and provided the timely tracking of novel SARS-CoV-2 variants. Therefore, the recession will have minimal impact on the growth of the overall RNA therapeutics market during the forecast period.

3 EXECUTIVE SUMMARY

The RNA therapeutics market is projected to reach USD 18,065.5 million by 2028 from USD 13,753.8 million in 2023, at a CAGR of 5.6% during the forecast period. Growth in this market is largely driven by factors such as the increasing number of partnerships and collaborations among market players and RNA technology manufacturers, expanding modalities for RNA therapeutics, and the rising number of EUAs and approvals for COVID-19 booster vaccines. On the other hand, the discontinuation/recalls of RNA therapeutic products is expected to hinder market growth.

FIGURE 9 RNA THERAPEUTICS MARKET, BY PRODUCT, 2023 VS. 2028 (USD MILLION)

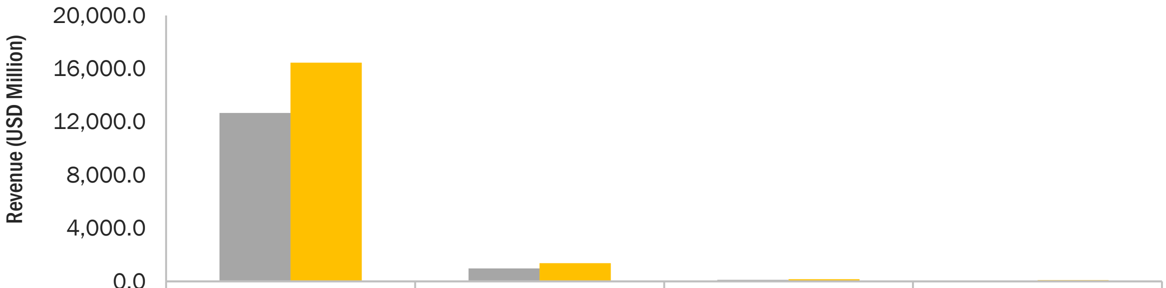


	Vaccines	Drugs
2023	10,264.4	3,489.4
2028	10,729.8	7,335.8

Source: National Institutes of Health (NIH), US Food and Drug Administration (US FDA), World Health Organization (WHO), European Pharmaceutical Review, American Society of Gene & Cell Therapy (ASGCT), European Society of Gene and Cell Therapy (ESGCT), RNA Society (US), European Medicines Agency (EMA), Alliance for Regenerative Medicine (ARM), Annual Reports, SEC Filings, Press Releases, Investor Presentations, Corporate Presentations, Journals, Interviews with Experts, and MarketsandMarkets Analysis

Based on the product segment, the RNA therapeutics market is segmented into vaccines and drugs. In 2022, the vaccines segment accounted for the largest share of the global RNA therapeutics market, with a market share of 92.3%. With the approval for the flagship RNA vaccines against SARS-COV-2 infection, such as Comirnaty (Pfizer, Inc.) and Spikevax (Moderna, Inc.), the revenue of this segment showcased considerable growth between 2021 and 2022. However, the recent recall/discontinuation of Spikevax (Moderna, Inc.) can hinder revenue growth from 2023 onwards. On the other hand, the high influx of innovative medicine companies in the market has rendered it with a fast growth rate during the forecast period.

FIGURE 10 RNA THERAPEUTICS MARKET, BY TYPE, 2023 VS. 2028 (USD MILLION)



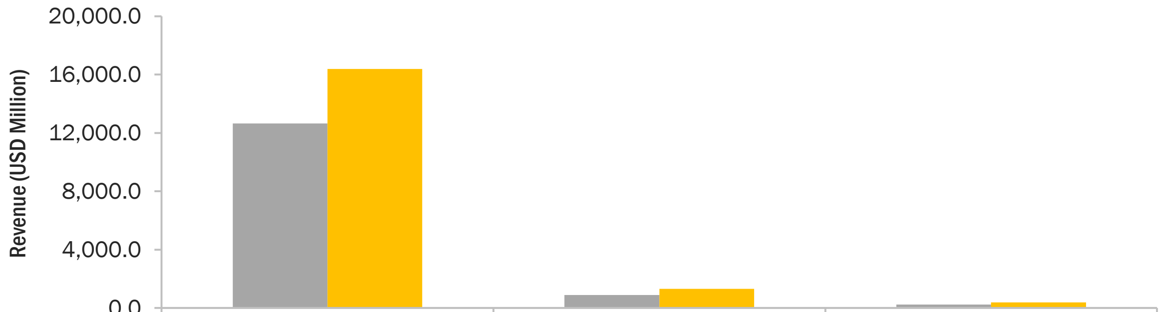
	mRNA Therapeutics	RNA Interference (RNAi) Therapeutics	Antisense Oligonucleotide (ASO) Therapeutics	Other Therapeutics
2023	12,657.4	971.4	125.1	
2028	16,438.0	1,370.6	156.9	100.0

Note: Other therapeutics include RNA aptamers and small-activating RNA.

Source: National Institutes of Health (NIH), US Food and Drug Administration (US FDA), World Health Organization (WHO), European Pharmaceutical Review, American Society of Gene & Cell Therapy (ASGCT), European Society of Gene and Cell Therapy (ESGCT), RNA Society (US), European Medicines Agency (EMA), Alliance for Regenerative Medicine (ARM), Annual Reports, SEC Filings, Press Releases, Investor Presentations, Corporate Presentations, Journals, Interviews with Experts, and MarketsandMarkets Analysis

Based on the type segment, the RNA therapeutics market is segmented into mRNA therapeutics, RNA interference (RNAi) therapeutics, antisense oligonucleotide (ASO) therapeutics, and other therapeutics. The mRNA therapeutics segment accounted for the largest share of 92.1% in the RNA therapeutics market in 2022. The large share of this segment can primarily be attributed to the advantages of mRNA over DNA or proteins/peptides and the advancements in vaccine development. On the other hand, the RNA interference (RNAi) therapeutics segment is expected to form the fastest-growing segment with a 7.1% growth rate in the RNA therapeutics market due to the increasing precision, versatility, and potential of RNAi for personalized medicine and growing successes in clinical applications.

FIGURE 11 RNA THERAPEUTICS MARKET, BY INDICATION, 2023 VS. 2028 (USD MILLION)



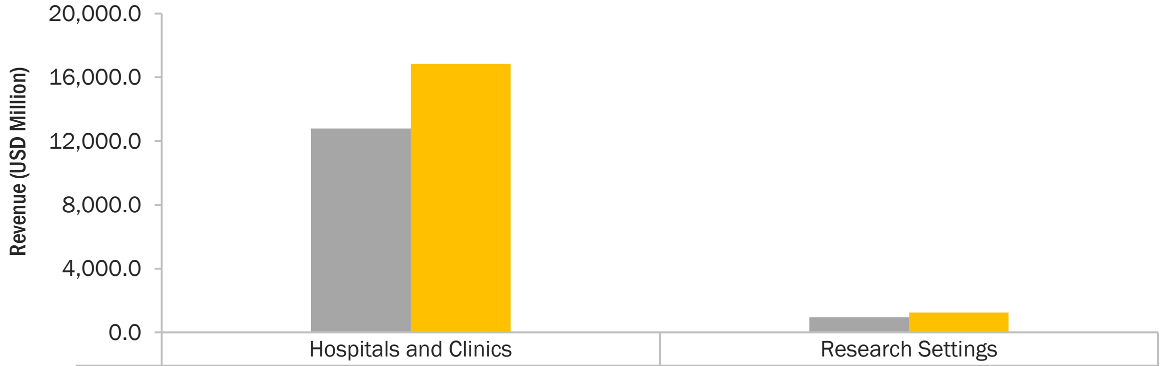
	Infectious Diseases	Rare Genetic/Hereditary Diseases	Other Indications
2023	12,643.7	877.8	232.3
2028	16,373.9	1,301.4	390.3

Note: Other indications include oncology, ophthalmology, and cardiovascular diseases.

Source: National Institutes of Health (NIH), US Food and Drug Administration (US FDA), World Health Organization (WHO), European Pharmaceutical Review, American Society of Gene & Cell Therapy (ASGCT), European Society of Gene and Cell Therapy (ESGCT), RNA Society (US), European Medicines Agency (EMA), Alliance for Regenerative Medicine (ARM), Annual Reports, SEC Filings, Press Releases, Investor Presentations, Corporate Presentations, Journals, Interviews with Experts, and MarketsandMarkets Analysis

Based on the indication segment, the RNA therapeutics market is segmented into infectious diseases, rare genetic/hereditary diseases, and other indications. The infectious diseases segment accounted for the largest share of 92.2% in the RNA therapeutics market in 2022. The large share of this segment can primarily be attributed to the rapid development capabilities of RNA vaccines for the treatment of infectious diseases and a higher number of epidemic outbreaks.

FIGURE 12 RNA THERAPEUTICS MARKET, BY END USER, 2023 VS. 2028 (USD MILLION)

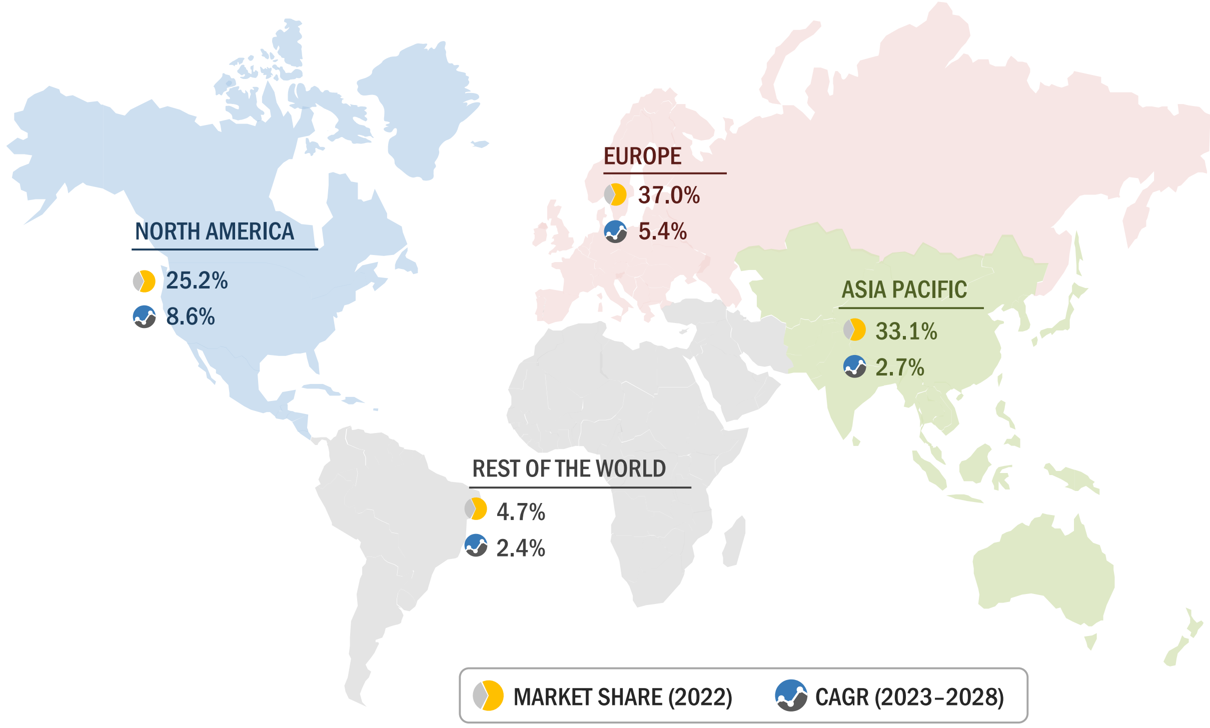


	Hospitals and Clinics	Research Settings
2023	12,795.9	958.0
2028	16,830.5	1,235.0

Source: National Institutes of Health (NIH), US Food and Drug Administration (US FDA), World Health Organization (WHO), European Pharmaceutical Review, American Society of Gene & Cell Therapy (ASGCT), European Society of Gene and Cell Therapy (ESGCT), RNA Society (US), European Medicines Agency (EMA), Alliance for Regenerative Medicine (ARM), Annual Reports, SEC Filings, Press Releases, Investor Presentations, Corporate Presentations, Journals, Interviews with Experts, and MarketsandMarkets Analysis

Based on the end user segment, the RNA therapeutics market is segmented into hospitals and clinics and research settings. The growing use of RNA therapeutics across hospitals and clinics is driven by factors such as the increasing understanding and advancements of RNA-based therapeutics, the growing number of trends in personalized medicine, the increasing prevalence of chronic diseases, and the rising research investments by governments, private foundations, and industry stakeholders.

FIGURE 13 REGIONAL SNAPSHOT: RNA THERAPEUTICS MARKET



Source: National Institutes of Health (NIH), US Food and Drug Administration (US FDA), World Health Organization (WHO), European Pharmaceutical Review, American Society of Gene & Cell Therapy (ASGCT), European Society of Gene and Cell Therapy (ESGCT), RNA Society (US), European Medicines Agency (EMA), Alliance for Regenerative Medicine (ARM), Annual Reports, SEC Filings, Press Releases, Investor Presentations, Corporate Presentations, Journals, Interviews with Experts, and MarketsandMarkets Analysis

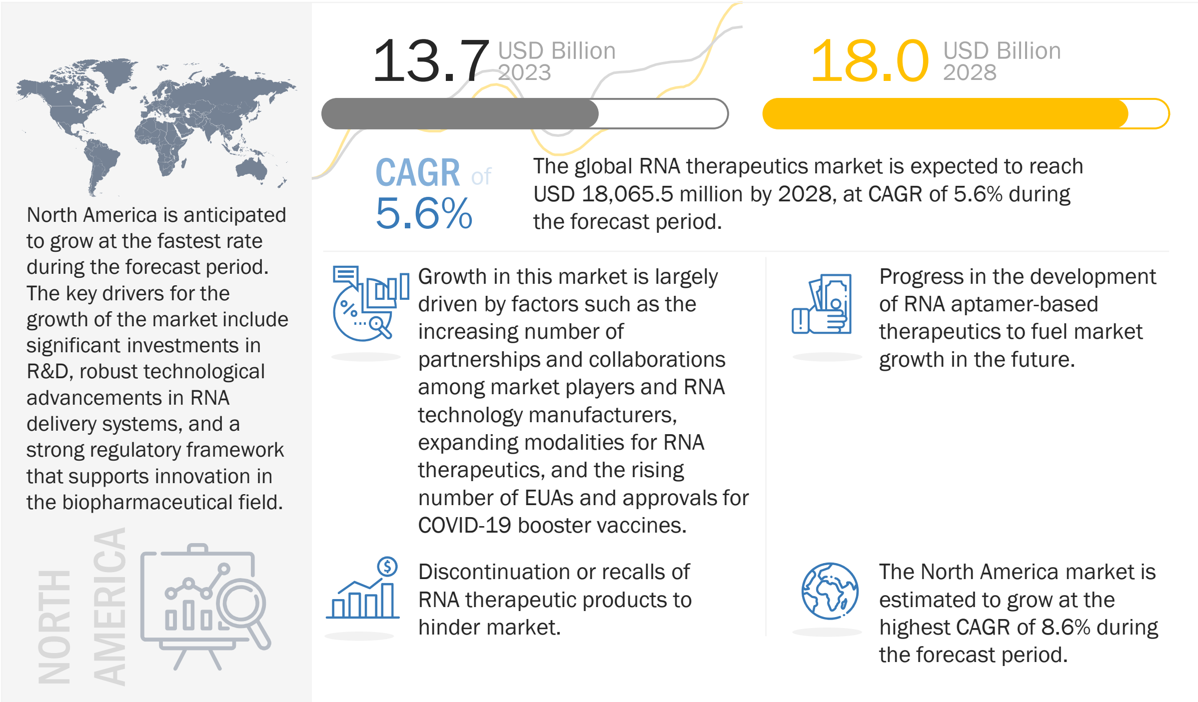
The global RNA therapeutics market is segmented into four major regions, namely, North America, Europe, the Asia Pacific, and the Rest of the World. In 2022, Europe accounted for the largest share OF 37% in the RNA therapeutics market. This can be attributed to the increased government funding for drug discovery and life science research, advances in RNA therapeutics techniques, and growth in the biotechnology and pharmaceutical industries in this region. North America has emerged as the fastest growing region from 2023 to 2028, with a CAGR of 8.6%.

The prominent players in the global RNA therapeutics market are Moderna, Inc. (US), Alnylam Pharmaceuticals, Inc. (US), Novartis AG (Switzerland), Ionis Pharmaceuticals, Inc. (US), Sarepta Therapeutics, Inc. (US), Sanofi (France), Pfizer Inc. (US), Arrowhead Pharmaceuticals, Inc. (US), BioNTech SE (Germany), Orna Therapeutics (US), CRISPR Therapeutics (Switzerland), Silence Therapeutics (UK), Astellas Pharma Inc. (Japan), CureVac SE (Germany), Sirnaomics (US), Arcturus Therapeutics Inc. (US), and Arbutus Biopharma (US)..

4 PREMIUM INSIGHTS

4.1 RNA THERAPEUTICS MARKET OVERVIEW

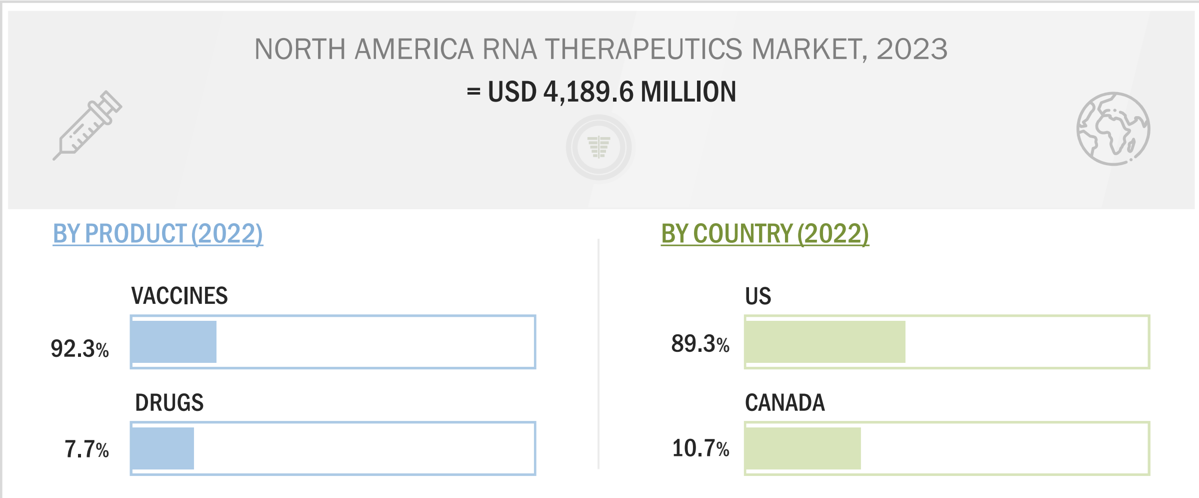
FIGURE 14 GROWING PARTNERSHIPS AND COLLABORATIONS AMONG KEY PLAYERS TO DRIVE MARKET



Source: National Institutes of Health (NIH), US Food and Drug Administration (US FDA), World Health Organization (WHO), European Pharmaceutical Review, American Society of Gene & Cell Therapy (ASGCT), European Society of Gene and Cell Therapy (ESGCT), RNA Society (US), European Medicines Agency (EMA), Alliance for Regenerative Medicine (ARM), Annual Reports, SEC Filings, Press Releases, Investor Presentations, Corporate Presentations, Journals, Interviews with Experts, and MarketsandMarkets Analysis

4.2 NORTH AMERICA: RNA THERAPEUTICS MARKET SHARE, BY PRODUCT AND COUNTRY (2022)

FIGURE 15 VACCINES ACCOUNTED FOR LARGEST SHARE OF NORTH AMERICAN RNA THERAPEUTICS MARKET IN 2022



Source: National Institutes of Health (NIH), US Food and Drug Administration (US FDA), World Health Organization (WHO), European Pharmaceutical Review, American Society of Gene & Cell Therapy (ASGCT), European Society of Gene and Cell Therapy (ESGCT), RNA Society (US), European Medicines Agency (EMA), Alliance for Regenerative Medicine (ARM), Annual Reports, SEC Filings, Press Releases, Investor Presentations, Corporate Presentations, Journals, Interviews with Experts, and MarketsandMarkets Analysis

4.3 NORTH AMERICA: RNA THERAPEUTICS MARKET, BY INDICATION, 2023 VS. 2028 (USD MILLION)

FIGURE 16 INFECTIOUS DISEASES TO DOMINATE NORTH AMERICAN RNA THERAPEUTICS MARKET DURING FORECAST PERIOD

	Infectious Diseases	Rare Genetic/Hereditary Diseases	Other Indications
2023	3,860.2	263.0	66.4
2028	5,793.3	430.1	110.6

Source: National Institutes of Health (NIH), US Food and Drug Administration (US FDA), World Health Organization (WHO), European Pharmaceutical Review, American Society of Gene & Cell Therapy (ASGCT), European Society of Gene and Cell Therapy (ESGCT), RNA Society (US), European Medicines Agency (EMA), Alliance for Regenerative Medicine (ARM), Annual Reports, SEC Filings, Press Releases, Investor Presentations, Corporate Presentations, Journals, Interviews with Experts, and MarketsandMarkets Analysis

5 MARKET OVERVIEW

5.1 INTRODUCTION

The global RNA therapeutics market is projected to grow at a CAGR of 5.6% from 2023 to 2028. Growth in this market is largely driven by factors such as the increasing number of partnerships and collaborations among market players and RNA technology manufacturers, expanding modalities for RNA therapeutics, and the rising number of emergency use authorizations and approvals for COVID-19 booster vaccines.

On the other hand, the discontinuation/recalls of RNA therapeutic products is expected to hinder market growth.

5.2 MARKET DYNAMICS

FIGURE 17 DRIVERS, RESTRAINTS, OPPORTUNITIES, AND CHALLENGES: RNA THERAPEUTICS MARKET

DRIVERS	▪ Increasing partnerships and collaborations between market players and RNA technology manufacturers ▪ Increasing number of novel modalities for RNA therapeutics ▪ Growing number of emergency use authorizations and approvals for COVID-19 booster vaccines
RESTRAINTS	▪ Discontinuation or recalls of RNA therapeutic products
OPPORTUNITIES	▪ Higher progress in development of RNA aptamer-based therapeutics
CHALLENGES	▪ Rapid degradation by ubiquitous RNases in environment and tissues with strong immunogenicity of exogenous RNA

Source: Secondary Literature, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 4 IMPACT ANALYSIS: RNA THERAPEUTICS MARKET

FACTOR	IMPACT		
	1-2 YEARS	3-4 YEARS	5 YEARS
DRIVERS			
Increasing partnerships and collaborations between market players and RNA technology manufacturers	High	High	High
Increasing number of novel modalities for RNA therapeutics	Moderate	High	High
Growing number of emergency use authorizations and approvals for COVID-19 booster vaccines	High	High	High
RESTRAINTS			
Discontinuation or recalls of RNA therapeutic products	Moderate	Moderate	Moderate
OPPORTUNITIES			
Higher progress in development of RNA aptamer-based therapeutics	Low	Moderate	High
CHALLENGES			
Rapid degradation by ubiquitous RNases in environment and tissues with strong immunogenicity of exogenous RNA	Moderate	Low	Low

5.2.1 DRIVERS

5.2.1.1 Increasing partnerships and collaborations between market players and RNA technology manufacturers

RNA therapeutics is a rapidly growing field that can revolutionize the treatment of a wide range of diseases. The rapid growth of this market has been fueled by strategic partnerships and collaborations, both among market players and RNA technology manufacturers.

Partnerships and collaborations between market players accelerate the development and commercialization of RNA therapeutics. By pooling their resources and expertise, companies can share the costs and risks of drug development and gain access to new technologies and markets. For instance, in January 2022, Pfizer Inc. and Beam Therapeutics Inc. entered a four-year exclusive research collaboration focused on in vivo base editing programs aimed at three targets for rare genetic diseases of the liver, muscle, and central nervous system. This collaboration can produce novel therapeutics for patients with rare diseases by combining Beam Therapeutics's in vivo delivery technologies, which deliver base editors to target organs via messenger RNA (mRNA) and lipid nanoparticles (LNPs), with Pfizer's expertise in the development of drugs and vaccines.

Moreover, increasing collaborations between pharmaceutical and biotechnology companies and RNA technology manufacturers ensure their access to cutting-edge technologies and platforms for RNA therapeutics synthesis, distribution, and optimization. The ability to create more tailored, stable, and effective RNA-based therapies is improved by this combination.

Some of the notable strategic collaborations and partnerships for RNA therapeutics include the following:

- In April 2023, Sensible Biotechnologies and Ginkgo Bioworks partnered to generate in vivo RNA for the production of RNA-based therapeutics.
- In April 2023, MiNA Therapeutics announced a multi-target research collaboration and option licensing agreement with BioMarin Pharmaceutical Inc. to identify, advance, and commercialize RNAa therapeutic candidates for treating several rare genetic diseases. Under the terms of the agreement, BioMarin will use MiNA Therapeutics' patented RNAa algorithm and technology platform to identify and characterize RNAa compounds, targeting a variety of genetic diseases that currently have no or few therapeutic options.
- In January 2023, Moderna, Inc. and CytomX Therapeutics, Inc. announced a partnership and licensing agreement to develop investigational mRNA-based conditionally activated therapies utilizing Moderna's mRNA technologies and CytomX's Probody therapeutic platform.
- In December 2022, Eli Lilly and Company and ProQR Therapeutics N.V. expanded their licensing and collaboration agreement to discover, develop, and commercialize new genetic medicines, utilizing ProQR's proprietary Axiomer RNA editing platform.
- In December 2022, GSK PLC and Wave Life Sciences Ltd. announced a strategic collaboration to advance oligonucleotide therapeutics, which will include Wave's preclinical RNA editing program targeting alpha-1 antitrypsin deficiency (AATD), WVE-006.
- In August 2022, Merck and Orna Therapeutics announced a collaboration agreement to discover, develop, and commercialize multiple programs, including vaccines and therapeutics, in the areas of infectious disease and oncology. Orna's proprietary oRNA technology self-circularizes linear RNAs to form circular RNAs (oRNAs).
- In February 2022, Moderna, Inc. and Thermo Fisher Scientific, Inc. announced a 15-year strategic collaboration agreement for large-scale manufacturing of Spikevax, Moderna's COVID-19 vaccine, and other investigational mRNA medicines in its pipeline in the US.
- In January 2022, Amgen and Arrakis Therapeutics announced a research collaboration to discover and develop RNA degrader therapeutics against a range of difficult-to-drug targets in multiple therapeutic areas. This new class of "targeted RNA degraders" consists of small molecule drugs that selectively destroy RNAs encoding disease-causing proteins by inducing their proximity to nucleases.

5.2.1.2 Increasing number of novel modalities for RNA therapeutics

Traditional RNA-based therapies, such as small interfering RNAs (siRNAs) and messenger RNAs (mRNAs), have paved the way for an array of novel modalities that are revolutionizing the treatment landscape. One of the emerging RNA-based technologies is the development of RNA aptamers, which are single-stranded RNA molecules that are engineered to bind to specific target molecules with high affinity. Aptamers have a significant potential to replace monoclonal antibodies in therapeutic and diagnostic applications since they can be made using chemical synthesis and are more affordable to manufacture, are easier to modify, and cause lower immunogenicity.

Additionally, the emergence of small activating RNAs (saRNAs) and circular RNAs (circRNAs) has expanded the scope of RNA therapeutics. saRNAs are 21-nucleotide long double-stranded RNAs (dsRNAs) complementary to the promoter region of a targeted gene. They can upregulate gene expression, potentially correcting deficiencies associated with genetic disorders and re-activating tumor suppressor genes in multiple types of cancers. For example, a drug named MTL-CEBPA is currently under Phase 2 clinical trials (Sponsor- Mina Alpha Limited) for patients with advanced hepatocellular carcinoma (HCC) and has been shown to be safe and with no maximum dose reached in the first-dose escalation study. CircRNAs, on the other hand, display remarkable stability and serve as natural microRNA sponges, offering a novel strategy for modulating cellular processes. These modalities extend the therapeutic options and provide new approaches for addressing previously challenging therapeutic targets.

Antisense oligonucleotides (ASOs) have emerged as a remarkable class of RNA therapeutics, driving transformative advancements in the treatment of various genetic disorders and diseases. ASO are 18–30 base pairs, single-stranded RNA/DNA molecules designed to specifically inhibit mRNA function. There are two classes of ASOs: RNase H-dependent ASO and RNase H-independent (steric block) ASO. There are three ASO drugs that have received FDA approval to date that includes the following—Nusinersen (by Ionis Pharmaceuticals), Eteplirsen (by Sarepta Therapeutics), and Inotersen (Ionis Pharmaceuticals and Akcea Therapeutics) to treat spinal muscular atrophy (SMA), Duchenne muscular dystrophy (DMD), and familial amyloid polyneuropathy respectively. Additionally, Volanesorsen (by Ionis Pharmaceuticals) had recently received the EU conditional marketing authorization to treat familial chylomicronemia syndrome.

As these modalities continue to evolve, the RNA therapeutics market is poised to witness accelerated growth and transformation in the years ahead.

5.2.1.3 Growing number of emergency use authorizations and approvals for COVID-19 booster vaccines

While both emergency use authorizations (EUA) and FDA approval involve regulatory authorization for medical products, they differ in terms of the purpose, process, level of evidence, duration, and post-authorization requirements. EUA provides a faster pathway for addressing emergencies, while FDA approval signifies comprehensive evaluation and authorization based on extensive clinical data. The EUAs and approvals for COVID-19 booster vaccines have demonstrated the safety and efficacy of RNA vaccines, which is likely to drive further investment in this area.

Advanced mRNA therapeutic applications include the development of vaccines against infectious diseases. So far, mRNA vaccines have created a barrier against a number of infectious diseases, such as respiratory syncytial virus, Zika, and influenza. mRNA-based vaccines showed results against the severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) during the COVID-19 pandemic. The first two FDA-approved SARS-CoV-2 vaccines were Pfizer-BioNTech's BNT162b2 (Comirnaty) and Moderna's mRNA-1273, both of which demonstrated >94% efficacy in a phase III clinical trial. Both vaccines used lipid nanoparticles (LNPs) that were modified with nucleoside and ionizable lipids. To enhance mRNA translation, N1-methylpseudouridine is also used to replace all of the uridines in the mRNA. The SARS-CoV-2 spike protein is encoded by the mRNA sequence with two proline modifications that give the protein a prefusion form.

In August 2022, US FDA amended the EUAs of the Moderna COVID-19 vaccine and the Pfizer-BioNTech COVID-19 vaccine to authorize bivalent formulations of the vaccines for use as a single booster dose at least two months following primary or booster vaccination. Furthermore, the Novavax COVID-19 vaccine, Adjuvanted, is available under EUA to provide a first booster dose to the following individuals 18 years of age and older at least 6 months after the completion of primary vaccination with an authorized or approved COVID-19 vaccine.

The RNA therapeutics market has experienced a transformative phase with the emergence of COVID-19 booster vaccines, granted EUAs, and subsequent approvals. Furthermore, the authorization of booster doses for these vaccines has helped to increase immunity and extend the reach of RNA therapeutics. The rigorous evaluation and authorization processes have underscored the safety and effectiveness of mRNA-based interventions, fostering greater acceptance and trust in RNA therapeutics.

Moreover, the collaborative efforts between pharmaceutical companies, research institutions, and regulatory bodies have set a precedent for expedited development and regulatory pathways. This experience is likely to influence future RNA therapeutics' development and regulatory timelines, streamlining the path to market for novel interventions.

5.2.2 RESTRAINTS

5.2.2.1 Discontinuation or recalls of RNA therapeutic products

Instances of discontinuation or recalls can arise due to various reasons, including unexpected adverse effects, safety concerns, efficacy issues, or regulatory non-compliance. These events can lead to a loss of consumer and healthcare provider confidence in RNA therapeutic products, resulting in reduced demand and adoption.

Furthermore, discontinuations or recalls can affect the reputation of the companies involved in the process and the broader perception of RNA therapeutics as a viable treatment avenue. Investors can also become cautious, decreasing the funding for R&D in the concerned field. Regulatory authorities can also intensify regulations, potentially resulting in more stringent approval processes.

Kynamro, a second-generation 2′- O -methoxyethyl chimeric antisense oligonucleotide (ASO) that can treat patients with homozygous familial hypercholesterolemia, has been discontinued due to some adverse reactions such as injection site reactions, alanine aminotransferase increased, flu-like symptoms, aspartate aminotransferase increased, and liver function test abnormal.

Additionally, some clinical trials have been terminated owing to a lack of efficacy.

TABLE 5 LIST OF RNA THERAPEUTICS FOR WHICH CLINICAL DEVELOPMENT WAS HALTED

THERAPEUTIC	TYPE	INDICATION	REASON FOR TERMINATION
Aganirsen (GS-101)	ASO	Ischaemic central retinal vein occlusion, neovascular glaucoma	Formulation stability issues
Suvodirsen (WVE-210201)	ASO	Duchenne muscular dystrophy	Lack of clinical efficacy, failure to increase dystrophin levels
PRO-040201 (TKM-ApoB, ApoB SNALP)	siRNA	Hypercholesterolaemia	Potential for immune stimulation
AGN 211745 (AGN-745, siRNA-027)	siRNA	Age-related macular degeneration, choroidal neovascularization	Lack of clinical efficacy, TLR3 stimulation
Oblimersen sodium (G3139, Genasense)	ASO	Various malignancies	Lack of clinical efficacy, insufficient delivery, primary endpoints not met

Source: Research Publications, Investor Presentations, and MarketsandMarkets Analysis

RNA therapies are still a relatively new field of medicine, and there is still a lot of research that needs to be done to improve their safety and efficacy. However, the potential benefits of RNA therapies are great, and the industry is working hard to overcome the challenges that it faces.

5.2.3 OPPORTUNITIES

5.2.3.1 Higher progress in development of RNA aptamer-based therapeutics

RNA aptamers are single-stranded nucleic acid molecules that can bind to specific targets with high affinity and specificity. They have been investigated for a variety of therapeutic applications, including cancer, cardiovascular diseases, and infectious diseases.

The progress in the development of RNA aptamer-based therapeutics is a major market opportunity for the RNA therapeutics market. This is because RNA aptamers have a number of potential advantages over traditional drug therapies, which include the following:

- High specificity and affinity: RNA aptamers can bind to their target molecules with high specificity and affinity, resulting in a lesser chance of side effects.
- Targeted delivery: RNA aptamers can be targeted to specific cells or tissues, which can improve their efficacy and reduce their toxicity.
- Scalability: RNA aptamers can be produced in large quantities using a variety of manufacturing methods.

The field of RNA therapeutics has undergone a significant transformation with the emergence of RNA aptamer-based therapeutics. There are currently more pipeline products than approved products. One such example includes pegaptanib sodium (Macugen), discovered by NeXstar Pharmaceuticals, which is an anti-vascular endothelial growth factor (anti-VEGF) RNA aptamer to treat all types of neovascular age-related macular degeneration (AMD). The rapid growth of pipeline products showcases the immense potential of this technology in addressing various diseases. These products are indicative of the expanding range of applications, including cancer treatment, autoimmune diseases, and infectious diseases. The diversity of targets and indications being pursued underscores the broad therapeutic potential of RNA aptamers.

Below is an indicative list of candidates/aptamers under clinical development for various therapeutic applications:

TABLE 6 RNA APTAMERS UNDER CLINICAL DEVELOPMENT

CANDIDATE	SPONSOR/COLLABORATOR	PHASE	ESTIMATED STUDY COMPLETION DATE
Olaptesed pegol	TME Pharma AG, Merck Sharp & Dohme LLC	II	October 2026
Zimura	IVERIC Bio, Inc.	II	September 2024
Avacincaptad pegol 2 mg intravitreal injection	IVERIC Bio, Inc.	III	January 2025
Pegcetacoplan	Swedish Orphan Biovitrum Apellis Pharmaceuticals, Inc.	II	March 31, 2024
Pegcetacoplan	Apellis Pharmaceuticals, Inc.	III	July 2027

Source: Clinicaltrialsl.gov

Leading participants in this segment are spread across different regions, contributing to the global growth of RNA aptamer-based therapeutics. Companies such as Archemix Corporation (US), NOXXON Pharma (Germany), IVERIC bio, Inc. (US) (part of Astellas Pharma, Inc.), Regado Biosciences (US), and SomaLogic (US) advance their RNA aptamer platforms by leveraging cutting-edge technologies to design and optimize aptamers with enhanced affinity and selectivity, driving the development of novel therapeutic options.

5.2.4 CHALLENGES

5.2.4.1 Rapid degradation by ubiquitous RNases in environment and tissues with strong immunogenicity of exogenous RNA

The progress of RNA therapeutics faces significant challenges that have led to the need for innovative solutions in their development and manufacturing. One of the primary obstacles pertains to the inherent susceptibility of exogenous RNA to rapid degradation by ubiquitous RNases present in both the environment and tissues.

Another critical challenge involves effectively delivering negatively charged RNA molecules across the hydrophobic cytoplasmic membrane of target cells. Additionally, the strong immunogenicity of exogenous RNA posed a significant obstacle, leading to cell toxicity and hampering the translation of RNA into therapeutic proteins.

Despite the numerous advantages of aptamers over antibodies, their clinical translation encountered complexities rooted in suboptimal pharmacokinetic properties and the intricacies of selection techniques. The unique properties of aptamers necessitated the innovation of novel approaches to enhance their stability, extend their presence within the body, and streamline the selection process for improved clinical success rates.

As RNA molecules, especially mRNA, are inherently unstable, the development of sophisticated delivery vehicles is paramount to safeguard the therapeutic cargo from RNAase degradation. Novel delivery vehicles capable of precisely delivering RNA drugs to their intended sites of action are imperative, enabling the RNA drug to effectively enter the cytoplasm and exert its therapeutic effects.

These challenges range from preventing RNA molecules from degrading or hindering their distribution into cells to reducing immunogenicity and improving selection methods for the best possible therapeutic results. The therapeutic potential of RNA-based therapies could be achieved with a sustained effort to find solutions to these challenges.

5.3 TECHNOLOGY ANALYSIS (RNA VACCINE AND THERAPY MANUFACTURING)

Viral delivery platforms such as adeno-associated virus (AAV) vectors have gained recognition and authorization for their application in vaccine development and gene therapies. Despite their widespread adoption, these viral systems can prompt immunogenic responses and are associated with more frequent systemic adverse effects compared to alternative methods. Furthermore, the manufacturing procedure for these systems can be intricate, requiring high levels of viral titers for effective gene therapy. In contrast, non-viral delivery methods are anticipated to exhibit superior safety profiles and simplified manufacturing processes, presenting the potential for a streamlined approach.

The utilization of mRNA technology, relying on non-viral delivery mechanisms, offers a remarkable level of adaptability. The introduction of mRNA into a cell's cytosol can trigger the synthesis of a desired protein, serving as a therapeutic or preventative agent. It can also serve as an immunogen, eliciting immune responses for vaccination, facilitating the replacement of malfunctioning proteins, or activating anti-tumor reactions. The promising initial outcomes observed in the context of three mRNA vaccines targeting SARS-CoV-2 hold implications that extend beyond the current pandemic. These outcomes act well for analogous strategies in combatting ailments such as cancer, cardiovascular disorders, and other infectious diseases.

5.3.1 PARAMETERS OF MANUFACTURING

Given the expedited timeline spanning from inception to clinical stages and eventual approval, mRNA technology presents a compelling proposition. This pertains not only to its responsiveness in tackling infectious disease outbreaks and pandemics but also extends to the creation of innovative therapeutic strategies to address conditions that currently lack effective treatments.

The generation of mRNA involves an enzymatic process conducted through in vitro synthesis, marking a departure from the traditional in vivo protein expression approach that necessitates laborious steps such as cloning and amplification. By employing an in vitro synthesis method, the requirement to eliminate cells or host cell proteins is obviated. This streamlined manufacturing process enables GMP facilities to swiftly transition to a fresh protein target (therapeutic candidate in development) with minimal adjustments to both process and formulation.

5.3.2 CIRCULAR RNA ENGINEERING

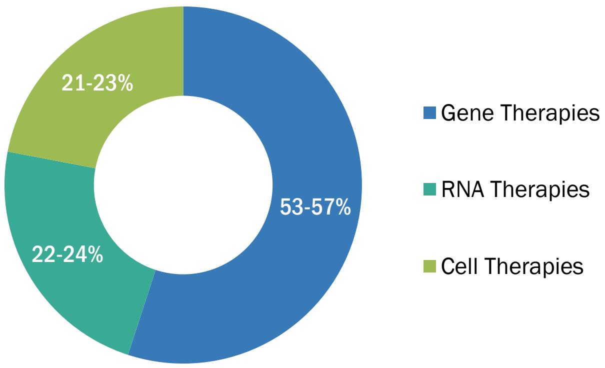
RNA circularization holds the potential to revolutionize RNA-based therapeutics, enhancing the stability of these inherently fleeting molecules. Nevertheless, due to the distinct mechanisms underlying circRNA and mRNA translation, the extensive insights garnered over the years regarding the optimization of mRNA translation might not necessarily be directly applicable to circRNAs. While prior experiments have confirmed the feasibility of protein expression from circRNAs, the precise rules governing circRNA translation remain incompletely understood. Several research studies have been performed to mitigate this issue. As per an activity published in July 2022, an effort was made to decipher these rules, resulting in the formulation of several universally applicable strategies for enhancing translation in circRNA engineering.

To facilitate this study, a modular cloning system for circRNA was established, facilitating the evaluation of various sequence alterations and the independent optimization of multiple parameters. While the majority of sequences were crafted through informed design, the platform also accommodated the creation of random libraries, as demonstrated through IRES shuffling. Additionally, distinct libraries can be flexibly combined to create comprehensive datasets of RNA elements. An illustrative example involves the juxtaposition of shuffled 5′ UTR and shuffled 3′ UTR segments flanking a reporter gene.

5.4 PIPELINE ANALYSIS

According to the recent report published by the American Society of Gene+Cell Therapy (ASCT) in 2022, the gene, cell, and RNA therapy pipeline displayed growth of 7% from the preclinical to the pre-registration phase. This growth resulted in a total number of 3,726 therapies in development, of which more than 50% accounted for gene therapies, including genetically modified therapies such as CAR T-cell therapies. Nearly ~21% to 23% accounted for non-genetically modified cell therapies, and RNA therapies represented the remaining 22–24% of the pipeline. In addition, almost 21 RNA therapies received approval in quarter 4 of 2022. The RNA pipeline showed a growth of 17% throughout 2022. Given below is the diagram for pipeline therapies by category as of 2022.

FIGURE 18 PIPELINE THERAPIES, BY CATEGORY, 2022



Source: Annual Reports, SEC Filings, Investor Presentations, Interviews with Experts, and MarketsandMarkets Analysis

5.4.1 PIPELINE ANALYSIS FOR KEY MARKET PLAYERS (TOP 5)

5.4.1.1 Moderna, Inc.

As of February 2023, Moderna, Inc. has four pivotal RNA therapy programs in Phase 3 development, including COVID-19, RSV, cytomegalovirus (CMV), and seasonal flu. The below table gives an indicative list of pipeline candidates for the company.

TABLE 7 PIPELINE CANDIDATES OF MODERNA, INC.

PROGRAM/APPLICATION	PRODUCT/CANDIDATE ID	PHASE (PRECLINICAL TO COMMERCIAL)
COVID-19 Vaccines	mRNA-1273/Spikevax	Commercial
	mRNA-1273.214	Commercial
	mRNA-1273.222	Commercial
	mRNA-1273.529	Phase 3
	mRNA-1273.351	Phase 2
	mRNA-1273.617	Phase 2
	mRNA-1273.211	Phase 2
	mRNA-1273.213	Phase 2
	mRNA-1283	Phase 2

Flu Vaccines	mRNA-1010	Phase 3
	mRNA-1020	Phase 1
	mRNA-1030	Phase 1
	mRNA-1011	Preclinical
	mRNA-1012	Preclinical
Older Adults RSV Vaccine	mRNA-1345	Phase 3
COVID-19+Flu Vaccine	mRNA-1073	Phase 1
COVID-19+Flu Vaccine+RSV Vaccine	mRNA-1230	Phase 1
Flu+RSV Vaccine	mRNA-1045	Phase 1
Endemic HCoV	mRNA-1287	Preclinical
COVID-19 Vaccine (Adolescent)	mRNA-1273/Spikevax	Commercial
COVID-19 Vaccine (Pediatric)	mRNA-1273/Spikevax	Commercial
Pediatric RSV Vaccine	mRNA-1345	Phase 1
Pediatric hMPV+PIV3 Vaccine	mRNA-1653	Phase 1
Pediatric hMPV+RSV Vaccine	mRNA-1365	Preclinical
CMV Vaccine	mRNA-1647	Phase 3
EBV Vaccine	mRNA-1189 mRNA-1195	Phase 1 Preclinical
HSV Vaccine	mRNA-1608	Preclinical
HIV Vaccine	mRNA-1644 mRNA-1574	Phase 1 Phase 1
Zika Vaccine	mRNA-1893	Phase 2
Nash Vaccine	mRNA-1215	Phase 1
Heart Failure	mRNA-0184	Phase 1
PD-L1 Autoimmune Hepatitis	mRNA-6981	Phase 1
Personalized Cancer Vaccine	mRNA-4157	Phase 3
KRAS Vaccine	mRNA-5671	Phase 1
Checkpoint Vaccine	mRNA-4359	Phase 1
Solid Tumor/Lymphoma Vaccine	mRNA-2752	Phase 1
IL-12 Solid Tumors	MEDI1191	Phase 1
VEGF-A Myocardial Ischemia	AZD-5601	Phase 2

Source: Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

5.4.1.2 Pfizer, Inc.

As of January 2023, Pfizer, Inc. has nearly 34 candidates in Phase 1, 37 candidates in Phase 2, 23 candidates in Phase 3, and 16 candidates in the registration phase, making it a total of more than 100 candidates in development. The below table gives an indicative list of pipeline candidates for the company.

TABLE 8 PIPELINE CANDIDATES OF PFIZER INC.

PROGRAM/APPLICATION	PRODUCT/CANDIDATE ID	PRODUCT TYPE	PHASE (PRECLINICAL TO COMMERCIAL)
COVID-19 Vaccine	COMIRNATY	mRNA	Registration
	BNT162b2 bivalent (BA.4/BA.5)	mRNA	Registration
	BNT162b2 bivalent (BA.4/BA.5)	mRNA	Registration
	Omicron XBB.1.5-Adapted Monovalent COVID-19 Vaccine	mRNA	Registration
Influenza Vaccines (Adult)	PF-07252220	mRNA	Phase 3
Influenza Vaccines (Adult)	PF-0784510	saRNA	Phase 1
COVID-19+Influenza	PF-07926307	mRNA	Phase 1
Varicella	PF-07911145	mRNA	Phase 1/2

Source: Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

5.4.1.3 Alnylam Pharmaceuticals, Inc.

Alnylam Pharmaceuticals, Inc. aims to add two to four new investigational drug (IND) products to their pipeline each year, using RNAi technology. The present pipeline candidates of the company are targeted toward 4 therapeutic applications- infectious diseases, genetic medicines, cardio-metabolic diseases, and central nervous system (CNS) and ocular diseases. The below table gives an indicative list of pipeline candidates for the company.

TABLE 9 PIPELINE CANDIDATES OF ALNYLAM PHARMACEUTICALS, INC.

PROGRAM/APPLICATION	PRODUCT/CANDIDATE ID	PRODUCT TYPE	PHASE (PRECLINICAL TO COMMERCIAL)
Genetic Medicine/ATTR	Vutrisiran	RNAi	Phase 2/3
Genetic Medicine/Heamophilia	Fitusiran	RNAi	Phase 2/3
Genetic Medicine/Complement-mediated Diseases	Cemdisiran	RNAi	Phase 2/3
Genetic Medicine/ATTR	ALN-TTRsc04	RNAi	Early Stage/Phase 1
Genetic Medicine/Alpha-1 Liver Disease	Belcesiran	RNAi	Early Stage/Phase 1
Infectious Diseases/Hepatitis B Virus Infection	ALN-HBV02	RNAi	Early Stage/Phase 1
Cardio-metabolic Diseases/Hypertension	Zilebisiran	RNAi	Early Stage/Phase 1

Cardio-metabolic Diseases/NASH	ALN-HSD	RNAi	Early Stage/Phase 1
CNS/Alzheimer's	ALN-APP	RNAi	Early Stage/Phase 1
Cardio-metabolic Diseases/NASH	ALN-PNP	RNAi	Early Stage/Phase 1
Cardio-metabolic Diseases/Type 2 Diabetes	ALN-KHK	RNAi	Early Stage/Phase 1

Source: Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

5.4.1.4 Novartis AG

As of 2022, Novartis has built a pipeline of nearly 150 candidates (clinical development). The below table gives an indicative list of pipeline candidates for the company.

TABLE 10 PIPELINE CANDIDATES OF NOVARTIS AG

PROGRAM/APPLICATION	PRODUCT/CANDIDATE ID	PRODUCT TYPE	PHASE (PRECLINICAL TO COMMERCIAL)
Cardiovascular	KJX839/ Leqvio	siRNA	Phase 3
Cardiovascular (Primary Prevention)	KJX839/ Leqvio	siRNA	Phase 3
Cardiovascular (Pediatric Hyperlipidemia)	KJX839/ Leqvio	siRNA	Phase 3
NSCLC/Solid Tumor	MGY825	KARS (Lysyl-tRNA synthetase) Inhibitor	Phase 1

Source: Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

5.4.1.5 Ionis Pharmaceuticals, Inc.

Ionis Pharmaceuticals, Inc. applies its proprietary antisense technology platform to discover and develop innovative medicines based on RNA technology. The company's pipeline as of 2022 is comprised of nearly 40 potential medicines designed to treat a broad range of diseases. The below table gives an indicative list of pipeline candidates for the company.

TABLE 11 PIPELINE CANDIDATES OF IONIS PHARMACEUTICALS, INC.

PROGRAM/APPLICATION	PRODUCT/CANDIDATE ID	PRODUCT TYPE	PHASE (PRECLINICAL TO COMMERCIAL)
Neurological	Ulefnersen	Antisense RNA	Phase 3
	Zilganersen	Antisense RNA	Phase 2
Cardiovascular	Olezarsen	Antisense RNA	Phase 3
	Fesomersen	Antisense RNA	Phase 2
	IONIS-AGT-LRx	Antisense RNA	Phase 2
	ION904	Antisense RNA	Phase 2
Specialty Rare	Donidalorsen	Antisense RNA	Phase 3

	Sapablursen	Antisense RNA	Phase 2
Non-Alcoholic Steatohepatitis	ION224	Antisense RNA	Phase 2

Note: The above table includes Ionis-owned pipeline products only.

Source: Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

5.5 PORTER'S FIVE FORCES ANALYSIS

Porter's Five Forces Analysis is a framework that determines the competitive intensity in the RNA therapeutics market. It helps with decision-making related to the entry or exit of players from the market. This section analyzes the market from five different perspectives, namely, the degree of competition within the industry, the threat of new entrants, the bargaining power of suppliers, the bargaining power of buyers, and the threat of substitutes.

TABLE 12 PORTER'S FIVE FORCES ANALYSIS: RNA THERAPEUTICS MARKET

PORTER'S FIVE FORCES	IMPACT
Intensity of Competitive Rivalry	High
Bargaining Power of Suppliers	Moderate
Bargaining Power of Buyers	High
Threat of Substitutes	Moderate
Threat of New Entrants	Low

5.5.1 THREAT OF NEW ENTRANTS

Many players in the RNA therapeutics market have candidates under development (in the pipeline) and very few approved products (FDA approved), which drive the market revenue. This indicates the stiff competition in this market, which restricts the entry of new businesses. Established companies have a competitive advantage over new entrants, as they operate efficiently due to the benefits of better economies of scale, thereby providing the established players with an opportunity to offer their products at lower prices. Thus, already established players operating in the market can pose a hindrance to the new entrants despite their ability to provide systems efficiently and charge low prices for their products.

New entrants in the market have less experience as compared to established players. It becomes difficult for start-ups to set up large manufacturing facilities; most start-up companies go for small-scale units, which in turn is not sufficient to produce items as per needs. These high entry barriers pose a competitive disadvantage for new entrants, limiting their entry into the global RNA therapeutics market.

To ensure a prominent position in the market, the new entrants must improve their brand value to compete effectively and expand their business operations, and for both of these, they require heavy investment, in turn increasing the overall cost of production capacity. Initial capital requirements represent an entry barrier, depending upon the type of product a new player is attempting to manufacture. Hence, the threat of new entrants in the RNA therapeutics market is low.

5.5.2 THREAT OF SUBSTITUTES

The threat of substitutes for RNA therapeutics is moderate, as there are other kinds of treatments available in the market compared to RNA therapeutics therapies, which require less stringent regulatory approvals and provide better results with greater efficacy and lower cost.

5.5.3 BARGAINING POWER OF BUYERS

Few companies in the market provide FDA/other regulatory-approved therapies. Business-oriented buyers prefer forms that provide end-to-end services to ensure efficient quality checks and improved compliance that have been approved and commercialized by all regulatory bodies. Hence, the bargaining power of buyers in this market is high.

5.5.4 BARGAINING POWER OF SUPPLIERS

The number of suppliers of raw materials and consumables can influence the prices of final products. The availability of major constituents of drug development is not uniform across the industry and depends upon the level of vertical integration, making the industry vulnerable to an exchange rate risk. It also affects the ability of competitors to source raw materials at a cheaper rate.

Moreover, the cost of materials also depends on the demand and supply balance. Following the supply shortage and higher demand, the cost of raw materials may increase, again increasing the bargaining power of the suppliers. Apart from this, the cost of the raw materials is also high, particularly the cost associated with raw materials used for developing, qualifying, and validating RNA therapeutics. The overall bargaining power of suppliers in this market is moderate.

5.5.5 INTENSITY OF COMPETITIVE RIVALRY

The market for RNA therapeutics is highly consolidated, with very few manufacturers competing with each other.

The key competitors such as Moderna, Inc. (US), Pfizer, Inc. (US), Alnylam Pharmaceuticals (US), Novartis AG (US), and Ionis Pharmaceuticals, Inc. (US) possess substantial market share, target attractive market segments, and evaluate the pricing of their products accordingly. Hence, the presence of these key companies has increased the intensity of rivalry in this market as all the companies have similar potential to attract customers. Therefore, the intensity of competitive rivalry in the RNA therapeutics therapy market is high.

5.6 REGULATORY ANALYSIS

TABLE 13 NORTH AMERICA: LIST OF REGULATORY BODIES, GOVERNMENT AGENCIES, AND OTHER ORGANIZATIONS

COUNTRY	ORGANIZATION NAME	ORGANIZATION TYPE	SHORT DESCRIPTION
US	US Food and Drug Administration	Government Regulatory Body	It is a cabinet-level department of the US government concerned with US policies regarding drugs and food safety. FDA regulates food, drugs, biologics, medical devices, electronic products, cosmetics, veterinary products, and tobacco products.
US	Center for Biologics Evaluation and Research (CBER)	Center within US FDA	It is a part of the US FDA that regulates biological products for human use under federal laws, including the Public Health Service Act and the Federal Food, Drug, and Cosmetic Act.
US	Center for Drug Evaluation and Research (CDER)	Center within US FDA	It is a part of the US FDA that regulates OTC and prescription drugs, including biological therapeutics and generic drugs. CDER covers more than just medicines—fluoride toothpaste, antiperspirants, dandruff shampoos, and sunscreens are all considered drugs.
US	Alliance of Advanced Biomedical Engineering (AABME)	Part of The American Society of Mechanical Engineers	It is a part of the American Society of Mechanical Engineers (ASME) initiative to stimulate biomedical innovation by bringing together and providing resources to the biomedical engineering community.

Source: Government websites and MnM analysis

TABLE 14 EUROPE: LIST OF REGULATORY BODIES, GOVERNMENT AGENCIES, AND OTHER ORGANIZATIONS

COUNTRY/REGION	ORGANIZATION NAME	ORGANIZATION TYPE	SHORT DESCRIPTION
Europe (HQ-Netherlands)	European Medicines Agency (EMA)	Government Regulatory Body	It protects and promotes human and animal health by evaluating and monitoring medicines within the European Union (EU) and the European Economic Area (EEA).
UK	Medicines and Healthcare Products Regulatory Agency (MHRA)	Government Regulatory Body	It is a government regulatory agency of the Department of Health and Social Care in the UK, which regulates the safety of medicines and medical devices.

Source: Government websites and MnM analysis

TABLE 15 ASIA PACIFIC: LIST OF REGULATORY BODIES, GOVERNMENT AGENCIES, AND OTHER ORGANIZATIONS

COUNTRY	ORGANIZATION NAME	ORGANIZATION TYPE	SHORT DESCRIPTION
China	NMPA (National Medical Product Administration)	Government Regulatory Body	Due to the reorganization of the Chinese government in 2018, CFDA (China Food and Drug Administration) was renamed NMPA, and its food supervision function was transferred to SAMR (State Administration for Market Regulation of China).
Japan	Pharmaceuticals and Medical Devices Agency (PMDA)	Government Regulatory Body	It is a regulatory agency in Japan that works with the Ministry of Health, Labour, and Welfare to ensure the safety, efficacy, and quality of pharmaceuticals and medical devices.
India	Central Drugs Standard Control Organization (CDSCO)	Government Regulatory Body	It functions under the Directorate General of Health Services, Ministry of Health & Family Welfare, Government of India. It is the National Regulatory Authority (NRA) that regulates medical products manufactured, imported, and distributed in the country.
Australia	Department of Health and Aged Care	Government Regulatory Body	The TGA regulates therapeutic goods consisting of gene, cell, and tissue therapies under the Therapeutic Goods Act 1989.
South Korea	Ministry of Food and Drug Safety (MFDS)	Government Regulatory Body	It is the regulatory authority that regulates therapeutic products, medical devices, and food in the country.

Source: Government websites and MnM analysis

TABLE 16 REST OF THE WORLD: LIST OF REGULATORY BODIES, GOVERNMENT AGENCIES, AND OTHER ORGANIZATIONS

COUNTRY	ORGANIZATION NAME	ORGANIZATION TYPE	SHORT DESCRIPTION
Brazil	Brazilian Health Regulatory Agency (ANVISA)	Government Regulatory Body	The Brazilian Health Regulatory Agency (ANVISA) is an autarchy linked to the Ministry of Health, part of the Brazilian National Health System (SUS) as the coordinator of the Brazilian Health Regulatory System (SNVS), present throughout the national territory.
UAE	Dubai Health Authority (DHA)	Government Regulatory Body	Dubai Health Authority (DHA), under the instructions of the Ministry of Health and Prevention (MOHAP), lists the medications in use in the UAE.

Source: Government websites and MnM analysis

5.7 VALUE CHAIN ANALYSIS

The RNA therapeutics market has grown significantly in the last decade. The value chain of RNA therapeutics begins with R&D. Extensive R&D is required to develop therapies. R&D activities are divided into in-house and outsourced tasks. In-house tasks focus on critical activities that deal with fundamental analysis and the electronic interpretation of testing parameters. During the R&D phase, basic research is conducted, and prototypes are built, adding <10% value to the product. The manufacturing and package & storage phases claim most of the addition (around 60–70%). Around 10% to 20% of the value is added during marketing & sales and post-sales activities, including geographical distribution and access. However, the gross profit earned by every stakeholder depends on the region and product type.

R&D

The R&D phase comprises collection and product development, process regulatory approval, and sustainable engineering. In the RNA therapeutics market, various players have invested significantly in R&D activities, which has led to the launch of advanced or unique products in the market.

MANUFACTURING AND PACKAGING & STORAGE

The manufacturing and packaging & storage phases involve the maintenance of viral vectors, DNA plasmids, CRISPR-Associated Protein 9 (CRISPR-Cas9), and manufacturing RNA therapy products. Companies in this market have expanded their activities, primarily to set up new sites to produce RNA therapeutics products.

DISTRIBUTION, MARKETING & SALES, AND POST-SALES

These are essential procedures in the value chain of any product in the market. Players have entered into partnerships to distribute RNA therapeutics therapies to reach a larger customer base.

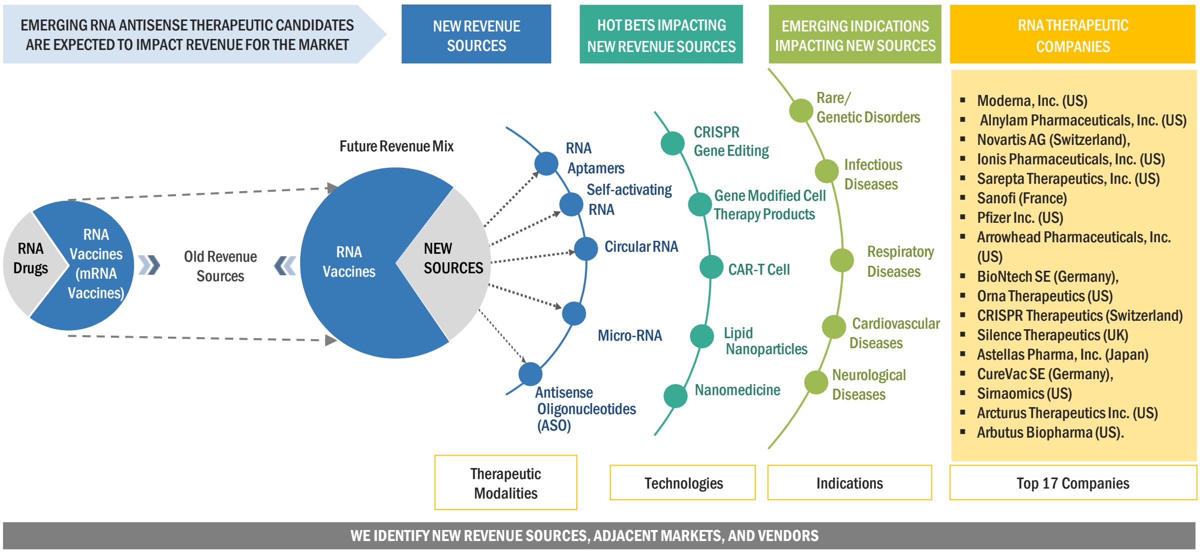
FIGURE 19 VALUE CHAIN ANALYSIS OF RNA THERAPEUTICS MARKET: R&D AND MANUFACTURING PHASES ADD MAXIMUM VALUE

STAGES					
R&D	MANUFACTURING	PACKAGING & STORAGE	DISTRIBUTION	MARKETING & SALES	POST-SALES
▪ Formulation Development ▪ Regulatory Approval	▪ Direct Suppliers ▪ Third-party Suppliers	▪ Packaging ▪ Sterilization	▪ Pharmaceutical and Biotechnology Companies, Specialty Pharmacies		▪ Consulting, Counselling

Source: Annual Reports, SEC Filings, Investor Presentations, Interviews with Experts, and MarketsandMarkets Analysis

5.8 TRENDS/DISRUPTIONS IMPACTING CUSTOMERS' BUSINESSES

FIGURE 20 TRENDS/DISRUPTIONS IMPACTING CUSTOMERS' BUSINESSES

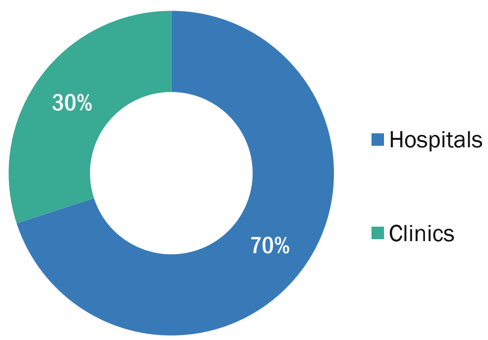


Source: Investor Presentation and MarketsandMarkets Analysis.

5.9 KEY STAKEHOLDERS AND BUYING CRITERIA

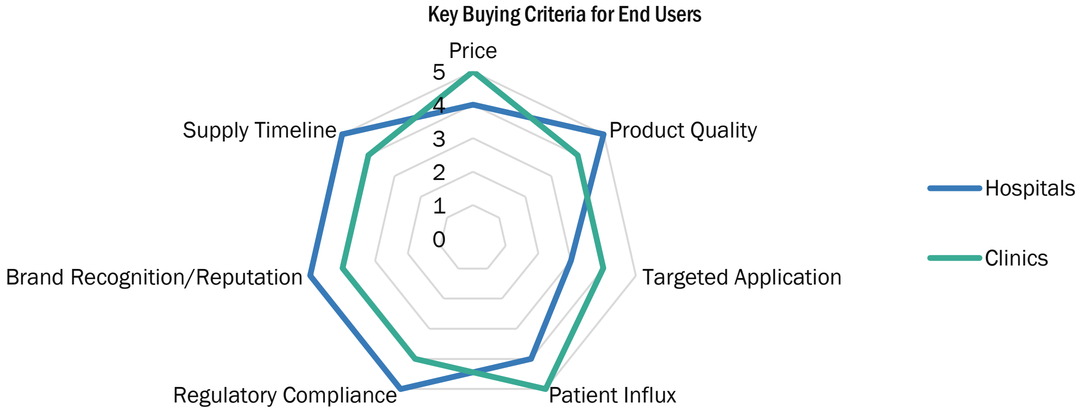
5.9.1 KEY STAKEHOLDERS IN BUYING PROCESS

FIGURE 21 INFLUENCE OF STAKEHOLDERS ON BUYING PROCESS OF RNA-BASED THERAPEUTICS



5.9.2 BUYING CRITERIA FOR RNA THERAPEUTIC PRODUCTS/MEDICINES

FIGURE 22 KEY BUYING CRITERIA FOR END USERS



Note: Respondents rated each criterion on a scale of one (least important) to ten (most important).

5.10 PRICING ANALYSIS

5.10.1 PRICING ANALYSIS FOR FDA-APPROVED RNA THERAPEUTICS, BY PRODUCT

TABLE 17 AVERAGE SELLING PRICE FOR RNA VACCINES AND DRUGS, BY REGION

PRODUCT CATEGORY	PRODUCT	FORM	QUANTITY	ASP (USD)
Drugs	Leqvio (Novartis AG)	Subcutaneous Solution	284 mg/1.5 mL	3,000-4,00
	Spinraza (Biogen)	Intrathecal Solution	12 mg/5 mL	140,000-145,000
	Tegsedi (Alnylam Pharmaceuticals, Inc.)	Subcutaneous Solution	284 mg/1.5 mL	39,000-40,000
	Amvuttra (Alnylam Pharmaceuticals, Inc.)	Subcutaneous Solution	25 mg/0.5 mL	125,000-126,00
	Onpattro (Alnylam Pharmaceuticals, Inc.)	Subcutaneous Solution	2 mg/mL	10,000-11,000 (5 mL)
	Oxlumo (Alnylam Pharmaceuticals, Inc.)	Subcutaneous Solution	94.5 mg/0.5 mL	60,000-62,000
Vaccines	Spikevax (Moderna, Inc.)		Per dose	110-130/per dose
	COMIRNATY (Pfizer, Inc.)		Per dose	30-32/per dose

Note: This is an indicative list of pricing, with ASPs mentioned at a global level.

Source: Company website and MarketsandMarkets Analysis

5.10.2 PRICING TREND ANALYSIS: RNA VACCINES AND DRUGS

Healthcare systems have gradually prepared for the upcoming commercial release of inclisiran (marketed as Leqvio by Novartis). This therapy, designed to reduce LDL-cholesterol levels, is administered twice a year in a medical setting. The drug's approval at the end of 2021, following a lengthy evaluation, has led to it being acclaimed as a possible revolutionary advancement in the field. The medication will be associated with a substantial cost; Novartis has specified that the wholesale acquisition cost (WAC) for each inclisiran injection is USD 3,250 USD, equating to an annual expense of $6,500 for two doses per year.

Earlier, Moderna indicated that it was contemplating a pricing range of USD 110 to USD 130 per dose for its COVID-19 vaccine in the US. This pricing range aligns with what Pfizer Inc. had also mentioned in October 2022 for its COVID-19 vaccines developed in collaboration with BioNTech. The government's Medicare health plan, which caters to seniors, covers a cost of USD 70 per dose for the seasonal influenza vaccine. The decision regarding the pricing of Moderna's vaccine took into account the fact that the past few months in 2022 alone witnessed two to three times more hospitalizations and fatalities due to COVID-19 compared to influenza.

Pfizer, Inc. announced in 2023 that it is considering charging a private-market price between USD 110 and USD 130 per dose for its COVID-19 shot Comirnaty, which was co-developed by BioNTech (Germany). The updated pricing is anticipated to become effective sometime in 2023, most likely within the first quarter. This adjustment will coincide with the depletion of the US's stockpile of COVID-19 vaccines, leading to a shift toward a commercial model, as explained by Pfizer, Inc.

Since the availability of vaccines in the US, the government has covered the expenses and made them available at no cost. In the most recent supply agreement signed between the US and Pfizer along with BioNTech in June of this year, the companies invoiced USD 30.48 per dose. This marked an increase from USD 24 per dose in July 2021 and USD 19.50 per dose in July 2020.

5.11 KEY CONFERENCES AND EVENTS (2023–2024)

TABLE 18 KEY CONFERENCES AND EVENTS IN RNA THERAPEUTICS MARKET (2023–2024)

COUNTRY/REGION	YEAR & MONTH	CONFERENCE/EVENT	DESCRIPTION/AGENDA
US	August 2023	The mRNA Conference 2023	To explore the latest innovations in mRNA-based therapeutics and vaccines for infectious diseases, oncology, CAR-T cell therapies, and beyond.
UK	October 2023	RNA Vaccines and Therapeutics Conference	Discussion focused on the most recent developments in the rapidly evolving science of encoded RNA medicines
Singapore	May 2023	28th Annual International Conference	Explore innovations in the field of RNA vaccines and drugs.
US	April 2024	2024 mRNA Technology Conference	To discuss topics such as rapid advancement of new modalities, groundbreaking mRNA applications for broad-spectrum mRNA vaccines, mRNA cancer treatments, and overcoming mRNA challenges.
US	June 2024	RNA Therapeutics From Concepts to Clinic	Discussion focused on the most recent developments in the RNA medicines industry.

Source: Org. Websites, Interviews with Experts, and MarketsandMarkets Analysis

5.12 ECOSYSTEM MARKET MAP

	PRODUCT/TYPES	INDICATION	END USER
CORE SEGMENT	▪ Vaccines ▪ Drugs ▪ mRNA Therapeutics ▪ RNA Interference Therapeutics ▪ Antisense Oligonucleotide Therapeutics	▪ Infectious Diseases ▪ Rare Genetic/Hereditary Diseases ▪ Cardiovascular Diseases ▪ Other Indications	▪ Hospitals and Clinics ▪ Research Settings

6 RNA THERAPEUTICS MARKET, BY PRODUCT

KEY FINDINGS

- Vaccines accounted for a larger share of 92.3% of the global RNA therapeutics market in 2022.
- The drugs segment is projected to reach USD 7,335.8 million by 2028 from USD 3,489.4 million in 2023, at a CAGR of 16%.
- The global demand for COVID-19 mRNA vaccines has boosted confidence in the potential of RNA-based vaccines to combat infectious diseases.
- The success of RNA-based drugs, such as Patisiran, Lumasiran, and Viltolarsen, has encouraged manufacturers to further develop and invest in the RNA therapeutics drug segment.

6.1 INTRODUCTION

Based on the product segment, the RNA therapeutics market is segmented into vaccines and drugs. In 2022, the vaccines segment accounted for the largest share of 92.3% in the global RNA therapeutics market. With the approval for the flagship RNA vaccines against SARS-COV-2 infection, such as Comirnaty (Pfizer, Inc.) and Spikevax (Moderna, Inc.), the revenue of this segment showcased considerable growth between 2021 and 2022. However, the recent recall/discontinuation of Spikevax (Moderna, Inc.) can hinder revenue growth from 2023 onwards. On the other hand, the high influx of innovative medicine companies in the RNA therapeutics market has rendered it the fastest growth rate of 16% during the forecast period.

TABLE 19 RNA THERAPEUTICS MARKET, BY PRODUCT, 2021–2028 (USD MILLION)

Product	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Vaccines	12,504.0	29,764.0	10,264.4	10,069.7	10,220.7	10,374.1	10,550.4	10,729.8	0.9%
Drugs	1,588.0	2,491.3	3,489.4	4,776.5	6,060.6	7,083.9	7,571.7	7,335.8	16.0%
Total	**14,092.0**	**32,255.3**	**13,753.8**	**14,846.2**	**16,281.4**	**17,458.0**	**18,122.1**	**18,065.5**	**5.6%**

Source: National Institutes of Health (NIH), US Food and Drug Administration (US FDA), World Health Organization (WHO), European Pharmaceutical Review, American Society of Gene & Cell Therapy (ASGCT), European Society of Gene and Cell Therapy (ESGCT), RNA Society (US), European Medicines Agency (EMA), Alliance for Regenerative Medicine (ARM), Annual Reports, SEC Filings, Press Releases, Investor Presentations, Corporate Presentations, Journals, Interviews with Experts, and MarketsandMarkets Analysis

6.2 VACCINES

6.2.1 INCREASED NUMBER OF APPROVALS AND HIGHER INVESTMENTS FOR COVID-19 BOOSTER VACCINES TO DRIVE MARKET

Increasing investments in R&D, advancements in RNA synthesis and delivery technologies, and a growing understanding of RNA-based therapeutics have propelled the growth of the RNA therapeutics market. The R&D departments of the major companies have focused on new vaccines due to the increasing investments in this market. For instance, in August 2023, CureVac announced that the first participant was dosed in the Phase 2 study of monovalent and bivalent modified mRNA COVID-19 vaccine candidates, developed in collaboration with GSK.

Moreover, in March 2022, Moderna announced it had begun its second phase 1 HIV vaccine trial. This latest phase 1 trial is an open-label, multicenter, randomized study (HVTN 302), which evaluates the safety and immunogenicity of their experimental HIV trimer mRNA vaccine (mRNA-1574).

Furthermore, the global demand for COVID-19 mRNA vaccines has boosted confidence in the potential of RNA-based vaccines to combat infectious diseases. mRNA COVID-19 vaccines have showcased the potential of RNA therapeutics in revolutionizing vaccine development and public health response. Government bodies and organizations worldwide have invested heavily in mRNA technology and infrastructure to develop vaccines for infectious diseases. The RNA therapeutics vaccine market has also witnessed continued growth and innovation, driven by the momentum and confidence gained through the COVID-19 pandemic response and advancements in RNA-based technologies. RNA therapeutics vaccines, including the COVID-19 vaccines developed by Pfizer-BioNTech, CureVac, NovaVax, and Moderna, have harnessed mRNA technology to create highly effective vaccines against the SARS-CoV-2 virus.

TABLE 20 RNA THERAPEUTICS MARKET FOR VACCINES, BY REGION, 2021–2028 (USD MILLION)

Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
North America	2,860.6	6,803.3	2,344.1	2,297.6	2,330.1	2,362.9	2,401.0	2,439.7	0.8%
Europe	4,644.0	11,063.4	3,818.4	3,749.0	3,808.3	3,868.5	3,937.5	4,007.6	1.0%
Asia Pacific	4,370.9	10,407.3	3,590.1	3,523.0	3,576.9	3,631.5	3,694.3	3,758.2	0.9%
Rest of the World	628.5	1,490.0	511.8	500.1	505.5	511.0	517.6	524.3	0.5%
Total	**12,504.0**	**29,764.0**	**10,264.4**	**10,069.7**	**10,220.7**	**10,374.1**	**10,550.4**	**10,729.8**	**0.9%**

Source: National Institutes of Health (NIH), US Food and Drug Administration (US FDA), World Health Organization (WHO), European Pharmaceutical Review, American Society of Gene & Cell Therapy (ASGCT), European Society of Gene and Cell Therapy (ESGCT), RNA Society (US), European Medicines Agency (EMA), Alliance for Regenerative Medicine (ARM), Annual Reports, SEC Filings, Press Releases, Investor Presentations, Corporate Presentations, Journals, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 21 NORTH AMERICA: RNA THERAPEUTICS MARKET FOR VACCINES, BY COUNTRY, 2021–2028 (USD MILLION)

Country	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
US	2,573.1	6,123.0	2,110.9	2,070.2	2,100.6	2,131.4	2,166.9	2,203.0	0.9%
Canada	287.5	680.3	233.2	227.5	229.5	231.6	234.1	236.6	0.3%
Total	**2,860.6**	**6,803.3**	**2,344.1**	**2,297.6**	**2,330.1**	**2,362.9**	**2,401.0**	**2,439.7**	**0.8%**

Source: National Institutes of Health (NIH), US Food and Drug Administration (US FDA), World Health Organization (WHO), European Pharmaceutical Review, American Society of Gene & Cell Therapy (ASGCT), European Society of Gene and Cell Therapy (ESGCT), RNA Society (US), European Medicines Agency (EMA), Alliance for Regenerative Medicine (ARM), Annual Reports, SEC Filings, Press Releases, Investor Presentations, Corporate Presentations, Journals, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 22 EUROPE: RNA THERAPEUTICS MARKET FOR VACCINES, BY COUNTRY, 2021–2028 (USD MILLION)

Country/Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Germany	1,390.0	3,319.0	1,148.2	1,129.9	1,150.5	1,171.4	1,195.0	1,219.1	1.2%
France	1,322.6	3,153.1	1,089.0	1,070.0	1,087.6	1,105.6	1,126.1	1,147.0	1.0%
UK	816.0	1,947.2	673.2	662.1	673.7	685.5	698.9	712.6	1.1%
Rest of Europe	1,115.5	2,644.2	908.0	887.0	896.5	906.0	917.4	929.0	0.5%
Total	**4,644.0**	**11,063.4**	**3,818.4**	**3,749.0**	**3,808.3**	**3,868.5**	**3,937.5**	**4,007.6**	**1.0%**

Source: National Institutes of Health (NIH), US Food and Drug Administration (US FDA), World Health Organization (WHO), European Pharmaceutical Review, American Society of Gene & Cell Therapy (ASGCT), European Society of Gene and Cell Therapy (ESGCT), RNA Society (US), European Medicines Agency (EMA), Alliance for Regenerative Medicine (ARM), Annual Reports, SEC Filings, Press Releases, Investor Presentations, Corporate Presentations, Journals, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 23 ASIA PACIFIC: RNA THERAPEUTICS MARKET FOR VACCINES, BY COUNTRY, 2021–2028 (USD MILLION)

Country/Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Japan	2,401.8	5,724.0	1,976.3	1,941.2	1,972.6	2,004.6	2,041.1	2,078.3	1.0%
China	978.6	2,331.2	804.5	789.9	802.3	814.9	829.4	844.1	1.0%
South Korea	448.0	1,061.5	364.4	355.8	359.5	363.2	367.6	372.1	0.4%
Rest of Asia Pacific	542.4	1,290.5	444.8	436.1	442.5	448.9	456.3	463.8	0.8%
Total	**4,370.9**	**10,407.3**	**3,590.1**	**3,523.0**	**3,576.9**	**3,631.5**	**3,694.3**	**3,758.2**	**0.9%**

Source: National Institutes of Health (NIH), US Food and Drug Administration (US FDA), World Health Organization (WHO), European Pharmaceutical Review, American Society of Gene & Cell Therapy (ASGCT), European Society of Gene and Cell Therapy (ESGCT), RNA Society (US), European Medicines Agency (EMA), Alliance for Regenerative Medicine (ARM), Annual Reports, SEC Filings, Press Releases, Investor Presentations, Corporate Presentations, Journals, Interviews with Experts, and MarketsandMarkets Analysis

6.3 DRUGS

6.3.1 GROWING FUNDING AND INVESTMENTS IN RNA THERAPEUTICS RESEARCH TO DRIVE MARKET

The growth of the RNA therapeutics drugs market is propelled by a surge in the prevalence of chronic and genetic diseases, increased funding and research investments in RNA therapeutics, technological advancements in RNA synthesis and drug delivery, and an increasing number of FDA approvals for RNA-based drugs. For instance, in November 2020, the FDA approved a supplemental new drug application for Alnylam Pharmaceuticals' Oxlumo (lumasiran) to treat patients with primary hyperoxaluria type 1 (PH1), a rare and serious metabolic disease that often presents with kidney stones. Furthermore, numerous clinical trials are underway for RNA-based drugs targeting various diseases, indicating the continued interest and investment in this transformative therapeutic approach. The RNA therapeutics drugs market is poised for substantial expansion, driven by scientific advancements, growing acceptance of RNA-based therapies, and their potential to revolutionize disease treatment. Moreover, the success and regulatory approvals of RNA-based drugs, such as Patisiran, Lumasiran, Viltolarsen, and other RNA therapeutics, have encouraged further development and investment in the RNA therapeutics drug segment.

However, the complexity of RNA-based drug development and delivery is hindering market growth. RNA molecules are inherently unstable, requiring sophisticated delivery systems to ensure their effective delivery to target cells. Overcoming these challenges is crucial to the widespread adoption of RNA therapeutics drugs in clinical settings.

TABLE 24 RNA THERAPEUTICS MARKET FOR DRUGS, BY REGION, 2021–2028 (USD MILLION)

Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
North America	838.6	1,316.6	1,845.4	2,528.0	3,210.1	3,755.0	4,016.5	3,894.3	16.1%
Europe	559.9	877.1	1,226.8	1,676.9	2,124.6	2,479.8	2,646.8	2,560.7	15.9%
Asia Pacific	171.1	267.9	374.5	511.6	648.0	756.0	806.5	779.9	15.8%
Rest of the World	18.5	29.8	42.7	59.9	77.9	93.1	101.8	100.8	18.7%
Total	**1,588.0**	**2,491.3**	**3,489.4**	**4,776.5**	**6,060.6**	**7,083.9**	**7,571.7**	**7,335.8**	**16.0%**

Source: National Institutes of Health (NIH), US Food and Drug Administration (US FDA), World Health Organization (WHO), European Pharmaceutical Review, American Society of Gene & Cell Therapy (ASGCT), European Society of Gene and Cell Therapy (ESGCT), RNA Society (US), European Medicines Agency (EMA), Alliance for Regenerative Medicine (ARM), Annual Reports, SEC Filings, Press Releases, Investor Presentations, Corporate Presentations, Journals, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 25 NORTH AMERICA: RNA THERAPEUTICS MARKET FOR DRUGS, BY COUNTRY, 2021–2028 (USD MILLION)

Country	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
US	716.5	1,125.7	1,578.8	2,164.0	2,749.5	3,218.0	3,444.2	3,341.3	16.2%
Canada	122.0	190.9	266.7	364.0	460.7	537.0	572.4	553.0	15.7%
Total	**838.6**	**1,316.6**	**1,845.4**	**2,528.0**	**3,210.1**	**3,755.0**	**4,016.5**	**3,894.3**	**16.1%**

Source: National Institutes of Health (NIH), US Food and Drug Administration (US FDA), World Health Organization (WHO), European Pharmaceutical Review, American Society of Gene & Cell Therapy (ASGCT), European Society of Gene and Cell Therapy (ESGCT), RNA Society (US), European Medicines Agency (EMA), Alliance for Regenerative Medicine (ARM), Annual Reports, SEC Filings, Press Releases, Investor Presentations, Corporate Presentations, Journals, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 26 EUROPE: RNA THERAPEUTICS MARKET FOR DRUGS, BY COUNTRY, 2021–2028 (USD MILLION)

Country/Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Germany	179.9	282.4	395.9	542.3	688.6	805.5	861.5	835.3	16.1%
UK	159.5	250.0	349.7	478.2	606.2	707.7	755.7	731.3	15.9%
France	112.7	176.8	247.7	339.1	430.2	502.9	537.6	520.8	16.0%
RoE	107.8	167.9	233.5	317.3	399.6	463.7	492.0	473.2	15.2%
Total	**559.9**	**877.1**	**1,226.8**	**1,676.9**	**2,124.6**	**2,479.8**	**2,646.8**	**2,560.7**	**15.9%**

Source: National Institutes of Health (NIH), US Food and Drug Administration (US FDA), World Health Organization (WHO), European Pharmaceutical Review, American Society of Gene & Cell Therapy (ASGCT), European Society of Gene and Cell Therapy (ESGCT), RNA Society (US), European Medicines Agency (EMA), Alliance for Regenerative Medicine (ARM), Annual Reports, SEC Filings, Press Releases, Investor Presentations, Corporate Presentations, Journals, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 27 ASIA PACIFIC: RNA THERAPEUTICS MARKET FOR DRUGS, BY COUNTRY, 2021–2028 (USD MILLION)

Country/Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
China	76.8	120.5	168.9	231.4	293.7	343.5	367.4	355.6	16.1%
Japan	38.2	60.0	84.1	115.1	146.1	170.9	182.7	176.7	16.0%
India	17.4	27.3	38.3	52.4	66.5	77.7	83.1	79.9	15.9%
Rest of Asia Pacific	38.6	60.0	83.2	112.8	141.7	163.9	173.4	167.6	15.0%
Total	**171.1**	**267.9**	**374.5**	**511.6**	**648.0**	**756.0**	**806.5**	**779.9**	**15.8%**

Source: National Institutes of Health (NIH), US Food and Drug Administration (US FDA), World Health Organization (WHO), European Pharmaceutical Review, American Society of Gene & Cell Therapy (ASGCT), European Society of Gene and Cell Therapy (ESGCT), RNA Society (US), European Medicines Agency (EMA), Alliance for Regenerative Medicine (ARM), Annual Reports, SEC Filings, Press Releases, Investor Presentations, Corporate Presentations, Journals, Interviews with Experts, and MarketsandMarkets Analysis

7 RNA THERAPEUTICS MARKET, BY TYPE

KEY FINDINGS

- The mRNA therapeutics segment accounted for the largest share of 92.1% of the RNA therapeutics market in 2022.
- The large share of the mRNA therapeutics segment can be attributed to the advantages of mRNA over DNA or proteins/peptides and the advancements in vaccine development.
- The mRNA therapeutics segment is projected to reach USD 16,438.0 million by 2028 from USD 12,657.4 million in 2023, at a CAGR of 5.4%.
- The RNA Interference (RNAi) therapeutics segment is anticipated to show the fastest growth rate of 7.1% in the RNA therapeutics market during the forecast period. This segment is projected to reach USD 1,370.6 million by 2028 from USD 971.4 million in 2023, at a CAGR of 7.1%.

7.1 INTRODUCTION

Based on the type segment, the RNA therapeutics market is segmented into mRNA therapeutics, RNA interference (RNAi) therapeutics, antisense oligonucleotide (ASO) therapeutics, and other therapeutics. The mRNA therapeutics segment for the largest share of 92.1% in the RNA therapeutics market in 2022. The large share of this segment can primarily be attributed to the advantages of mRNA over DNA or proteins/peptides and the advancements in vaccine development.

On the other hand, the RNA interference (RNAi) therapeutics segment is expected to form the fastest-growing segment of the RNA therapeutics market due to the increasing precision, versatility, and potential of RNAi for personalized medicine and growing successes in clinical applications.

TABLE 28 RNA THERAPEUTICS MARKET, BY TYPE, 2021–2028 (USD MILLION)

Type	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
mRNA Therapeutics	12,996.4	29,712.5	12,657.4	13,648.6	14,952.6	15,923.6	16,509.8	16,438.0	5.4%
RNA Interference (RNAi) Therapeutics	964.3	2,244.6	971.4	1,064.1	1,183.7	1,287.2	1,355.2	1,370.6	7.1%
Antisense Oligonucleotide (ASO) Therapeutics	131.2	298.1	125.1	133.5	145.0	154.1	158.7	156.9	4.6%
Other Therapeutics	0.0	0.0	0.0	0.0	0.0	93.1	98.4	100.0	3.6%
Total	**14,092.0**	**32,255.3**	**13,753.8**	**14,846.2**	**16,281.4**	**17,458.0**	**18,122.1**	**18,065.5**	**5.6%**

Source: National Institutes of Health (NIH), US Food and Drug Administration (US FDA), World Health Organization (WHO), European Pharmaceutical Review, American Society of Gene & Cell Therapy (ASGCT), European Society of Gene and Cell Therapy (ESGCT), RNA Society (US), European Medicines Agency (EMA), Alliance for Regenerative Medicine (ARM), Annual Reports, SEC Filings, Press Releases, Investor Presentations, Corporate Presentations, Journals, Interviews with Experts, and MarketsandMarkets Analysis

7.2 MRNA THERAPEUTICS

7.2.1 INCREASING INVESTMENTS FOR COVID-19 MRNA VACCINES TO DRIVE MARKET

Messenger RNA (abbreviated mRNA) is a type of single-stranded RNA involved in protein synthesis. In contrast to DNA-based medications, mRNA transcripts exhibit relatively high transfection efficiency and low toxicity as they do not penetrate the nucleus for functionality. Importantly, mRNA presents no risk of accidental infection or opportunistic insertional mutagenesis. Nevertheless, it showcases a broad potential for treating diseases that necessitate protein expression, thereby offering heightened therapeutic efficacy due to its sustained translation into encoded proteins and peptides and enduring expression compared to conventional transient protein or peptide drugs. Evidently, these advantages of mRNA over DNA or protein/peptide introduce mRNA-based technology and products across various biomedical applications.

The success of mRNA-based COVID-19 vaccines has also demonstrated the potential of mRNA technology in developing effective vaccines against infectious diseases, leading to increased interest and investment in mRNA therapeutics.

Major industry players such as Moderna, Inc. (US), CureVac SE (Germany), BioNTech SE (Germany), Argos Therapeutics Inc. (US), Ethris GmbH (Germany), Arcturus Therapeutics (US), and Acuitas Therapeutics (Canada) develop novel mRNA therapy for a variety of indications.

Some developments related to mRNA therapeutics include the following:

- In August 2023, CARsgen Therapeutics Holdings collaborated with Moderna, Inc. to investigate CARsgen's investigational Claudin18.2 CAR T-cell product candidate (CT041) in combination with Moderna's investigational Claudin18.2 mRNA cancer vaccine.
- In August 2023, Arcturus Therapeutics Holdings Inc. announced that ARCALIS Co., Ltd., its manufacturing joint venture in Japan to support the production of mRNA vaccines and therapeutics, has been awarded USD 115 million in two separate grants from the Japanese government. The grants will fund the construction of a factory and the purchase of capital equipment to support the current Good Manufacturing Practice (cGMP) production of mRNA drug substances and mRNA drug product operations.
- In March 2023, BioNTech SE announced that the company signed a Memorandum of Understanding (MoU) with the Weizmann Institute of Science. As part of this MoU, both of them will collaborate in basic and applied research to better understand various diseases, including cancer, infectious diseases, and neurodegenerative diseases. The joint research will be conducted at BioNTech's newly established mRNA Excellence Center and in Weizmann Institute laboratories.
- In January 2023, Moderna, Inc. and CytomX Therapeutics, Inc. announced a collaboration and licensing agreement to develop investigational mRNA-based therapies utilizing Moderna's mRNA technologies and CytomX's Probody therapeutic platform.

However, a lack of complete understanding of mRNA structure instability and immunogenicity has hampered the growth of mRNA-based therapeutics to combat diseases to some extent.

TABLE 29 **RNA THERAPEUTICS MARKET FOR MRNA THERAPEUTICS, BY REGION, 2021–2028 (USD MILLION)**

Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
North America	3,415.9	7,492.4	3,862.8	4,445.8	5,100.1	5,596.8	5,866.2	5,785.3	8.4%
Europe	4,799.9	11,002.3	4,644.1	4,989.5	5,450.3	5,792.3	6,000.5	5,978.9	5.2%
Asia Pacific	4,185.0	9,821.6	3,642.1	3,700.9	3,869.5	3,987.6	4,083.7	4,110.6	2.4%
Rest of the World	595.6	1,396.3	508.4	512.4	532.7	546.9	559.4	563.3	2.1%
Total	**12,996.4**	**29,712.5**	**12,657.4**	**13,648.6**	**14,952.6**	**15,923.6**	**16,509.8**	**16,438.0**	**5.4%**

Source: National Institutes of Health (NIH), US Food and Drug Administration (US FDA), World Health Organization (WHO), European Pharmaceutical Review, American Society of Gene & Cell Therapy (ASGCT), European Society of Gene and Cell Therapy (ESGCT), RNA Society (US), European Medicines Agency (EMA), Alliance for Regenerative Medicine (ARM), Annual Reports, SEC Filings, Press Releases, Investor Presentations, Corporate Presentations, Journals, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 30 NORTH AMERICA: RNA THERAPEUTICS MARKET FOR MRNA THERAPEUTICS, BY COUNTRY, 2021–2028 (USD MILLION)

Country	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
US	3,037.9	6,688.8	3,402.1	3,901.2	4,465.2	4,894.5	5,130.0	5,065.1	8.3%
Canada	378.1	803.6	460.7	544.6	634.8	702.3	736.2	720.1	9.3%
Total	**3,415.9**	**7,492.4**	**3,862.8**	**4,445.8**	**5,100.1**	**5,596.8**	**5,866.2**	**5,785.3**	**8.4%**

Source: National Institutes of Health (NIH), US Food and Drug Administration (US FDA), World Health Organization (WHO), European Pharmaceutical Review, American Society of Gene & Cell Therapy (ASGCT), European Society of Gene and Cell Therapy (ESGCT), RNA Society (US), European Medicines Agency (EMA), Alliance for Regenerative Medicine (ARM), Annual Reports, SEC Filings, Press Releases, Investor Presentations, Corporate Presentations, Journals, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 31 EUROPE: RNA THERAPEUTICS MARKET FOR MRNA THERAPEUTICS, BY COUNTRY, 2021–2028 (USD MILLION)

Country/ Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Germany	1,448.7	3,320.4	1,422.3	1,538.9	1,690.9	1,805.6	1,876.3	1,872.3	5.7%
UK	1,367.1	3,136.1	1,324.7	1,424.1	1,556.6	1,655.4	1,715.9	1,710.9	5.2%
France	856.2	1,956.5	847.5	920.5	1,014.1	1,084.3	1,126.9	1,122.8	5.8%
Rest of Europe	1,127.8	2,589.2	1,049.7	1,106.0	1,188.8	1,247.1	1,281.4	1,272.8	3.9%
Total	**4,799.9**	**11,002.3**	**4,644.1**	**4,989.5**	**5,450.3**	**5,792.3**	**6,000.5**	**5,978.9**	**5.2%**

Source: National Institutes of Health (NIH), US Food and Drug Administration (US FDA), World Health Organization (WHO), European Pharmaceutical Review, American Society of Gene & Cell Therapy (ASGCT), European Society of Gene and Cell Therapy (ESGCT), RNA Society (US), European Medicines Agency (EMA), Alliance for Regenerative Medicine (ARM), Annual Reports, SEC Filings, Press Releases, Investor Presentations, Corporate Presentations, Journals, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 32 ASIA PACIFIC: RNA THERAPEUTICS MARKET FOR MRNA THERAPEUTICS, BY COUNTRY, 2021–2028 (USD MILLION)

Country/Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Japan	2,284.4	5,379.1	1,971.6	1,993.9	2,077.0	2,135.8	2,187.2	2,206.8	2.3%
China	936.8	2,199.8	816.3	830.2	868.8	896.1	918.6	925.0	2.5%
South Korea	428.7	1,001.3	369.7	374.2	389.8	400.2	408.3	408.8	2.0%
Rest of Asia Pacific	535.1	1,241.3	484.4	502.7	533.9	555.4	569.6	570.0	3.3%
Total	**4,185.0**	**9,821.6**	**3,642.1**	**3,700.9**	**3,869.5**	**3,987.6**	**4,083.7**	**4,110.6**	**2.4%**

Source: National Institutes of Health (NIH), US Food and Drug Administration (US FDA), World Health Organization (WHO), European Pharmaceutical Review, American Society of Gene & Cell Therapy (ASGCT), European Society of Gene and Cell Therapy (ESGCT), RNA Society (US), European Medicines Agency (EMA), Alliance for Regenerative Medicine (ARM), Annual Reports, SEC Filings, Press Releases, Investor Presentations, Corporate Presentations, Journals, Interviews with Experts, and MarketsandMarkets Analysis

7.3 RNA INTERFERENCE (RNAI) THERAPEUTICS

7.3.1 INCREASING NUMBER OF APPROVALS FOR SIRNA THERAPEUTICS BY US FDA TO DRIVE MARKET

RNA interference (RNAi) is an endogenous cellular process that causes double-stranded (ds) RNAs to degrade specific RNA targets, providing an inbuilt defense mechanism against invading viruses and transposable elements. The major forms of RNAi molecules are genome-derived microRNAs (miRNAs) and exogenous small interfering RNAs (siRNAs). siRNAs are short dsRNAs (20-24 nt) containing 5'-phosphate/3'-hydroxyl endings and two 3'-overhang ribonucleotides on each duplex strand. miRNAs are endogenous single-stranded small ncRNAs influencing gene expression via RNAi.

RNAi is the fastest growing segment, by type, in the RNA therapeutics market, which is attributed to the adaptable and versatile nature of RNAi. The growth is further supported by the approval from US FDA for four siRNA therapeutics namely patisiran (Onpattro, 2018), givosiran (Givlaari, 2019), lumasiran (Oxlumo, 2020), and inclisiran (Leqvio, 2021); each siRNA drug selectively acts on a target mRNA transcript to combat diseases including hereditary transthyretin-mediated amyloidosis (hATTR), acute hepatic porphyria (AHP), primary hyperoxaluria type 1 (PH1), and heterozygous familial hypercholesterolemia (HeFH), respectively. In addition, several siRNA agents (e.g., fitusiran, nedosiran, teprasiran, tivanisiran, and vutrisiran) have entered phase III clinical trials with many other RNAi-based therapeutics progressing through early-stage clinical trials or preclinical development. Hence, the ongoing research in this domain has propelled numerous RNAi-based therapies into clinical trials.

While most of the RNAi therapeutics function as siRNAs or anti-miRs, the arena of miRNA replacement therapy remains relatively untapped. This holds a substantial opportunity for key players as miRNAs hold a distinct ability to simultaneously target multiple mRNA transcripts. This unique attribute paves paths for a single drug to regulate multiple targets or biological pathways that are often dysregulated in a disease. miRNA-based therapeutics are studied further to combat various types of cancers (human papillomavirus (HPV)-related cancers, colorectal cancers, lung cancers, and acute myeloid leukemias, among others. miRNAs also show potential in the treatment of cardiovascular disease. Further, miRNA-based therapies are also used in combination with other well-established therapies toward optimal outcomes.

Some companies involved in the development of RNAi-based therapeutics include Alnylam Pharmaceuticals (US), Novartis AG (Switzerland), F. Hoffmann-La Roche AG (Switzerland), Santaris Pharma A/S (Denmark), Cardior Pharmaceuticals GmbH (Germany), Sanofi (France), AstraZeneca (UK), MiRagen Therapeutics (US), Dicerna Pharmaceuticals (US), Sylentis S.A (Spain)., Arbutus Biopharma (US), Quark Pharmaceuticals (US) and Silence Therapeutics (UK).

TABLE 33 RNA THERAPEUTICS MARKET FOR RNA INTERFERENCE (RNAI) THERAPEUTICS, BY REGION, 2021–2028 (USD MILLION)

Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
North America	252.2	558.7	291.0	338.2	391.8	436.6	462.1	460.1	9.6%
Europe	355.6	829.4	356.1	389.1	432.2	469.6	494.5	500.7	7.1%
Asia Pacific	311.9	748.9	284.0	295.0	315.2	333.9	349.2	358.8	4.8%
Rest of the World	44.6	107.6	40.3	41.7	44.4	47.1	49.4	51.0	4.0%
Total	964.3	2,244.6	971.4	1,064.1	1,183.7	1,287.2	1,355.2	1,370.6	7.1%

Source: National Institutes of Health (NIH), US Food and Drug Administration (US FDA), World Health Organization (WHO), European Pharmaceutical Review, American Society of Gene & Cell Therapy (ASGCT), European Society of Gene and Cell Therapy (ESGCT), RNA Society (US), European Medicines Agency (EMA), Alliance for Regenerative Medicine (ARM), Annual Reports, SEC Filings, Press Releases, Investor Presentations, Corporate Presentations, Journals, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 34 NORTH AMERICA: RNA THERAPEUTICS MARKET FOR RNA INTERFERENCE (RNAI) THERAPEUTICS, BY COUNTRY, 2021–2028 (USD MILLION)

Country	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
US	224.3	498.6	256.0	296.4	342.4	380.8	402.8	401.4	9.4%
Canada	27.9	60.1	34.9	41.9	49.5	55.8	59.3	58.7	11.0%
Total	**252.2**	**558.7**	**291.0**	**338.2**	**391.8**	**436.6**	**462.1**	**460.1**	**9.6%**

Source: National Institutes of Health (NIH), US Food and Drug Administration (US FDA), World Health Organization (WHO), European Pharmaceutical Review, American Society of Gene & Cell Therapy (ASGCT), European Society of Gene and Cell Therapy (ESGCT), RNA Society (US), European Medicines Agency (EMA), Alliance for Regenerative Medicine (ARM), Annual Reports, SEC Filings, Press Releases, Investor Presentations, Corporate Presentations, Journals, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 35 EUROPE: RNA THERAPEUTICS MARKET FOR RNA INTERFERENCE (RNAI) THERAPEUTICS, BY COUNTRY, 2021–2028 (USD MILLION)

Country/Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Germany	107.0	249.2	108.4	119.0	132.8	144.7	152.6	154.5	7.3%
UK	101.2	236.1	101.4	110.8	123.1	133.8	140.9	142.7	7.1%
France	63.5	147.8	65.2	72.1	80.8	88.4	93.5	94.7	7.8%
Rest of Europe	83.8	196.3	81.1	87.2	95.5	102.7	107.5	108.8	6.0%
Total	**355.6**	**829.4**	**356.1**	**389.1**	**432.2**	**469.6**	**494.5**	**500.7**	**7.1%**

Source: National Institutes of Health (NIH), US Food and Drug Administration (US FDA), World Health Organization (WHO), European Pharmaceutical Review, American Society of Gene & Cell Therapy (ASGCT), European Society of Gene and Cell Therapy (ESGCT), RNA Society (US), European Medicines Agency (EMA), Alliance for Regenerative Medicine (ARM), Annual Reports, SEC Filings, Press Releases, Investor Presentations, Corporate Presentations, Journals, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 36 ASIA PACIFIC: RNA THERAPEUTICS MARKET FOR RNA INTERFERENCE (RNAI) THERAPEUTICS, BY COUNTRY, 2021–2028 (USD MILLION)

Country/Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Japan	170.0	409.1	153.2	158.1	168.1	177.5	185.4	190.8	4.5%
China	69.9	167.8	63.7	66.2	70.8	75.1	78.6	80.8	4.9%
South Korea	32.0	76.6	29.0	30.0	32.0	33.9	35.3	36.2	4.5%
Rest of Asia Pacific	40.0	95.3	38.2	40.6	44.2	47.4	49.8	51.0	6.0%
Total	**311.9**	**748.9**	**284.0**	**295.0**	**315.2**	**333.9**	**349.2**	**358.8**	**4.8%**

Source: National Institutes of Health (NIH), US Food and Drug Administration (US FDA), World Health Organization (WHO), European Pharmaceutical Review, American Society of Gene & Cell Therapy (ASGCT), European Society of Gene and Cell Therapy (ESGCT), RNA Society (US), European Medicines Agency (EMA), Alliance for Regenerative Medicine (ARM), Annual Reports, SEC Filings, Press Releases, Investor Presentations, Corporate Presentations, Journals, Interviews with Experts, and MarketsandMarkets Analysis

7.4 ANTISENSE OLIGONUCLEOTIDE (ASO) THERAPEUTICS

7.4.1 INCREASED NUMBER OF CHEMICAL MODIFICATIONS AND TECHNOLOGICAL ADVANCEMENTS TO DRIVE MARKET

Antisense oligonucleotide (ASO) is a single-stranded DNA oligonucleotide with 12–25 nucleotides in length. Advancement in technologies such as artificial intelligence, advanced computer modeling, and machine learning for the identification of new genetic targets; the development of chemical modifications of ASOs to improve their pharmacokinetic properties; increasing research to study the effect of ASO for a wide range of therapeutics applications; and the ease and accuracy of customization of ASO sequences are some factors attributed to the growth of this segment in RNA therapeutics market. A few examples of chemical modifications on different parts of the ASO include the inter-nucleotide linkages, the ring that connects the ASO backbone to the nucleobase, and the nucleobases that form the complementary Watson–Crick DNA base pairing with target RNA.

The clinical development of new ASOs for the treatment of a range of human diseases is progressing at a rapid pace. Some of the recently ASO-approved products include Macugen, Amondys 45, Viltepso, Vyondys 53, and Tegsedi. Additionally, in clinical trials, several aerosolized ASOs developed for CF therapy had favorable safety profiles and showed exposure in the intended lung tissues. Furthermore, RNA sequencing, or RNA-seq, has provided insight into previously unknown open reading frames (ORFs) and revealed a result of discontinuous transcription in the complex transcriptome of SARS-CoV-2. By employing antisense oligonucleotides (ASOs) to target newly discovered viral ORFs or host factors, this discovery expands the potential for the development of antiviral therapies.

However, ASOs may degrade in plasma, which can lead to off-target effects or immune system activation. Additionally, ASOs have a strong tendency to accumulate in the liver and kidney, among other organs, which might lead to hepatotoxicity or renal failure. These factors may restrain the growth of ASO in the market during the forecast period.

Some leading companies offering ASO therapeutics include Ionis Pharmaceuticals, Inc. (US), OSI Pharmaceuticals, Inc. (Acquired by Astellas Pharma Inc.), Sarepta Therapeutics, Inc. (US), Jazz Pharmaceuticals plc (Ireland), and NS Pharma (US), among others.

TABLE 37 RNA THERAPEUTICS MARKET FOR ANTISENSE OLIGONUCLEOTIDE (ASO) THERAPEUTICS, BY REGION, 2021–2028 (USD MILLION)

Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
North America	31.0	68.7	35.8	41.7	48.3	53.8	57.0	56.7	9.6%
Europe	48.5	108.8	44.9	47.2	50.4	52.6	53.2	51.7	2.8%
Asia Pacific	45.0	104.7	38.5	38.7	40.1	41.2	41.7	41.6	1.6%
Rest of the World	6.7	15.9	5.8	6.0	6.3	6.6	6.8	6.9	2.7%
Total	**131.2**	**298.1**	**125.1**	**133.5**	**145.0**	**154.1**	**158.7**	**156.9**	**4.6%**

Source: National Institutes of Health (NIH), US Food and Drug Administration (US FDA), World Health Organization (WHO), European Pharmaceutical Review, American Society of Gene & Cell Therapy (ASGCT), European Society of Gene and Cell Therapy (ESGCT), RNA Society (US), European Medicines Agency (EMA), Alliance for Regenerative Medicine (ARM), Annual Reports, SEC Filings, Press Releases, Investor Presentations, Corporate Presentations, Journals, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 38 NORTH AMERICA: RNA THERAPEUTICS MARKET FOR ANTISENSE OLIGONUCLEOTIDE (ASO) THERAPEUTICS, BY COUNTRY, 2021–2028 (USD MILLION)

Country	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
US	27.5	61.2	31.5	36.6	42.4	47.3	50.2	50.1	9.7%
Canada	3.6	7.5	4.3	5.1	5.9	6.5	6.8	6.6	8.9%
Total	**31.0**	**68.7**	**35.8**	**41.7**	**48.3**	**53.8**	**57.0**	**56.7**	**9.6%**

Source: National Institutes of Health (NIH), US Food and Drug Administration (US FDA), World Health Organization (WHO), European Pharmaceutical Review, American Society of Gene & Cell Therapy (ASGCT), European Society of Gene and Cell Therapy (ESGCT), RNA Society (US), European Medicines Agency (EMA), Alliance for Regenerative Medicine (ARM), Annual Reports, SEC Filings, Press Releases, Investor Presentations, Corporate Presentations, Journals, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 39 EUROPE: RNA THERAPEUTICS MARKET FOR ANTISENSE OLIGONUCLEOTIDE (ASO) THERAPEUTICS, BY COUNTRY, 2021–2028 (USD MILLION)

Country/Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Germany	14.1	31.9	13.4	14.3	15.4	16.3	16.6	16.3	4.0%
UK	13.8	30.8	12.7	13.2	14.1	14.6	14.7	14.2	2.3%
France	8.9	19.6	8.2	8.6	9.0	9.3	9.3	8.8	1.5%
Rest of Europe	11.7	26.6	10.7	11.1	11.9	12.4	12.6	12.4	3.1%
Total	**48.5**	**108.8**	**44.9**	**47.2**	**50.4**	**52.6**	**53.2**	**51.7**	**2.8%**

Source: National Institutes of Health (NIH), US Food and Drug Administration (US FDA), World Health Organization (WHO), European Pharmaceutical Review, American Society of Gene & Cell Therapy (ASGCT), European Society of Gene and Cell Therapy (ESGCT), RNA Society (US), European Medicines Agency (EMA), Alliance for Regenerative Medicine (ARM), Annual Reports, SEC Filings, Press Releases, Investor Presentations, Corporate Presentations, Journals, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 40 ASIA PACIFIC: RNA THERAPEUTICS MARKET FOR ANTISENSE OLIGONUCLEOTIDE (ASO) THERAPEUTICS, BY COUNTRY, 2021–2028 (USD MILLION)

Country/Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Japan	24.2	56.4	20.5	20.5	21.2	21.7	22.0	22.0	1.5%
China	10.2	23.5	8.6	8.5	8.8	8.9	9.0	8.8	0.6%
South Korea	4.7	10.9	4.0	4.0	4.2	4.3	4.3	4.3	1.3%
Rest of Asia Pacific	6.0	13.8	5.4	5.6	6.0	6.3	6.5	6.5	3.6%
Total	**45.0**	**104.7**	**38.5**	**38.7**	**40.1**	**41.2**	**41.7**	**41.6**	**1.6%**

Source: National Institutes of Health (NIH), US Food and Drug Administration (US FDA), World Health Organization (WHO), European Pharmaceutical Review, American Society of Gene & Cell Therapy (ASGCT), European Society of Gene and Cell Therapy (ESGCT), RNA Society (US), European Medicines Agency (EMA), Alliance for Regenerative Medicine (ARM), Annual Reports, SEC Filings, Press Releases, Investor Presentations, Corporate Presentations, Journals, Interviews with Experts, and MarketsandMarkets Analysis

7.5 OTHER THERAPEUTICS

The other therapeutics segment includes circular RNA therapeutics, RNA aptamers, and small activating RNA. Circular RNAs (circRNA) are a class of natural (biogenic) or synthetic closed RNA without 5′ or 3′ ends. circRNA has a prolonged shelf life, ameliorates the stringent conditions for their storage and transportation, and pharmacokinetically promotes their accumulation in various types of cells or tissues. Employing various chemical and enzymatic synthetic methodologies, circRNAs have been successfully synthesized for use as protein-coding and non-coding therapeutic agents and vaccines. However, there are some challenges associated with circRNA, including low circularization efficiency and the high cost of reagents such as enzymes.

In 2022, Orna Therapeutics completed its USD 221 million Series B funding round. With the support of its joint venture with ReNAgade Therapeutics, Orna can further develop its circular RNA + LNP (lipid nanoparticle) delivery technology.

RNA aptamers are generated through an in vitro selection process called SELEX (Systematic Evolution of Ligands by Exponential Enrichment), where a large pool of random RNA sequences is iteratively refined to isolate those that exhibit strong binding to the desired target. RNA aptamers hold therapeutic potential by interfering with disease-related molecules and pathways, making them promising candidates for various applications in the RNA therapeutics market. Furthermore, the growth of RNA aptamers in the RNA therapeutics market is driven by their specificity, adaptability, and potential for diverse applications. With ongoing research, technological advancements, and strategic collaborations, RNA aptamers can contribute significantly to the evolution of personalized and precise therapies.

Small activating RNAs (saRNAs) are small double-stranded RNAs that could mediate the target-specific gene expression by targeting selected sequences in gene promoters at both the transcriptional and epigenetic levels. Owing to the advantages of gene upregulating using saRNA, it is anticipated that more saRNA drugs will be developed to the market and transform patients' health. Further understanding of the saRNA chemistry and the gene upregulation mechanism will improve the selection of saRNA with minimal off-target effect and immunogenicity, with enhanced potency and stability.

8 RNA THERAPEUTICS MARKET, BY INDICATION

KEY FINDINGS

- The infectious diseases segment accounted for the largest share of 92.2% of the RNA therapeutics market in 2022.
- The large share of the infectious diseases segment can be attributed to the advantages of mRNA over DNA or proteins/peptides and the advancements in vaccine development.
- The infectious diseases segment is projected to reach USD 16,373.9 million by 2028 from USD 12,643.7 million in 2023, at a CAGR of 5.3%
- The rare genetic/hereditary diseases segment is anticipated to reach USD 1,301.4 million by 2028 from USD 877.8 million in 2023, at a CAGR of 8.2%.

8.1 INTRODUCTION

Based on the indication segment, the RNA therapeutics market is segmented into infectious diseases, rare genetic diseases/hereditary diseases, and other indications. The infectious diseases segment accounted for the largest share of 92.2% of the RNA therapeutics market in 2022. The large share of this segment can primarily be attributed to the rapid development capabilities of RNA vaccines for the treatment of infectious diseases and a higher number of epidemic outbreaks.

TABLE 41 RNA THERAPEUTICS MARKET, BY INDICATION, 2021–2028 (USD MILLION)

Indication	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Infectious Diseases	13,032.9	29,738.3	12,643.7	13,609.1	14,883.2	15,914.7	16,473.6	16,373.9	5.3%
Rare Genetic/ Hereditary Diseases	850.4	2,004.1	877.8	972.1	1,092.5	1,199.7	1,274.9	1,301.4	8.2%
Other Indications	208.7	512.9	232.3	264.9	305.6	343.6	373.6	390.3	10.9%
Total	**14,092.0**	**32,255.3**	**13,753.8**	**14,846.2**	**16,281.4**	**17,458.0**	**18,122.1**	**18,065.5**	**5.6%**

Source: National Institutes of Health (NIH), US Food and Drug Administration (US FDA), World Health Organization (WHO), European Pharmaceutical Review, American Society of Gene & Cell Therapy (ASGCT), European Society of Gene and Cell Therapy (ESGCT), RNA Society (US), European Medicines Agency (EMA), Alliance for Regenerative Medicine (ARM), Annual Reports, SEC Filings, Press Releases, Investor Presentations, Corporate Presentations, Journals, Interviews with Experts, and MarketsandMarkets Analysis

8.2 INFECTIOUS DISEASES

8.2.1 INCREASING PREVALENCE OF INFECTIOUS DISEASES AND GROWING INVESTMENTS IN R&D TO DRIVE MARKET

Infectious diseases are caused by pathogenic microorganisms, such as bacteria, viruses, parasites, or fungi; the diseases can be spread directly or indirectly from one person to another. Some infectious diseases can pass from person to person, while others are transmitted by insects or other animals. mRNA vaccines offer a compelling alternative to traditional vaccine strategies due to their remarkable potency, rapid development capabilities, cost-effective manufacturing potential, and safe administration. These vaccines have emerged as a promising solution for combatting various infectious diseases, including influenza, Zika virus, and respiratory syncytial virus. Influenza viruses are responsible for an estimated 290,000–650,000 deaths annually worldwide, making it essential to develop vaccines against this virus. Notably, the effectiveness of mRNA vaccines extends to the ongoing COVID-19 pandemic, where these have successfully provided protection against SARS-CoV-2.

The growth of this segment is being driven by the increasing prevalence of infectious diseases, the development of new RNA-based vaccines and therapeutics, and the growing investment in RNA therapeutics R&D.

There is a growing pipeline of RNA-based vaccines and therapeutics in development by pharmaceutical and biopharmaceutical companies to treat infectious diseases. These therapies have the potential to be more effective and safer than traditional therapies. In January 2023, Agilent Technologies Inc. invested approximately USD 725 million to double the manufacturing capacity of therapeutic nucleic acids for a growing number of diseases, including cancer, cardiovascular diseases, and rare and infectious diseases.

TABLE 42 RNA THERAPEUTICS MARKET FOR INFECTIOUS DISEASES, BY REGION, 2021–2028 (USD MILLION)

Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
North America	3,418.3	7,492.5	3,860.2	4,439.8	5,089.7	5,612.2	5,878.4	5,793.3	8.5%
Europe	4,812.2	11,010.3	4,638.9	4,974.6	5,424.0	5,787.2	5,985.0	5,953.3	5.1%
Asia Pacific	4,203.2	9,836.1	3,637.0	3,685.0	3,841.6	3,971.8	4,056.2	4,071.5	2.3%
Rest of the World	599.1	1,399.4	507.7	509.7	527.9	543.5	553.9	555.7	1.8%
Total	**13,032.9**	**29,738.3**	**12,643.7**	**13,609.1**	**14,883.2**	**15,914.7**	**16,473.6**	**16,373.9**	**5.3%**

Source: National Institutes of Health (NIH), US Food and Drug Administration (US FDA), World Health Organization (WHO), European Pharmaceutical Review, American Society of Gene & Cell Therapy (ASGCT), European Society of Gene and Cell Therapy (ESGCT), RNA Society (US), European Medicines Agency (EMA), Alliance for Regenerative Medicine (ARM), Annual Reports, SEC Filings, Press Releases, Investor Presentations, Corporate Presentations, Journals, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 43 NORTH AMERICA: RNA THERAPEUTICS MARKET FOR INFECTIOUS DISEASES, BY COUNTRY, 2021–2028 (USD MILLION)

Country	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
US	3,039.8	6,688.8	3,399.9	3,896.1	4,456.5	4,908.4	5,141.2	5,072.9	8.3%
Canada	378.5	803.8	460.3	543.7	633.2	703.8	737.2	720.4	9.4%
Total	**3,418.3**	**7,492.5**	**3,860.2**	**4,439.8**	**5,089.7**	**5,612.2**	**5,878.4**	**5,793.3**	**8.5%**

Source: National Institutes of Health (NIH), US Food and Drug Administration (US FDA), World Health Organization (WHO), European Pharmaceutical Review, American Society of Gene & Cell Therapy (ASGCT), European Society of Gene and Cell Therapy (ESGCT), RNA Society (US), European Medicines Agency (EMA), Alliance for Regenerative Medicine (ARM), Annual Reports, SEC Filings, Press Releases, Investor Presentations, Corporate Presentations, Journals, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 44 EUROPE: RNA THERAPEUTICS MARKET FOR INFECTIOUS DISEASES, BY COUNTRY, 2021–2028 (USD MILLION)

Country/ Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Germany	1,451.3	3,321.8	1,421.0	1,535.4	1,684.7	1,806.8	1,875.3	1,869.0	5.6%
UK	1,370.5	3,138.2	1,323.2	1,420.0	1,549.3	1,654.1	1,711.8	1,703.9	5.2%
France	858.9	1,958.2	846.3	917.2	1,008.2	1,081.9	1,122.0	1,115.7	5.7%
Rest of Europe	1,131.6	2,592.0	1,048.4	1,102.1	1,181.9	1,244.5	1,275.9	1,264.7	3.8%
Total	**4,812.2**	**11,010.3**	**4,638.9**	**4,974.6**	**5,424.0**	**5,787.2**	**5,985.0**	**5,953.3**	**5.1%**

Source: National Institutes of Health (NIH), US Food and Drug Administration (US FDA), World Health Organization (WHO), European Pharmaceutical Review, American Society of Gene & Cell Therapy (ASGCT), European Society of Gene and Cell Therapy (ESGCT), RNA Society (US), European Medicines Agency (EMA), Alliance for Regenerative Medicine (ARM), Annual Reports, SEC Filings, Press Releases, Investor Presentations, Corporate Presentations, Journals, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 45 ASIA PACIFIC: RNA THERAPEUTICS MARKET FOR INFECTIOUS DISEASES, BY COUNTRY, 2021–2028 (USD MILLION)

Country/Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Japan	2,293.4	5,386.1	1,969.1	1,986.1	2,063.4	2,129.2	2,175.0	2,189.0	2.1%
China	941.1	2,203.2	815.1	826.4	862.2	892.1	911.7	915.4	2.3%
South Korea	430.8	1,003.0	369.1	372.4	386.6	398.2	405.0	404.2	1.8%
Rest of Asia Pacific	538.0	1,243.8	483.7	500.2	529.4	552.3	564.5	562.9	3.1%
Total	**4,203.2**	**9,836.1**	**3,637.0**	**3,685.0**	**3,841.6**	**3,971.8**	**4,056.2**	**4,071.5**	**2.3%**

Source: National Institutes of Health (NIH), US Food and Drug Administration (US FDA), World Health Organization (WHO), European Pharmaceutical Review, American Society of Gene & Cell Therapy (ASGCT), European Society of Gene and Cell Therapy (ESGCT), RNA Society (US), European Medicines Agency (EMA), Alliance for Regenerative Medicine (ARM), Annual Reports, SEC Filings, Press Releases, Investor Presentations, Corporate Presentations, Journals, Interviews with Experts, and MarketsandMarkets Analysis

8.3 RARE GENETIC/HEREDITARY DISEASES

8.3.1 TECHNOLOGICAL INNOVATION AND INCREASED DEMAND FOR PERSONALIZED MEDICINES TO DRIVE MARKET

Rare genetic diseases, often characterized by their limited prevalence and challenging treatment paradigms, create opportunities for the innovative potential of RNA-based interventions. RNA therapeutics offer a unique avenue to address these rare genetic disorders by precisely targeting the underlying genetic mutations that are responsible for the diseases. Through techniques such as gene silencing using small interfering RNAs (siRNAs) or gene editing using CRISPR/Cas9, RNA therapeutics correct or mitigate the genetic abnormalities causing these conditions.

The segment's growth is fueled by several key factors, including the advent of personalized medicine enabled by the precision of RNA-based approaches that offer tailored solutions for individuals afflicted by rare genetic diseases. Moreover, advancements in delivery technologies facilitate the transport of RNA molecules to target cells or tissues, enhancing the therapeutic potential of these interventions.

The efforts made by biotechnology companies and research institutions to address unmet needs in rare genetic or hereditary diseases further catalyze the segment's expansion. Milasen was developed in less than a year to treat Mila Makovec's CLN7 gene mutation that developed Batten disease. In April 2023, MiNA Therapeutics announced a multi-target research collaboration and option licensing agreement with BioMarin Pharmaceutical Inc. to discover, develop, and commercialize RNAa therapeutic vaccines for patients with rare genetic diseases. In January 2022, Acadia Pharmaceuticals Inc. and Stoke Therapeutics, Inc. entered a collaboration to develop and commercialize novel RNA-based therapeutics for the treatment of severe and rare genetic neurodevelopmental diseases of the central nervous system.

TABLE 46 RNA THERAPEUTICS MARKET FOR RARE GENETIC/HEREDITARY DISEASES, BY REGION, 2021–2028 (USD MILLION)

Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
North America	224.6	501.4	263.0	307.9	359.1	402.9	429.2	430.1	10.3%
Europe	314.3	741.1	321.5	354.8	397.8	436.2	463.3	473.1	8.0%
Asia Pacific	272.8	666.3	256.8	270.9	293.7	315.4	334.3	347.8	6.2%
Rest of the World	38.6	95.3	36.5	38.5	41.8	45.2	48.2	50.5	5.7%
Total	**850.4**	**2,004.1**	**877.8**	**972.1**	**1,092.5**	**1,199.7**	**1,274.9**	**1,301.4**	**8.2%**

Source: National Institutes of Health (NIH), US Food and Drug Administration (US FDA), World Health Organization (WHO), European Pharmaceutical Review, American Society of Gene & Cell Therapy (ASGCT), European Society of Gene and Cell Therapy (ESGCT), RNA Society (US), European Medicines Agency (EMA), Alliance for Regenerative Medicine (ARM), Annual Reports, SEC Filings, Press Releases, Investor Presentations, Corporate Presentations, Journals, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 47 NORTH AMERICA: RNA THERAPEUTICS MARKET FOR RARE GENETIC/HEREDITARY DISEASES, BY COUNTRY, 2021–2028 (USD MILLION)

Country	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
US	199.8	447.5	231.5	269.9	314.0	351.6	374.5	375.5	10.2%
Canada	24.8	53.9	31.5	38.0	45.2	51.2	54.7	54.5	11.6%
Total	**224.6**	**501.4**	**263.0**	**307.9**	**359.1**	**402.9**	**429.2**	**430.1**	**10.3%**

Source: National Institutes of Health (NIH), US Food and Drug Administration (US FDA), World Health Organization (WHO), European Pharmaceutical Review, American Society of Gene & Cell Therapy (ASGCT), European Society of Gene and Cell Therapy (ESGCT), RNA Society (US), European Medicines Agency (EMA), Alliance for Regenerative Medicine (ARM), Annual Reports, SEC Filings, Press Releases, Investor Presentations, Corporate Presentations, Journals, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 48 EUROPE: RNA THERAPEUTICS MARKET FOR RARE GENETIC/HEREDITARY DISEASES, BY COUNTRY, 2021–2028 (USD MILLION)

Country/Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Germany	95.0	223.1	97.8	108.3	121.6	133.5	141.8	144.5	8.1%
UK	89.6	211.1	91.6	101.0	113.2	124.1	131.8	134.6	8.0%
France	56.0	132.0	58.9	65.8	74.6	82.4	88.0	90.0	8.8%
Rest of Europe	73.7	175.0	73.3	79.8	88.4	96.2	101.8	104.1	7.3%
Total	**314.3**	**741.1**	**321.5**	**354.8**	**397.8**	**436.2**	**463.3**	**473.1**	**8.0%**

Source: National Institutes of Health (NIH), US Food and Drug Administration (US FDA), World Health Organization (WHO), European Pharmaceutical Review, American Society of Gene & Cell Therapy (ASGCT), European Society of Gene and Cell Therapy (ESGCT), RNA Society (US), European Medicines Agency (EMA), Alliance for Regenerative Medicine (ARM), Annual Reports, SEC Filings, Press Releases, Investor Presentations, Corporate Presentations, Journals, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 49 ASIA PACIFIC: RNA THERAPEUTICS MARKET FOR RARE GENETIC/HEREDITARY DISEASES, BY COUNTRY, 2021–2028 (USD MILLION)

Country/Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Japan	149.1	364.3	138.4	145.0	156.2	167.0	176.6	183.8	5.8%
China	61.0	149.3	57.6	60.8	66.0	71.0	75.3	78.4	6.4%
South Korea	27.9	68.1	26.2	27.7	30.0	32.2	34.0	35.3	6.1%
Rest of Asia Pacific	34.8	84.6	34.6	37.5	41.5	45.2	48.3	50.2	7.7%
Total	**272.8**	**666.3**	**256.8**	**270.9**	**293.7**	**315.4**	**334.3**	**347.8**	**6.2%**

Source: National Institutes of Health (NIH), US Food and Drug Administration (US FDA), World Health Organization (WHO), European Pharmaceutical Review, American Society of Gene & Cell Therapy (ASGCT), European Society of Gene and Cell Therapy (ESGCT), RNA Society (US), European Medicines Agency (EMA), Alliance for Regenerative Medicine (ARM), Annual Reports, SEC Filings, Press Releases, Investor Presentations, Corporate Presentations, Journals, Interviews with Experts, and MarketsandMarkets Analysis

8.4 OTHER INDICATIONS

Other indications include oncology, respiratory diseases, ophthalmology, and cardiovascular diseases. The increasing prevalence of cancer has increased the demand for the development of RNA therapeutics as its treatment option. According to the American Cancer Society (ACS), in 2023, 1,958,310 new cancer cases and 609,820 cancer deaths are projected to occur in the US. Owing to this, companies have focused on developing novel RNA therapeutics for cancer treatment, which is expected to drive the growth of this indication segment. According to a report by IQVIA, oncology trials reached historically high levels in 2022. These were up by 22% from 2018 and mostly focused on rare cancer indications. Emerging biopharmaceutical companies were responsible for 71% of the oncology pipeline in 2022.

Furthermore, ASOs and siRNAs have shown early clinical efficacy in the treatment of cancer patients, and these can act as potential therapeutic candidates against cancer in the future. The development of the next generation of ASO chemistry, or generation 2.5 cEt-containing ASOs, has significantly improved potency and activity in extrahepatic tissues, including tumors, which are more challenging to target; this is a particularly encouraging step in this direction. Moreover, gene-silencing therapies have also demonstrated promising responses in ocular disorders, including glaucoma, dry eye disease, age-related macular degeneration, diabetic macular oedema, and various inherited retinal diseases.

Chronic respiratory diseases (CRDs) are diseases associated with the airways and other parts of the lungs. Some of the most common CRDs include asthma, chronic obstructive pulmonary disease (COPD), pulmonary hypertension, and occupational lung diseases. Cardiovascular diseases (CVDs) are a leading cause of death worldwide. CVDs include a wide range of conditions, including hypertension, dyslipidemia, stroke, atherosclerosis, thrombosis, coronary heart disease, cerebrovascular disease, and rheumatic heart diseases. The growing prevalence of CRDs and CVDs has increased the demand for drugs, which, in turn, has led to an increase in the number of clinical trials and approvals by regulatory bodies.

Some developments pertaining to RNA therapeutics for the above-mentioned indications include the following:

- As of March 2023, researchers from MIT and the University of Massachusetts Medical School have collaborated to create an innovative nanoparticle design with the potential to revolutionize lung-related treatments. These nanoparticles have the unique capability to transport messenger RNA (mRNA) containing beneficial protein-coding to the lungs. This breakthrough offers a path to inhalable therapies for diseases such as cystic fibrosis and other pulmonary conditions.
- In January 2023, Solid Biosciences Inc. and Phlox Therapeutics announced a strategic collaboration focused on genetic cardiac diseases. This collaboration allows Phlox Therapeutics to leverage Solid Biosciences' vector biology and manufacturing capacities to deliver its RNA therapies.
- In August 2022, Orna Therapeutics and Merck announced a collaboration agreement to discover, develop, and commercialize vaccines and therapeutics in the oncology and infectious disease areas.

TABLE 50 RNA THERAPEUTICS MARKET FOR OTHER INDICATIONS, BY REGION, 2021–2028 (USD MILLION)

Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
North America	56.2	126.0	66.4	78.0	91.3	102.9	110.0	110.6	10.8%
Europe	77.4	189.1	84.8	96.4	111.1	124.9	135.9	141.8	10.8%
Asia Pacific	65.9	172.7	70.8	78.8	89.5	100.3	110.4	118.9	10.9%
Rest of the World	9.2	25.1	10.4	11.8	13.7	15.5	17.3	18.9	10.4%
Total	**208.7**	**512.9**	**232.3**	**264.9**	**305.6**	**343.6**	**373.6**	**390.3**	**10.9%**

Source: National Institutes of Health (NIH), US Food and Drug Administration (US FDA), World Health Organization (WHO), European Pharmaceutical Review, American Society of Gene & Cell Therapy (ASGCT), European Society of Gene and Cell Therapy (ESGCT), RNA Society (US), European Medicines Agency (EMA), Alliance for Regenerative Medicine (ARM), Annual Reports, SEC Filings, Press Releases, Investor Presentations, Corporate Presentations, Journals, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 51 NORTH AMERICA: RNA THERAPEUTICS MARKET FOR OTHER INDICATIONS, BY COUNTRY, 2021–2028 (USD MILLION)

Country	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
US	50.0	112.4	58.3	68.2	79.5	89.3	95.4	95.9	10.5%
Canada	6.2	13.6	8.0	9.8	11.8	13.5	14.6	14.7	12.8%
Total	**56.2**	**126.0**	**66.4**	**78.0**	**91.3**	**102.9**	**110.0**	**110.6**	**10.8%**

Source: National Institutes of Health (NIH), US Food and Drug Administration (US FDA), World Health Organization (WHO), European Pharmaceutical Review, American Society of Gene & Cell Therapy (ASGCT), European Society of Gene and Cell Therapy (ESGCT), RNA Society (US), European Medicines Agency (EMA), Alliance for Regenerative Medicine (ARM), Annual Reports, SEC Filings, Press Releases, Investor Presentations, Corporate Presentations, Journals, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 52 EUROPE: RNA THERAPEUTICS MARKET FOR OTHER INDICATIONS, BY COUNTRY, 2021–2028 (USD MILLION)

Country/Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Germany	23.5	56.5	25.3	28.6	32.7	36.6	39.5	40.9	10.1%
UK	22.1	53.8	24.0	27.3	31.3	35.2	38.2	39.8	10.6%
France	13.7	33.8	15.7	18.1	21.2	24.1	26.5	27.8	12.1%
Rest of Europe	18.0	45.0	19.7	22.4	25.8	29.0	31.7	33.4	11.1%
Total	**77.4**	**189.1**	**84.8**	**96.4**	**111.1**	**124.9**	**135.9**	**141.8**	**10.8%**

Source: National Institutes of Health (NIH), US Food and Drug Administration (US FDA), World Health Organization (WHO), European Pharmaceutical Review, American Society of Gene & Cell Therapy (ASGCT), European Society of Gene and Cell Therapy (ESGCT), RNA Society (US), European Medicines Agency (EMA), Alliance for Regenerative Medicine (ARM), Annual Reports, SEC Filings, Press Releases, Investor Presentations, Corporate Presentations, Journals, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 53 ASIA PACIFIC: RNA THERAPEUTICS MARKET FOR OTHER INDICATIONS, BY COUNTRY, 2021–2028 (USD MILLION)

Country/Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Japan	36.2	94.1	37.8	41.5	46.7	51.9	56.8	61.1	10.1%
China	14.7	38.7	15.9	17.7	20.2	22.7	25.0	27.0	11.1%
South Korea	6.7	17.7	7.3	8.2	9.4	10.5	11.6	12.5	11.3%
Rest of Asia Pacific	8.3	22.2	9.8	11.3	13.3	15.2	16.9	18.3	13.4%
Total	**65.9**	**172.7**	**70.8**	**78.8**	**89.5**	**100.3**	**110.4**	**118.9**	**10.9%**

Source: National Institutes of Health (NIH), US Food and Drug Administration (US FDA), World Health Organization (WHO), European Pharmaceutical Review, American Society of Gene & Cell Therapy (ASGCT), European Society of Gene and Cell Therapy (ESGCT), RNA Society (US), European Medicines Agency (EMA), Alliance for Regenerative Medicine (ARM), Annual Reports, SEC Filings, Press Releases, Investor Presentations, Corporate Presentations, Journals, Interviews with Experts, and MarketsandMarkets Analysis

9 RNA THERAPEUTICS MARKET, BY END USER

KEY FINDINGS

- The hospitals and clinics segment accounted for the largest share of 93% of the total market in 2022.
- This segment is projected to reach USD 16,830.5 million by 2028 from USD 12,795.9 million in 2023, registering the fastest growth of 5.6%.
- The employment of mRNA-based vaccines in hospitals and clinics to manage the immunization of people against COVID-19 is one of the key growth contributors.
- The research settings segment accounted for 7% of the RNA therapeutics market in 2022. This segment is projected to reach USD 1,235 million by 2028 from USD 958 million in 2023, at a CAGR of 5.2% from 2023 to 2028.
- The therapeutic potential of RNA technology to create new opportunities in medical applications is anticipated to propel the use of RNA therapeutics in clinical research settings.

9.1 INTRODUCTION

Based on the end user segment, the RNA therapeutics market is segmented into hospitals and clinics and research settings. The growing use of RNA therapeutics across hospitals and clinics is driven by factors such as the increasing understanding and advancements of RNA-based therapeutics, the growing number of trends in personalized medicine, the increasing prevalence of chronic diseases, and the rising research investments by governments, private foundations, and industry stakeholders.

TABLE 54 RNA THERAPEUTICS MARKET, BY END USER, 2021–2028 (USD MILLION)

End User	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Hospitals and Clinics	13,103.6	30,001.9	12,795.9	13,815.8	15,155.5	16,255.2	16,878.3	16,830.5	5.6%
Research Settings	988.4	2,253.4	958.0	1,030.4	1,125.9	1,202.8	1,243.7	1,235.0	5.2%
Total	**14,092.0**	**32,255.3**	**13,753.8**	**14,846.2**	**16,281.4**	**17,458.0**	**18,122.1**	**18,065.5**	**5.6%**

Source: National Institutes of Health (NIH), US Food and Drug Administration (US FDA), World Health Organization (WHO), European Pharmaceutical Review, American Society of Gene & Cell Therapy (ASGCT), European Society of Gene and Cell Therapy (ESGCT), RNA Society (US), European Medicines Agency (EMA), Alliance for Regenerative Medicine (ARM), Annual Reports, SEC Filings, Press Releases, Investor Presentations, Corporate Presentations, Journals, Interviews with Experts, and MarketsandMarkets Analysis

9.2 HOSPITALS AND CLINICS

9.2.1 INCREASED PREVALENCE OF CHRONIC DISEASES AND HIGHER EFFICACY OF RNA-BASED THERAPIES TO DRIVE MARKET

There are several RNA therapeutic-based medicines that are employed in medical settings (hospitals and clinics), including Inclisiran, Volanesorsen, Givosiran, and Nusinersen, among others. In addition, mRNA-based COVID-19 vaccines are also employed since the COVID-19 breakout for widespread immunization against the SARS-CoV-2 virus.

The increasing prevalence of chronic diseases and genetic disorders, the proven efficacy of RNA-based therapies, and the increasing potential for the use of personalized medicine, which tailors treatments to individual patients, cumulatively propel the segment growth. Furthermore, the advancements in RNA technology and the success of RNA-based COVID-19 vaccines have increased the confidence of healthcare providers to adopt RNA therapeutics within their treatment protocols. Also, they have exemplified the transformative potential of RNA therapeutics within the hospitals and clinics segment. Healthcare providers have increasingly integrated RNA-based products into their treatment regimens, thereby enhancing patient care and contributing to market growth.

RNA-based treatments have proved their therapeutic potential for diseases such as diabetes, cancer, genetic disorders, and Huntington's disease. RNA therapies have also provided a better treatment option to target the pathophysiological mechanisms of these disorders, which leads to better outcomes for patients. Additionally, many RNA therapies have been approved by US FDA, and more therapies are there in various phases of clinical trials, demonstrating the validity of RNA therapies for various diseases. For instance, Patisiran has emerged as a transformative treatment for hereditary transthyretin-mediated amyloidosis (hATTR).

However, the need for specialized delivery systems to ensure the stability and targeted delivery of RNA molecules to specific cells has hindered the growth of the market. Additionally, safety concerns and the complexity of RNA-based therapies have posed obstacles to seamless adoption.

TABLE 55 RNA THERAPEUTICS MARKET FOR HOSPITALS AND CLINICS, BY REGION, 2021–2028 (USD MILLION)

Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
North America	3,437.7	7,548.4	3,896.0	4,489.1	5,155.5	5,695.0	5,975.9	5,900.1	8.7%
Europe	4,838.2	11,103.7	4,692.6	5,047.8	5,520.7	5,908.5	6,129.4	6,115.9	5.4%
Asia Pacific	4,225.8	9,934.4	3,690.3	3,756.5	3,934.5	4,087.0	4,193.5	4,229.2	2.8%
Rest of the World	601.9	1,415.3	516.9	522.5	544.8	564.7	579.5	585.3	2.5%
Total	**13,103.6**	**30,001.9**	**12,795.9**	**13,815.8**	**15,155.5**	**16,255.2**	**16,878.3**	**16,830.5**	**5.6%**

Source: National Institutes of Health (NIH), US Food and Drug Administration (US FDA), World Health Organization (WHO), European Pharmaceutical Review, American Society of Gene & Cell Therapy (ASGCT), European Society of Gene and Cell Therapy (ESGCT), RNA Society (US), European Medicines Agency (EMA), Alliance for Regenerative Medicine (ARM), Annual Reports, SEC Filings, Press Releases, Investor Presentations, Corporate Presentations, Journals, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 56 NORTH AMERICA: RNA THERAPEUTICS MARKET FOR HOSPITALS AND CLINICS, BY COUNTRY, 2021–2028 (USD MILLION)

Country	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
US	3,057.2	6,738.7	3,431.2	3,938.9	4,513.2	4,979.5	5,224.8	5,164.3	8.5%
Canada	380.4	809.8	464.8	550.2	642.3	715.5	751.2	735.8	9.6%
Total	**3,437.7**	**7,548.4**	**3,896.0**	**4,489.1**	**5,155.5**	**5,695.0**	**5,975.9**	**5,900.1**	**8.7%**

Source: National Institutes of Health (NIH), US Food and Drug Administration (US FDA), World Health Organization (WHO), European Pharmaceutical Review, American Society of Gene & Cell Therapy (ASGCT), European Society of Gene and Cell Therapy (ESGCT), RNA Society (US), European Medicines Agency (EMA), Alliance for Regenerative Medicine (ARM), Annual Reports, SEC Filings, Press Releases, Investor Presentations, Corporate Presentations, Journals, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 57 EUROPE: RNA THERAPEUTICS MARKET FOR HOSPITALS AND CLINICS, BY COUNTRY, 2021–2028 (USD MILLION)

Country/Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Germany	1,458.9	3,348.1	1,435.9	1,555.6	1,711.4	1,840.1	1,915.0	1,913.6	5.9%
UK	1,377.9	3,164.3	1,338.1	1,440.2	1,576.0	1,687.6	1,751.6	1,748.8	5.5%
France	863.6	1,975.4	856.5	931.3	1,027.0	1,105.8	1,150.6	1,147.9	6.0%
Rest of Europe	1,137.8	2,615.9	1,062.0	1,120.6	1,206.3	1,275.0	1,312.2	1,305.6	4.2%
Total	**4,838.2**	**11,103.7**	**4,692.6**	**5,047.8**	**5,520.7**	**5,908.5**	**6,129.4**	**6,115.9**	**5.4%**

Source: National Institutes of Health (NIH), US Food and Drug Administration (US FDA), World Health Organization (WHO), European Pharmaceutical Review, American Society of Gene & Cell Therapy (ASGCT), European Society of Gene and Cell Therapy (ESGCT), RNA Society (US), European Medicines Agency (EMA), Alliance for Regenerative Medicine (ARM), Annual Reports, SEC Filings, Press Releases, Investor Presentations, Corporate Presentations, Journals, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 58 ASIA PACIFIC: RNA THERAPEUTICS MARKET FOR HOSPITALS AND CLINICS, BY COUNTRY, 2021–2028 (USD MILLION)

Country/Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Japan	2,306.1	5,438.1	1,996.2	2,021.7	2,109.1	2,185.3	2,241.6	2,265.4	2.6%
China	946.1	2,225.4	827.2	842.7	883.4	918.4	943.1	951.6	2.8%
South Korea	433.0	1,013.6	375.0	380.4	397.1	411.2	420.5	422.0	2.4%
Rest of Asia Pacific	540.6	1,257.4	492.0	511.8	545.0	572.1	588.3	590.3	3.7%
Total	**4,225.8**	**9,934.4**	**3,690.3**	**3,756.5**	**3,934.5**	**4,087.0**	**4,193.5**	**4,229.2**	**2.8%**

Source: National Institutes of Health (NIH), US Food and Drug Administration (US FDA), World Health Organization (WHO), European Pharmaceutical Review, American Society of Gene & Cell Therapy (ASGCT), European Society of Gene and Cell Therapy (ESGCT), RNA Society (US), European Medicines Agency (EMA), Alliance for Regenerative Medicine (ARM), Annual Reports, SEC Filings, Press Releases, Investor Presentations, Corporate Presentations, Journals, Interviews with Experts, and MarketsandMarkets Analysis

9.3 RESEARCH SETTINGS

9.3.1 HIGHER THERAPEUTIC POTENTIAL OF RNA TECHNOLOGY TO CREATE NEW OPPORTUNITIES IN MEDICAL RESEARCH

Research organizations employ small interfering RNA (siRNA) molecules to target gene silencing and messenger RNA (mRNA) constructs for vaccine development. Advances in gene regulation and protein synthesis have led to a rise in research activities in the RNA-based therapeutics domain. The versatility of RNA therapies, enabling gene replacement, suppression, correction, and elimination, has been used by researchers to work on innovative therapies for disease treatment.

The Human Genome Project (HGP) has revealed vast information about the human genome and has greatly enhanced its role in developing biomedical research. As a result of the advancements in next-generation sequencing (NGS) technology, researchers have been able to reveal the role of some genetic factors in diseases such as cancer, rheumatoid arthritis, Parkinson's, and Alzheimer's. Moreover, studies have revealed the importance of coding and non-coding RNAs (ncRNAs), such as microRNAs (miRNA), long ncRNA (IncRNA), circular RNA (circRNA), and small interfering RNAs (siRNAs), in various diseases. Additionally, the success of mRNA-based vaccines against COVID-19 has catalyzed research organizations to harness RNA technology for diverse applications, such as addressing infectious diseases.

However, the complex nature of RNA molecules poses challenges related to stability, delivery, and potential immunogenicity. Funding constraints and the complexity of translating research findings into clinical applications can also impede progress in market growth.

TABLE 59 RNA THERAPEUTICS MARKET FOR RESEARCH SETTINGS, BY REGION, 2021–2028 (USD MILLION)

Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
North America	261.5	571.4	293.5	336.6	384.7	422.9	441.6	433.9	8.1%
Europe	365.7	836.8	352.6	378.1	412.2	439.8	454.8	452.4	5.1%
Asia Pacific	316.1	740.7	274.2	278.2	290.4	300.6	307.3	308.9	2.4%
Rest of the World	45.0	104.5	37.7	37.5	38.6	39.5	40.0	39.8	0.4%
Total	**988.4**	**2,253.4**	**958.0**	**1,030.4**	**1,125.9**	**1,202.8**	**1,243.7**	**1,235.0**	**5.2%**

Source: National Institutes of Health (NIH), US Food and Drug Administration (US FDA), World Health Organization (WHO), European Pharmaceutical Review, American Society of Gene & Cell Therapy (ASGCT), European Society of Gene and Cell Therapy (ESGCT), RNA Society (US), European Medicines Agency (EMA), Alliance for Regenerative Medicine (ARM), Annual Reports, SEC Filings, Press Releases, Investor Presentations, Corporate Presentations, Journals, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 60 NORTH AMERICA: RNA THERAPEUTICS MARKET FOR RESEARCH SETTINGS, BY COUNTRY, 2021–2028 (USD MILLION)

Country	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
US	232.4	509.9	258.5	295.3	336.8	369.9	386.3	380.1	8.0%
Canada	29.1	61.5	35.1	41.3	47.9	53.0	55.3	53.8	8.9%
Total	**261.5**	**571.4**	**293.5**	**336.6**	**384.7**	**422.9**	**441.6**	**433.9**	**8.1%**

Source: National Institutes of Health (NIH), US Food and Drug Administration (US FDA), World Health Organization (WHO), European Pharmaceutical Review, American Society of Gene & Cell Therapy (ASGCT), European Society of Gene and Cell Therapy (ESGCT), RNA Society (US), European Medicines Agency (EMA), Alliance for Regenerative Medicine (ARM), Annual Reports, SEC Filings, Press Releases, Investor Presentations, Corporate Presentations, Journals, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 61 EUROPE: RNA THERAPEUTICS MARKET FOR RESEARCH SETTINGS, BY COUNTRY, 2021–2028 (USD MILLION)

Country/Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Germany	110.9	253.4	108.2	116.6	127.7	136.7	141.6	140.8	5.4%
UK	104.3	238.7	100.6	108.0	117.8	125.8	130.1	129.5	5.2%
France	65.1	148.6	64.3	69.8	76.9	82.7	85.9	85.5	5.9%
Rest of Europe	85.5	196.1	79.4	83.6	89.8	94.7	97.3	96.5	4.0%
Total	**365.7**	**836.8**	**352.6**	**378.1**	**412.2**	**439.8**	**454.8**	**452.4**	**5.1%**

Source: National Institutes of Health (NIH), US Food and Drug Administration (US FDA), World Health Organization (WHO), European Pharmaceutical Review, American Society of Gene & Cell Therapy (ASGCT), European Society of Gene and Cell Therapy (ESGCT), RNA Society (US), European Medicines Agency (EMA), Alliance for Regenerative Medicine (ARM), Annual Reports, SEC Filings, Press Releases, Investor Presentations, Corporate Presentations, Journals, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 62 ASIA PACIFIC: RNA THERAPEUTICS MARKET FOR RESEARCH SETTINGS, BY COUNTRY, 2021–2028 (USD MILLION)

Country/Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Japan	172.5	406.5	149.1	150.9	157.3	162.8	166.9	168.6	2.5%
China	70.8	165.8	61.4	62.3	65.1	67.4	68.9	69.3	2.4%
South Korea	32.4	75.3	27.7	27.9	28.9	29.7	30.1	30.0	1.7%
Rest of Asia Pacific	40.4	93.1	36.1	37.1	39.1	40.7	41.4	41.1	2.6%
Total	**316.1**	**740.7**	**274.2**	**278.2**	**290.4**	**300.6**	**307.3**	**308.9**	**2.4%**

Source: National Institutes of Health (NIH), US Food and Drug Administration (US FDA), World Health Organization (WHO), European Pharmaceutical Review, American Society of Gene & Cell Therapy (ASGCT), European Society of Gene and Cell Therapy (ESGCT), RNA Society (US), European Medicines Agency (EMA), Alliance for Regenerative Medicine (ARM), Annual Reports, SEC Filings, Press Releases, Investor Presentations, Corporate Presentations, Journals, Interviews with Experts, and MarketsandMarkets Analysis

10 RNA THERAPEUTICS MARKET, BY REGION

KEY FINDINGS

- North America accounted for a share of 25.2% of the global RNA therapeutics market in 2022.
- The RNA therapeutics market in North America is projected to reach USD 6,334 million by 2028 from USD 4,189.6 million in 2023, at a CAGR of 8.6% during the forecast period.
- In 2022, the US accounted for the largest share of 89.3% of the RNA therapeutics market in North America. The market in the US is projected to reach USD 5,544.4 million by 2028 from USD 3,689.7 million in 2023, at a CAGR of 8.5% during the forecast period.
- Europe accounted for the largest share of 37% of the global RNA therapeutics market in 2022.
- The Asia Pacific region is projected to grow at a CAGR of 2.7% in the RNA therapeutics market during the forecast period.

10.1 INTRODUCTION

The global RNA therapeutics market is segmented into four major regions, namely, North America, Europe, the Asia Pacific, and the Rest of the World. In 2022, Europe accounted for the largest share of 37% of the RNA therapeutics market. This can be attributed to the government funding for drug discovery and life science research, advances in RNA therapeutics techniques, and growth in the biotechnology and pharmaceutical industries in this region. North America has emerged as the fastest growing region, registering a CAGR of 8.6% from 2023 to 2028. Europe leads the RNA vaccines market, accounting for more than 37% market share in 2022, whereas North America leads the RNA therapeutics/drugs market, accounting for over 52% in 2022. Cumulatively, Europe is anticipated to lead the revenue of RNA therapeutics (vaccines + drugs) through 2021 to 2028.

TABLE 63 RNA THERAPEUTICS MARKET, BY REGION, 2021–2028 (USD MILLION)

Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
North America	3,699.2	8,119.9	4,189.6	4,825.7	5,540.2	6,117.9	6,417.5	6,334.0	8.6%
Europe	5,203.9	11,940.5	5,045.2	5,425.8	5,932.9	6,348.4	6,584.2	6,568.3	5.4%
Asia Pacific	4,542.0	10,675.2	3,964.6	4,034.6	4,224.8	4,387.5	4,500.9	4,538.1	2.7%
Rest of the World	647.0	1,519.8	554.5	560.0	583.4	604.2	619.4	625.1	2.4%
Total	**14,092.0**	**32,255.3**	**13,753.8**	**14,846.2**	**16,281.4**	**17,458.0**	**18,122.1**	**18,065.5**	**5.6%**

Source: National Institutes of Health (NIH), US Food and Drug Administration (US FDA), World Health Organization (WHO), European Pharmaceutical Review, American Society of Gene & Cell Therapy (ASGCT), European Society of Gene and Cell Therapy (ESGCT), RNA Society (US), European Medicines Agency (EMA), Alliance for Regenerative Medicine (ARM), Annual Reports, SEC Filings, Press Releases, Investor Presentations, Corporate Presentations, Journals, Interviews with Experts, and MarketsandMarkets Analysis

10.2 NORTH AMERICA

North America (US and Canada) is anticipated to grow at the fastest rate of 8.6% during the forecast period. The key drivers for the growth of the RNA therapeutics market in North America include significant investments in R&D, robust technological advancements in RNA delivery systems, and a strong regulatory framework that supports innovation in the biopharmaceutical field. The region hosts a multitude of pharmaceutical and biotechnology companies that develop RNA-based therapies across various therapeutic areas, including oncology, rare diseases, infectious diseases, and neurological disorders.

North America's well-established healthcare infrastructure, coupled with a high prevalence of chronic diseases and a growing aging population, creates a conducive environment for the adoption of advanced therapeutic approaches such as RNA-based interventions. Furthermore, partnerships between academic and research institutions and industry players drive collaborations that accelerate the translation of cutting-edge scientific discoveries into clinical applications.

In January 2022, Pfizer Inc. (US) and Acuitas Therapeutics (Canada), a company that develops lipid nanoparticle (LNP) delivery systems to support messenger RNA (mRNA)-based therapeutics, announced that they have entered into a Development and Option agreement. Under this agreement, Pfizer can license Acuitas' LNP technology for up to 10 targets for vaccine or therapeutic development on a non-exclusive basis.

Some of the major players operating in the North American RNA therapeutics market are Sarepta Therapeutics (US), Ionis Pharmaceuticals, Inc. (US), Alnylam Pharmaceuticals (US), Moderna, Inc. (US), Arcturus Therapeutics (US), Novavax, Inc. (US), Pfizer Inc. (US), Arrowhead Pharmaceuticals, Inc. (US), Orna Therapeutics (US), CRISPR Therapeutics (US), Sirnaomics Inc (US), Arbutus Biopharma (US), and Atalanta Therapeutics (US).

FIGURE 23 NORTH AMERICA: RNA THERAPEUTICS MARKET SNAPSHOT

NORTH AMERICA

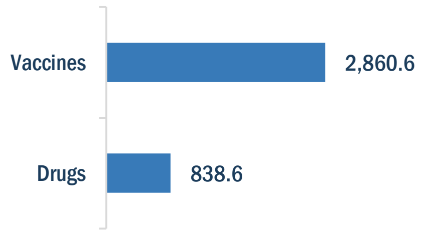
8.6%
CAGR (2023–2028)

USD 4,189.6 Million
Market Size in 2023

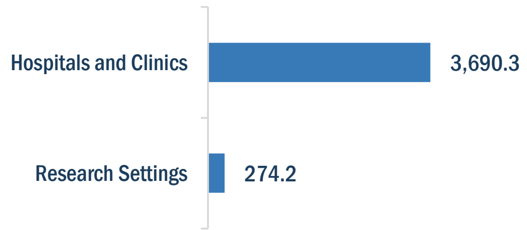
US
The Fastest-Growing Country in the Region

25.2%
Share of the Region in the Global Market in 2022

BY PRODUCT, 2023 (USD MILLION)



BY END USER, 2023 (USD MILLION)



BY COUNTRY, 2023 (USD MILLION)

COUNTRY	MARKET SIZE	CAGR (2023–2028)
US	3,689.7	8.5%
Canada	499.9	9.6%

DRIVING FACTORS FOR GROWTH IN NORTH AMERICA

- Rising availability of research grants and funding
- Funding initiatives for novel medicines, gene, and RNA therapies
- Rapid approvals for RNA-based vaccines

TABLE 64 NORTH AMERICA: RNA THERAPEUTICS MARKET, BY COUNTRY, 2021–2028 (USD MILLION)

Country	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
US	3,289.7	7,248.6	3,689.7	4,234.2	4,850.0	5,349.4	5,611.1	5,544.4	8.5%
Canada	409.5	871.2	499.9	591.5	690.2	768.5	806.5	789.6	9.6%
Total	**3,699.2**	**8,119.9**	**4,189.6**	**4,825.7**	**5,540.2**	**6,117.9**	**6,417.5**	**6,334.0**	**8.6%**

Source: National Institutes of Health (NIH), US Food and Drug Administration (US FDA), World Health Organization (WHO), European Pharmaceutical Review, American Society of Gene & Cell Therapy (ASGCT), European Society of Gene and Cell Therapy (ESGCT), RNA Society (US), European Medicines Agency (EMA), Alliance for Regenerative Medicine (ARM), Annual Reports, SEC Filings, Press Releases, Investor Presentations, Corporate Presentations, Journals, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 65 NORTH AMERICA: RNA THERAPEUTICS MARKET, BY PRODUCT, 2021–2028 (USD MILLION)

Product	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Vaccines	2,860.6	6,803.3	2,344.1	2,297.6	2,330.1	2,362.9	2,401.0	2,439.7	0.8%
Drugs	838.6	1,316.6	1,845.4	2,528.0	3,210.1	3,755.0	4,016.5	3,894.3	16.1%
Total	**3,699.2**	**8,119.9**	**4,189.6**	**4,825.7**	**5,540.2**	**6,117.9**	**6,417.5**	**6,334.0**	**8.6%**

Source: National Institutes of Health (NIH), US Food and Drug Administration (US FDA), World Health Organization (WHO), European Pharmaceutical Review, American Society of Gene & Cell Therapy (ASGCT), European Society of Gene and Cell Therapy (ESGCT), RNA Society (US), European Medicines Agency (EMA), Alliance for Regenerative Medicine (ARM), Annual Reports, SEC Filings, Press Releases, Investor Presentations, Corporate Presentations, Journals, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 66 NORTH AMERICA: RNA THERAPEUTICS MARKET, BY TYPE, 2021–2028 (USD MILLION)

Type	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
mRNA Therapeutics	3,415.9	7,478.9	3,862.6	4,445.8	5,100.1	5,596.8	5,866.2	5,785.3	8.4%
RNA Interference (RNAi) Therapeutics	252.2	558.7	291.0	338.2	391.8	436.6	462.1	460.1	9.6%
Antisense Oligonucleotide (ASO) Therapeutics	31.0	68.7	35.8	41.7	48.3	53.8	57.0	56.7	9.6%
Total	**3,699.1**	**8,106.3**	**4,189.4**	**4,825.7**	**5,540.2**	**6,087.2**	**6,385.3**	**6,302.1**	**8.5%**

Source: National Institutes of Health (NIH), US Food and Drug Administration (US FDA), World Health Organization (WHO), European Pharmaceutical Review, American Society of Gene & Cell Therapy (ASGCT), European Society of Gene and Cell Therapy (ESGCT), RNA Society (US), European Medicines Agency (EMA), Alliance for Regenerative Medicine (ARM), Annual Reports, SEC Filings, Press Releases, Investor Presentations, Corporate Presentations, Journals, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 67 NORTH AMERICA: RNA THERAPEUTICS MARKET, BY INDICATION, 2021–2028 (USD MILLION)

Indication	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Infectious Diseases	3,418.3	7,492.5	3,860.2	4,439.8	5,089.7	5,612.2	5,878.4	5,793.3	8.5%
Rare Genetic/Hereditary Diseases	224.6	501.4	263.0	307.9	359.1	402.9	429.2	430.1	10.3%
Other Indications	56.2	126.0	66.4	78.0	91.3	102.9	110.0	110.6	10.8%
Total	**3,699.2**	**8,119.9**	**4,189.6**	**4,825.7**	**5,540.2**	**6,117.9**	**6,417.5**	**6,334.0**	**8.6%**

Source: National Institutes of Health (NIH), US Food and Drug Administration (US FDA), World Health Organization (WHO), European Pharmaceutical Review, American Society of Gene & Cell Therapy (ASGCT), European Society of Gene and Cell Therapy (ESGCT), RNA Society (US), European Medicines Agency (EMA), Alliance for Regenerative Medicine (ARM), Annual Reports, SEC Filings, Press Releases, Investor Presentations, Corporate Presentations, Journals, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 68 NORTH AMERICA: RNA THERAPEUTICS MARKET, BY END USER, 2021–2028 (USD MILLION)

End User	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Hospitals and Clinics	3,437.7	7,548.4	3,896.0	4,489.1	5,155.5	5,695.0	5,975.9	5,900.1	8.7%
Research Settings	261.5	571.4	293.5	336.6	384.7	422.9	441.6	433.9	8.1%
Total	**3,699.2**	**8,119.9**	**4,189.6**	**4,825.7**	**5,540.2**	**6,117.9**	**6,417.5**	**6,334.0**	**8.6%**

Source: World Health Organization (WHO), American Society for Clinical Laboratory Science (ASCLS), Centers for Disease Control and Prevention (CDC), National Center for Biotechnology Information (NCBI), American Association for Clinical Chemistry (AACC), American Society for Pharmacology and Experimental Therapeutics (ASPET), National Human Genome Research Institute (NHGRI), Annual Reports, SEC Filings, Investor Presentations, Company Websites, Press Releases, Interviews with Experts, and MarketsandMarkets Analysis

10.2.1 NORTH AMERICA: RECESSION IMPACT

North America is very likely to enter a period of recession in mid-2023, with higher-than-expected inflation, especially in the US, triggering a tightening of global financial conditions. According to the International Monetary Fund (IMF) World Economic Outlook 2022, in the US, reduced household purchasing power and tighter monetary policies will drive down the GDP from 2.3% in 2022 to 1% in 2023. Similarly, in Canada, growth in real GDP is projected to slow from 3.2% in 2022 to 1.0% in 2023 before strengthening to 1.3% in 2024.

However, people's spending on healthcare is less likely to be affected by macroeconomic headwinds as diseases will always exist, and treatments and medicines will always be in demand, irrespective of economic turbulence. Health expenditure in North America could exceed economic growth by up to 5.9% in 2023. National health expenditure could grow at a rate of 7.1% over the next five years, from 2022 to 2027, compared with an expected economic growth rate of 4.7%.

The US government has taken action to combat the upcoming recession. In August 2022, the US government passed the Inflation Reduction Act, which features significant changes to federal prescription drug pricing policies that can reduce costs for individuals receiving care through Medicare. The bill also includes a three-year extension of expanded Affordable Care Act (ACA) healthcare benefits through 2025.

Additionally, more mergers and acquisitions are expected in 2023 in the US, with a deal value of USD 225 billion to USD 275 billion across all subsectors. Companies such as Johnson & Johnson and Pfizer have entered into deals with Abiomed and Biohaven Pharmaceuticals for USD 18.9 billion and USD 12.2 billion, respectively. In the pharmaceutical and life science sector, the US market is projected to exhibit higher growth in 2023, which can be attributed to the increasing number of mergers and acquisitions in the oncology and immunology streams in the pharmaceutical and biotechnology sectors.

However, the spillover effect of ongoing political and economic disturbances may affect the funding and grants offered to progress the drug discovery and development initiatives in the region, which could reduce outsourcing to CROs.

10.2.2 US

10.2.2.1 Growing technological innovation in RNA delivery systems and rising investments in biopharmaceutical research to drive market

The US holds a major share of global biomedical innovations. Owing to the scientific shift towards more complex biologic treatments, innovations in the US have remained relatively steady due to strong financial incentives to invest in R&D. Investments in medical health R&D in the US increased at a rate of 11.1% from 2019 to 2020 to reach USD 245.1 billion in 2020. Biopharmaceutical companies were the biggest contributors to R&D spending. Hence, technological advancements in RNA delivery systems, coupled with substantial investments in biopharmaceutical research, drive the development of innovative therapies targeting a wide range of medical conditions, including cancer, genetic disorders, and infectious diseases.

For instance, in March 2023, Evonik expanded its USD 220 million production facility for pharmaceutical specialty lipids in Indiana, US. The new facility secures the industry's access to critical excipients needed for mRNA vaccines and other nucleic acid therapies.

In March 2022, Nutcracker Therapeutics, a US-based biotech company, announced that it had raised USD 167 million in a Series C financing round led by ARCH Venture Partners. The company intends to use the funding to expand its portfolio of mRNA therapies and advance its biochip-based RNA manufacturing platform technology.

The increasing number of clinical trial studies is a significant driver for the advancement of the US RNA therapeutics market, ultimately leading to better regulatory approvals and improved patient care. Clinical trials play a pivotal role in demonstrating the safety, efficacy, and potential benefits of RNA-based therapies. The US benefits from a well-established regulatory framework that promotes innovation while ensuring patient safety. Regulatory agencies such as FDA play a pivotal role in evaluating and approving RNA therapeutics, providing a pathway for these therapies to reach patients in need. The presence of academic institutions, research centers, and a skilled workforce also contributes to the growth of the US RNA therapeutics market.

TABLE 69 LIST OF RECENT RNA THERAPEUTICS APPROVED BY FDA

PRODUCT	TYPE	MANUFACTURER	APPROVAL YEAR	INDICATION
Comirnaty (tozinameran)	mRNA	Pfizer-BioNTech	2021	COVID-19
Spikevax (elasomeran)	mRNA	Moderna	2022	COVID-19
Givlaari (givosiran)	siRNA	Alnylam	2019	Acute hepatic porphyria
Oxlumo (lumasiran)	siRNA	Alnylam	2020	Primary hyperoxaluria type 1
Leqvio (inclisiran)	siRNA	Novartis	2021	Heterozygous familial hypercholesterolemia (HeFH) or clinical atherosclerotic cardiovascular disease (ASCVD)
Vyondys 53 (golodisen)	ASO	Sarepta	2019	Duchenne muscular dystrophy (DMD)
Viltepso (viltolarsen)	ASO	NS Pharma	2020	Duchenne muscular dystrophy (DMD)
Amondys 45 (casimersen)	ASO	Sarepta	2021	Duchenne muscular dystrophy (DMD)

The key players operating in the US include Sarepta Therapeutics (US), Ionis Pharmaceuticals, Inc. (US), Alnylam Pharmaceuticals (US), Moderna, Inc. (US), Arcturus Therapeutics (US), Novavax, Inc. (US), Pfizer Inc. (US), Arrowhead Pharmaceuticals, Inc. (US), Orna Therapeutics (US), CRISPR Therapeutics (US), Sirnaomics Inc (US), Arbutus Biopharma (US), and Atalanta Therapeutics (US).

TABLE 70 US: RNA THERAPEUTICS MARKET, BY PRODUCT, 2021–2028 (USD MILLION)

Product	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Vaccines	2,573.1	6,123.0	2,110.9	2,070.2	2,100.6	2,131.4	2,166.9	2,203.0	0.9%
Drugs	716.5	1,125.7	1,578.8	2,164.0	2,749.5	3,218.0	3,444.2	3,341.3	16.2%
Total	**3,289.7**	**7,248.6**	**3,689.7**	**4,234.2**	**4,850.0**	**5,349.4**	**5,611.1**	**5,544.4**	**8.5%**

Source: National Institutes of Health (NIH), US Food and Drug Administration (US FDA), World Health Organization (WHO), European Pharmaceutical Review, American Society of Gene & Cell Therapy (ASGCT), European Society of Gene and Cell Therapy (ESGCT), RNA Society (US), European Medicines Agency (EMA), Alliance for Regenerative Medicine (ARM), Annual Reports, SEC Filings, Press Releases, Investor Presentations, Corporate Presentations, Journals, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 71 US: RNA THERAPEUTICS MARKET, BY TYPE, 2021–2028 (USD MILLION)

Type	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
mRNA Therapeutics	3,037.9	6,688.8	3,402.1	3,901.2	4,465.2	4,894.5	5,130.0	5,065.1	8.3%
RNA Interference (RNAi) Therapeutics	224.3	498.6	256.0	296.4	342.4	380.8	402.8	401.4	9.4%
Antisense Oligonucleotide (ASO) Therapeutics	27.5	61.2	31.5	36.6	42.4	47.3	50.2	50.1	9.7%
Total	**3,289.7**	**7,248.6**	**3,689.6**	**4,234.2**	**4,850.0**	**5,322.6**	**5,583.0**	**5,516.6**	**8.4%**

Source: National Institutes of Health (NIH), US Food and Drug Administration (US FDA), World Health Organization (WHO), European Pharmaceutical Review, American Society of Gene & Cell Therapy (ASGCT), European Society of Gene and Cell Therapy (ESGCT), RNA Society (US), European Medicines Agency (EMA), Alliance for Regenerative Medicine (ARM), Annual Reports, SEC Filings, Press Releases, Investor Presentations, Corporate Presentations, Journals, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 72 US: RNA THERAPEUTICS MARKET, BY INDICATION, 2021–2028 (USD MILLION)

Indication	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Infectious Diseases	3,039.8	6,688.8	3,399.9	3,896.1	4,456.5	4,908.4	5,141.2	5,072.9	8.3%
Rare Genetic/Hereditary Diseases	199.8	447.5	231.5	269.9	314.0	351.6	374.5	375.5	10.2%
Other Indications	50.0	112.4	58.3	68.2	79.5	89.3	95.4	95.9	10.5%
Total	**3,289.7**	**7,248.6**	**3,689.7**	**4,234.2**	**4,850.0**	**5,349.4**	**5,611.1**	**5,544.4**	**8.5%**

Source: National Institutes of Health (NIH), US Food and Drug Administration (US FDA), World Health Organization (WHO), European Pharmaceutical Review, American Society of Gene & Cell Therapy (ASGCT), European Society of Gene and Cell Therapy (ESGCT), RNA Society (US), European Medicines Agency (EMA), Alliance for Regenerative Medicine (ARM), Annual Reports, SEC Filings, Press Releases, Investor Presentations, Corporate Presentations, Journals, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 73 US: RNA THERAPEUTICS MARKET, BY END USER, 2021–2028 (USD MILLION)

End User	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Hospitals and Clinics	3,057.2	6,738.7	3,431.2	3,938.9	4,513.2	4,979.5	5,224.8	5,164.3	8.5%
Research Settings	232.4	509.9	258.5	295.3	336.8	369.9	386.3	380.1	8.0%
Total	**3,289.7**	**7,248.6**	**3,689.7**	**4,234.2**	**4,850.0**	**5,349.4**	**5,611.1**	**5,544.4**	**8.5%**

Source: World Health Organization (WHO), American Society for Clinical Laboratory Science (ASCLS), Centers for Disease Control and Prevention (CDC), National Center for Biotechnology Information (NCBI), American Association for Clinical Chemistry (AACC), American Society for Pharmacology and Experimental Therapeutics (ASPET), National Human Genome Research Institute (NHGRI), Annual Reports, SEC Filings, Investor Presentations, Company Websites, Press Releases, Interviews with Experts, and MarketsandMarkets Analysis

10.2.3 CANADA

10.2.3.1 Increasing government initiatives and investments for life science research to drive market

Government initiatives and investments play a key role in market growth. The Canadian government, through agencies such as the Canadian Institutes of Health Research (CIHR) and the Natural Sciences and Engineering Research Council of Canada (NSERC), provides substantial funding for various life science, drug/vaccine discovery, and other applications. Government support encourages innovation, facilitates the development of technologies, and drives research in cellular biology and disease mechanisms. In January 2023, CIHR funded USD 23 million to establish the Canadian Pediatric Cancer Consortium (CPCC). In April 2023, the Government of Canada invested USD 1.4 billion to support 11 large-scale research initiatives through the Canada First Research Excellence Fund (CFREF). One such initiative includes an investment of USD 165 million to McGill University for DNA to RNA: An Inclusive Canadian Approach to Genomic-based RNA Therapeutics (D2R). In April 2022, Moderna, Inc. announced that it is building a state-of-the-art mRNA vaccine manufacturing facility in Quebec, Canada, to support a long-term strategic partnership with the Government of Canada to improve pandemic preparedness.

Canada is home to renowned research institutions and universities that excel in life sciences and oligonucleotide research. Institutions such as the University of Toronto, McGill University, and the University of British Columbia have state-of-the-art research facilities and attract top scientists and researchers. These institutions contribute to advancements in RNA therapeutics techniques and their applications.

The favorable clinical trial environment in Canada has also shown significant opportunities for CROs to provide R&D support services, including drug discovery services. According to the Canadian government, 4% of the global clinical trials are conducted in the country. Canada is the second or third-most important location for clinical trials for some companies. As of January 2022, 24,631 clinical trials were being performed in Canada.

Some players involved in RNA therapeutics in the Canadian market include Providence Therapeutics (Canada), Deep Genomics (Canada), and Acuitas Therapeutics (Canada)

TABLE 74 CANADA: RNA THERAPEUTICS MARKET, BY PRODUCT, 2021–2028 (USD MILLION)

Product	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Vaccines	287.5	680.3	233.2	227.5	229.5	231.6	234.1	236.6	0.3%
Drugs	122.0	190.9	266.7	364.0	460.7	537.0	572.4	553.0	15.7%
Total	**409.5**	**871.2**	**499.9**	**591.5**	**690.2**	**768.5**	**806.5**	**789.6**	**9.6%**

Source: National Institutes of Health (NIH), US Food and Drug Administration (US FDA), World Health Organization (WHO), European Pharmaceutical Review, American Society of Gene & Cell Therapy (ASGCT), European Society of Gene and Cell Therapy (ESGCT), RNA Society (US), European Medicines Agency (EMA), Alliance for Regenerative Medicine (ARM), Annual Reports, SEC Filings, Press Releases, Investor Presentations, Corporate Presentations, Journals, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 75 CANADA: RNA THERAPEUTICS MARKET, BY TYPE, 2021–2028 (USD MILLION)

Type	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
mRNA Therapeutics	378.1	790.1	460.5	544.6	634.8	702.3	736.2	720.1	9.4%
RNA Interference (RNAi) Therapeutics	27.9	60.1	34.9	41.9	49.5	55.8	59.3	58.7	11.0%
Antisense Oligonucleotide (ASO) Therapeutics	3.6	7.5	4.3	5.1	5.9	6.5	6.8	6.6	8.9%
Total	**409.6**	**857.7**	**499.7**	**591.6**	**690.2**	**764.6**	**802.3**	**785.4**	**9.5%**

Source: National Institutes of Health (NIH), US Food and Drug Administration (US FDA), World Health Organization (WHO), European Pharmaceutical Review, American Society of Gene & Cell Therapy (ASGCT), European Society of Gene and Cell Therapy (ESGCT), RNA Society (US), European Medicines Agency (EMA), Alliance for Regenerative Medicine (ARM), Annual Reports, SEC Filings, Press Releases, Investor Presentations, Corporate Presentations, Journals, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 76 CANADA: RNA THERAPEUTICS MARKET, BY INDICATION, 2021–2028 (USD MILLION)

Indication	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Infectious Diseases	378.5	803.8	460.3	543.7	633.2	703.8	737.2	720.4	9.4%
Rare Genetic/Hereditary Diseases	24.8	53.9	31.5	38.0	45.2	51.2	54.7	54.5	11.6%
Other Indications	6.2	13.6	8.0	9.8	11.8	13.5	14.6	14.7	12.8%
Total	**409.5**	**871.2**	**499.9**	**591.5**	**690.2**	**768.5**	**806.5**	**789.6**	**9.6%**

Source: National Institutes of Health (NIH), US Food and Drug Administration (US FDA), World Health Organization (WHO), European Pharmaceutical Review, American Society of Gene & Cell Therapy (ASGCT), European Society of Gene and Cell Therapy (ESGCT), RNA Society (US), European Medicines Agency (EMA), Alliance for Regenerative Medicine (ARM), Annual Reports, SEC Filings, Press Releases, Investor Presentations, Corporate Presentations, Journals, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 77 CANADA: RNA THERAPEUTICS MARKET, BY END USER, 2021–2028 (USD MILLION)

End User	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Hospitals and Clinics	380.4	809.8	464.8	550.2	642.3	715.5	751.2	735.8	9.6%
Research Settings	29.1	61.5	35.1	41.3	47.9	53.0	55.3	53.8	8.9%
Total	**409.5**	**871.2**	**499.9**	**591.5**	**690.2**	**768.5**	**806.5**	**789.6**	**9.6%**

Source: World Health Organization (WHO), American Society for Clinical Laboratory Science (ASCLS), Centers for Disease Control and Prevention (CDC), National Center for Biotechnology Information (NCBI), American Association for Clinical Chemistry (AACC), American Society for Pharmacology and Experimental Therapeutics (ASPET), National Human Genome Research Institute (NHGRI), Annual Reports, SEC Filings, Investor Presentations, Company Websites, Press Releases, Interviews with Experts, and MarketsandMarkets Analysis

10.3 EUROPE

Factors such as the growing demand for RNA therapeutics for various chronic diseases, rising government investments in the pharmaceutical sector, and the growing pharmaceutical industry drive the growth of the European RNA therapeutics market. In 2022, Novartis invested USD 300 million to increase the capacity and capabilities for next-generation biotherapeutics across Switzerland, Slovenia, and Austria.

Moreover, the high degree of urbanization and mobility has increased the risk of epidemics such as SARS and H5N1 avian flu, which is expected to drive the growth of the RNA therapeutics market in the region. This will drive pharmaceutical and biopharmaceutical clinical trial activities for novel drug discovery and directly contribute to the demand for RNA-based therapies. Additionally, Europe's regulatory frameworks, including agencies such as the European Medicines Agency (EMA), support innovation while ensuring patient safety.

After North America, Europe is the second-largest regional operating in the pharmaceutical market. The research-based pharmaceutical industry can play a critical role in ensuring future competitiveness in an advancing global economy (Source: EFPIA). The development of new chemical and biological molecules, alongside increased R&D activities, will drive the growth of the RNA therapeutics market in Europe.

TABLE 78 EUROPE: RNA THERAPEUTICS MARKET, BY COUNTRY, 2021–2028 (USD MILLION)

Country/Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Germany	1,569.8	3,601.4	1,544.1	1,672.2	1,839.1	1,976.8	2,056.5	2,054.4	5.9%
UK	1,482.1	3,403.0	1,438.8	1,548.2	1,693.8	1,813.4	1,881.8	1,878.3	5.5%
France	928.7	2,124.0	920.9	1,001.1	1,103.9	1,188.4	1,236.5	1,233.4	6.0%
Rest of Europe	1,223.3	2,812.0	1,141.5	1,204.3	1,296.1	1,369.7	1,409.5	1,402.2	4.2%
Total	**5,203.9**	**11,940.5**	**5,045.2**	**5,425.8**	**5,932.9**	**6,348.4**	**6,584.2**	**6,568.3**	**5.4%**

Source: National Institutes of Health (NIH), US Food and Drug Administration (US FDA), World Health Organization (WHO), European Pharmaceutical Review, American Society of Gene & Cell Therapy (ASGCT), European Society of Gene and Cell Therapy (ESGCT), RNA Society (US), European Medicines Agency (EMA), Alliance for Regenerative Medicine (ARM), Annual Reports, SEC Filings, Press Releases, Investor Presentations, Corporate Presentations, Journals, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 79 EUROPE: RNA THERAPEUTICS MARKET, BY PRODUCT, 2021–2028 (USD MILLION)

Product	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Vaccines	4,644.0	11,063.4	3,818.4	3,749.0	3,808.3	3,868.5	3,937.5	4,007.6	1.0%
Drugs	559.9	877.1	1,226.8	1,676.9	2,124.6	2,479.8	2,646.8	2,560.7	15.9%
Total	**5,203.9**	**11,940.5**	**5,045.2**	**5,425.8**	**5,932.9**	**6,348.4**	**6,584.2**	**6,568.3**	**5.4%**

Source: National Institutes of Health (NIH), US Food and Drug Administration (US FDA), World Health Organization (WHO), European Pharmaceutical Review, American Society of Gene & Cell Therapy (ASGCT), European Society of Gene and Cell Therapy (ESGCT), RNA Society (US), European Medicines Agency (EMA), Alliance for Regenerative Medicine (ARM), Annual Reports, SEC Filings, Press Releases, Investor Presentations, Corporate Presentations, Journals, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 80 EUROPE: RNA THERAPEUTICS MARKET, BY TYPE, 2021–2028 (USD MILLION)

Type	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
mRNA Therapeutics	4,799.9	11,002.3	4,644.1	4,989.5	5,450.3	5,792.3	6,000.5	5,978.9	5.2%
RNA Interference (RNAi) Therapeutics	355.6	829.4	356.1	389.1	432.2	469.6	494.5	500.7	7.1%
Antisense Oligonucleotide (ASO) Therapeutics	48.5	108.8	44.9	47.2	50.4	52.6	53.2	51.7	2.8%
Total	**5,204.0**	**11,940.5**	**5,045.1**	**5,425.8**	**5,932.9**	**6,314.5**	**6,548.2**	**6,531.3**	**5.3%**

Source: National Institutes of Health (NIH), US Food and Drug Administration (US FDA), World Health Organization (WHO), European Pharmaceutical Review, American Society of Gene & Cell Therapy (ASGCT), European Society of Gene and Cell Therapy (ESGCT), RNA Society (US), European Medicines Agency (EMA), Alliance for Regenerative Medicine (ARM), Annual Reports, SEC Filings, Press Releases, Investor Presentations, Corporate Presentations, Journals, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 81 EUROPE: RNA THERAPEUTICS MARKET, BY INDICATION, 2021–2028 (USD MILLION)

Indication	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Infectious Diseases	4,812.2	11,010.3	4,638.9	4,974.6	5,424.0	5,787.2	5,985.0	5,953.3	5.1%
Rare Genetic/Hereditary Diseases	314.3	741.1	321.5	354.8	397.8	436.2	463.3	473.1	8.0%
Other Indications	77.4	189.1	84.8	96.4	111.1	124.9	135.9	141.8	10.8%
Total	**5,203.9**	**11,940.5**	**5,045.2**	**5,425.8**	**5,932.9**	**6,348.4**	**6,584.2**	**6,568.3**	**5.4%**

Source: National Institutes of Health (NIH), US Food and Drug Administration (US FDA), World Health Organization (WHO), European Pharmaceutical Review, American Society of Gene & Cell Therapy (ASGCT), European Society of Gene and Cell Therapy (ESGCT), RNA Society (US), European Medicines Agency (EMA), Alliance for Regenerative Medicine (ARM), Annual Reports, SEC Filings, Press Releases, Investor Presentations, Corporate Presentations, Journals, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 82 EUROPE: RNA THERAPEUTICS MARKET, BY END USER, 2021–2028 (USD MILLION)

End User	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Hospitals and Clinics	4,838.2	11,103.7	4,692.6	5,047.8	5,520.7	5,908.5	6,129.4	6,115.9	5.4%
Research Settings	365.7	836.8	352.6	378.1	412.2	439.8	454.8	452.4	5.1%
Total	**5,203.9**	**11,940.5**	**5,045.2**	**5,425.8**	**5,932.9**	**6,348.4**	**6,584.2**	**6,568.3**	**5.4%**

Source: World Health Organization (WHO), American Society for Clinical Laboratory Science (ASCLS), Centers for Disease Control and Prevention (CDC), National Center for Biotechnology Information (NCBI), American Association for Clinical Chemistry (AACC), American Society for Pharmacology and Experimental Therapeutics (ASPET), National Human Genome Research Institute (NHGRI), Annual Reports, SEC Filings, Investor Presentations, Company Websites, Press Releases, Interviews with Experts, and MarketsandMarkets Analysis

10.3.1 EUROPE: RECESSION IMPACT

Europe and most EU countries will head into an economic recession in the last quarter of 2022, according to the European Commission's autumn economic forecast. In the region, inflation is expected to have a higher impact on all the major economies, including the UK, Germany, France, Italy, and Spain. The region is expected to experience recession till mid-2024.

Healthcare and pharmaceutical spending by European nations has been on the rise. According to OECD, health spending continued to grow strongly in 2021 by 6%. Health spending was 9.7% and 9.6% of the GDP in 2020 and 2021, respectively. However, the proportion of health spending is not expected to grow any further. Additionally, the coming recession is expected to impact the healthcare and pharmaceutical industry with reduced health budgets, workers' wages, healthcare provision, pharmaceutical spending, and increased hours worked by healthcare professionals. Thus, there will be a slight impact on the healthcare market, including the RNA therapeutics market, with slow growth in 2023. The market, however, is expected to recover in 2025–2026.

This impact will be balanced by other factors, such as the expansion of business operations in Europe and increasing clinical trials and drug pipelines. International pharmaceutical companies have worked towards expanding their operations in Europe. For instance, in December 2022, Pfizer invested USD 1.2 billion in its manufacturing site in Puurs, Belgium. This expansion will occur over the next three years, with some projects already started while others are scheduled to begin in early 2023. Many pharmaceutical companies have a number of clinical trials registered for 2023 and 2024. This includes companies such as Merk, which has cancer drugs in the pipeline and for which clinical trial data is expected in 2023 and 2024. Such developments will help European countries overcome the economic downturn.

10.3.2 GERMANY

10.3.2.1 Rising number of market players and growing public and private funding for research activities to drive market

The strong presence of key market players, increasing public and private funding for advanced research activities, and the presence of a large number of academic research institutes are some significant factors that contribute to market growth in Germany. In July 2022, the German Federal Ministry of Education and Research (BMBF) announced that it would fund around USD 5.1 million (around 5 million Euros) to the Cluster for Nucleic Acid Therapeutics Munich (C-NATM) for RNA-based therapeutic R&D. The network is headed by scientists from Ludwig-Maximilians-Universität München (LMU) and the Technical University of Munich (TUM) to bring together start-ups and fundamental research organizations with pharmaceutical companies in the Munich area.

Companies such as CureVac N.V. and BioNTech have a strong presence in the country and offer a wide range of RNA therapeutics. Some other German-based players, including PRAMOMOLECULAR, Cardior Pharmaceuticals, and Ethris, also foster technological advancements and support the growth of the RNA therapeutics market.

Germany is known for its world-class research institutions, universities, and scientific infrastructure. Institutions such as the Max Planck Society, Helmholtz Association, and Leibniz Association undertake cutting-edge research in the life sciences. The availability of state-of-the-art facilities and a strong research ecosystem drive the growth of RNA therapeutics in Germany.

The National Center for Tumor Diseases (NCT) is a long-term cooperation between the German Cancer Research Center (DKFZ), university medical centers, and other outstanding research partners at various sites in Germany. On February 2, 2023, the German Federal Ministry of Education and Research (BMBF) confirmed the NCT expansion with four new sites: Berlin, SouthWest (Tübingen/Stuttgart-Ulm), WERA (Würzburg with partners Erlangen, Regensburg, and Augsburg), and West (Essen/Cologne). With the existing sites in Heidelberg and Dresden, a total of six NCT sites have now cooperated with the DKFZ to sustainably advance the state-of-the-art clinical cancer research in Germany and thereby improve the treatment outcomes and quality of life of cancer patients.

TABLE 83 GERMANY: RNA THERAPEUTICS MARKET, BY PRODUCT, 2021–2028 (USD MILLION)

Product	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Vaccines	1,390.0	3,319.0	1,148.2	1,129.9	1,150.5	1,171.4	1,195.0	1,219.1	1.2%
Drugs	179.9	282.4	395.9	542.3	688.6	805.5	861.5	835.3	16.1%
Total	**1,569.8**	**3,601.4**	**1,544.1**	**1,672.2**	**1,839.1**	**1,976.8**	**2,056.5**	**2,054.4**	**5.9%**

Source: National Institutes of Health (NIH), US Food and Drug Administration (US FDA), World Health Organization (WHO), European Pharmaceutical Review, American Society of Gene & Cell Therapy (ASGCT), European Society of Gene and Cell Therapy (ESGCT), RNA Society (US), European Medicines Agency (EMA), Alliance for Regenerative Medicine (ARM), Annual Reports, SEC Filings, Press Releases, Investor Presentations, Corporate Presentations, Journals, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 84 GERMANY: RNA THERAPEUTICS MARKET, BY TYPE, 2021–2028 (USD MILLION)

Type	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
mRNA Therapeutics	3,415.9	7,478.9	3,862.6	4,445.8	5,100.1	5,596.8	5,866.2	5,785.3	8.4%
RNA Interference (RNAi) Therapeutics	107.0	249.2	108.4	119.0	132.8	144.7	152.6	154.5	7.3%
Antisense Oligonucleotide (ASO) Therapeutics	14.1	31.9	13.4	14.3	15.4	16.3	16.6	16.3	4.0%
Total	**3,537.0**	**7,760.0**	**3,984.4**	**4,579.1**	**5,248.3**	**5,757.8**	**6,035.4**	**5,956.1**	**8.4%**

Source: National Institutes of Health (NIH), US Food and Drug Administration (US FDA), World Health Organization (WHO), European Pharmaceutical Review, American Society of Gene & Cell Therapy (ASGCT), European Society of Gene and Cell Therapy (ESGCT), RNA Society (US), European Medicines Agency (EMA), Alliance for Regenerative Medicine (ARM), Annual Reports, SEC Filings, Press Releases, Investor Presentations, Corporate Presentations, Journals, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 85 GERMANY: RNA THERAPEUTICS MARKET, BY INDICATION, 2021–2028 (USD MILLION)

Indication	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Infectious Diseases	1,451.3	3,321.8	1,421.0	1,535.4	1,684.7	1,806.8	1,875.3	1,869.0	5.6%
Rare Genetic/Hereditary Diseases	95.0	223.1	97.8	108.3	121.6	133.5	141.8	144.5	8.1%
Other Indications	23.5	56.5	25.3	28.6	32.7	36.6	39.5	40.9	10.1%
Total	**1,569.8**	**3,601.4**	**1,544.1**	**1,672.2**	**1,839.1**	**1,976.8**	**2,056.5**	**2,054.4**	**5.9%**

Source: National Institutes of Health (NIH), US Food and Drug Administration (US FDA), World Health Organization (WHO), European Pharmaceutical Review, American Society of Gene & Cell Therapy (ASGCT), European Society of Gene and Cell Therapy (ESGCT), RNA Society (US), European Medicines Agency (EMA), Alliance for Regenerative Medicine (ARM), Annual Reports, SEC Filings, Press Releases, Investor Presentations, Corporate Presentations, Journals, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 86 GERMANY: RNA THERAPEUTICS MARKET, BY END USER, 2021–2028 (USD MILLION)

End User	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Hospitals and Clinics	1,458.9	3,348.1	1,435.9	1,555.6	1,711.4	1,840.1	1,915.0	1,913.6	5.9%
Research Settings	110.9	253.4	108.2	116.6	127.7	136.7	141.6	140.8	5.4%
Total	**1,569.8**	**3,601.4**	**1,544.1**	**1,672.2**	**1,839.1**	**1,976.8**	**2,056.5**	**2,054.4**	**5.9%**

Source: World Health Organization (WHO), American Society for Clinical Laboratory Science (ASCLS), Centers for Disease Control and Prevention (CDC), National Center for Biotechnology Information (NCBI), American Association for Clinical Chemistry (AACC), American Society for Pharmacology and Experimental Therapeutics (ASPET), National Human Genome Research Institute (NHGRI), Annual Reports, SEC Filings, Investor Presentations, Company Websites, Press Releases, Interviews with Experts, and MarketsandMarkets Analysis

10.3.3 FRANCE

10.3.3.1 Growing pharmaceutical manufacturing industry and increasing number of clinical trials for oncology to drive market

France has a large pharmaceutical manufacturing industry owing to the presence of many global pharmaceutical manufacturers. In June 2021, Sanofi invested approximately USD 435 million (€400 million) annually in a vaccines mRNA Center of Excellence across sites at Cambridge, MA (US) and Marcy l'Etoile, Lyon (France). In the EU, France has the third-highest public expenditure on R&D, after Germany and the UK. The French National Research Agency provides funds for scientific research projects (both basic and applied research) to public research organizations, universities, and private companies. An increasing number of clinical trials for oncology and rare diseases, favorable government policies, growing funding for research projects, and an increase in drug discovery activities are the major factors expected to drive the growth of the RNA therapeutics market in France.

France is a competitive and attractive location for clinical trials. In 2022, 2,176 clinical trials were ongoing in France. According to WHO ICTRP, the number of clinical trial registrations in France went from 2,726 in 2017 to 3,317 in 2021, at a CAGR of 5% between 2017 and 2021. The country's main strength in clinical trials is its focus on cancer therapy, with 473 ongoing trials for oncology in 2022. Besides, according to Pharma R&D Review 2022, ~10.2% of the R&D pipeline is under development in France. Such factors will likely boost the market for RNA therapeutics in the region.

TABLE 87 FRANCE: RNA THERAPEUTICS MARKET, BY PRODUCT, 2021–2028 (USD MILLION)

Product	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Vaccines	816.0	1,947.2	673.2	662.1	673.7	685.5	698.9	712.6	1.1%
Drugs	112.7	176.8	247.7	339.1	430.2	502.9	537.6	520.8	16.0%
Total	**928.7**	**2,124.0**	**920.9**	**1,001.1**	**1,103.9**	**1,188.4**	**1,236.5**	**1,233.4**	**6.0%**

Source: National Institutes of Health (NIH), US Food and Drug Administration (US FDA), World Health Organization (WHO), European Pharmaceutical Review, American Society of Gene & Cell Therapy (ASGCT), European Society of Gene and Cell Therapy (ESGCT), RNA Society (US), European Medicines Agency (EMA), Alliance for Regenerative Medicine (ARM), Annual Reports, SEC Filings, Press Releases, Investor Presentations, Corporate Presentations, Journals, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 88 FRANCE: RNA THERAPEUTICS MARKET, BY TYPE, 2021–2028 (USD MILLION)

Type	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
mRNA Therapeutics	856.2	1,956.5	847.5	920.5	1,014.1	1,084.3	1,126.9	1,122.8	5.8%
RNA Interference (RNAi) Therapeutics	63.5	147.8	65.2	72.1	80.8	88.4	93.5	94.7	7.8%
Antisense Oligonucleotide (ASO) Therapeutics	8.9	19.6	8.2	8.6	9.0	9.3	9.3	8.8	1.5%
Total	**928.6**	**2,123.9**	**920.9**	**1,001.2**	**1,103.9**	**1,182.0**	**1,229.7**	**1,226.3**	**5.9%**

Source: National Institutes of Health (NIH), US Food and Drug Administration (US FDA), World Health Organization (WHO), European Pharmaceutical Review, American Society of Gene & Cell Therapy (ASGCT), European Society of Gene and Cell Therapy (ESGCT), RNA Society (US), European Medicines Agency (EMA), Alliance for Regenerative Medicine (ARM), Annual Reports, SEC Filings, Press Releases, Investor Presentations, Corporate Presentations, Journals, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 89 FRANCE: RNA THERAPEUTICS MARKET, BY INDICATION, 2021–2028 (USD MILLION)

Indication	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Infectious Diseases	858.9	1,958.2	846.3	917.2	1,008.2	1,081.9	1,122.0	1,115.7	5.7%
Rare Genetic/Hereditary Diseases	56.0	132.0	58.9	65.8	74.6	82.4	88.0	90.0	8.8%
Other Indications	13.7	33.8	15.7	18.1	21.2	24.1	26.5	27.8	12.1%
Total	**928.7**	**2,124.0**	**920.9**	**1,001.1**	**1,103.9**	**1,188.4**	**1,236.5**	**1,233.4**	**6.0%**

Source: National Institutes of Health (NIH), US Food and Drug Administration (US FDA), World Health Organization (WHO), European Pharmaceutical Review, American Society of Gene & Cell Therapy (ASGCT), European Society of Gene and Cell Therapy (ESGCT), RNA Society (US), European Medicines Agency (EMA), Alliance for Regenerative Medicine (ARM), Annual Reports, SEC Filings, Press Releases, Investor Presentations, Corporate Presentations, Journals, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 90 FRANCE: RNA THERAPEUTICS MARKET, BY END USER, 2021–2028 (USD MILLION)

End User	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Hospitals and Clinics	863.6	1,975.4	856.5	931.3	1,027.0	1,105.8	1,150.6	1,147.9	6.0%
Research Settings	65.1	148.6	64.3	69.8	76.9	82.7	85.9	85.5	5.9%
Total	**928.7**	**2,124.0**	**920.9**	**1,001.1**	**1,103.9**	**1,188.4**	**1,236.5**	**1,233.4**	**6.0%**

Source: World Health Organization (WHO), American Society for Clinical Laboratory Science (ASCLS), Centers for Disease Control and Prevention (CDC), National Center for Biotechnology Information (NCBI), American Association for Clinical Chemistry (AACC), American Society for Pharmacology and Experimental Therapeutics (ASPET), National Human Genome Research Institute (NHGRI), Annual Reports, SEC Filings, Investor Presentations, Company Websites, Press Releases, Interviews with Experts, and MarketsandMarkets Analysis

10.3.4 UK

10.3.4.1 Higher R&D investment by pharmaceutical companies for RNA-based drug discovery to drive market

In the UK, pharmaceuticals remain the highest R&D spending sector. In 2020, the total spending on R&D by the pharmaceutical sector amounted to ~USD 6.76 billion (GBP 5.02 billion). A relatively R&D-intensive country for pharmaceuticals, the UK aims to increase the GDP expenditure on R&D to 2.4% by 2027.

The strength of the partnership between the government, the NHS, academia, and the industry players makes the UK a global standard-bearer for discovery research and advanced manufacturing. In June 2023, the Centre for Process Innovation (CPI) launched a new UK Intracellular Drug Delivery Centre to create innovative lipid nanoparticle (LNP) formulations for RNA drug delivery and a framework to improve next-generation nano-delivery systems. Through the partnership, CPI has collaborated with the Universities of Strathclyde, Liverpool, Imperial College London, and Medicines Discovery Catapult using a USD 12.7 million (£10 million) three-year grant from Innovate UK's Transforming Medicines Manufacturing program. Additionally, in June 2022, the UK government and Moderna, Inc. announced an initial agreement to establish an mRNA Innovation and Technology Centre in the UK. Moderna's respiratory virus vaccine candidates, along with rapid pandemic response capabilities, should be accessible from this mRNA vaccine manufacturing facility. Moderna also intends to increase its R&D expenditures to strengthen its position in the UK.

In June 2023, Innovate UK announced a grant of over USD 892 K (£700 K) to VaxEquity, a company that develops transformative RNA vaccines and medicines based on its next generation self-amplifying RNA (saRNA) platform, and CPI, the UK's largest independent technology innovation center. A joint project improves the production of RNA vaccines and treatments.

Such collaborations between the government and private stakeholders accelerate and increase drug discovery opportunities in the country, which will significantly drive the RNA therapeutics market in the UK.

VaxEquity, MiNA Therapeutics, Storm Therapeutics, Roquefort Therapeutics, and Silence Therapeutics are some players that are operating in the UK RNA therapeutics market.

TABLE 91 UK: RNA THERAPEUTICS MARKET, BY PRODUCT, 2021–2028 (USD MILLION)

Product	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Vaccines	1,322.6	3,153.1	1,089.0	1,070.0	1,087.6	1,105.6	1,126.1	1,147.0	1.0%
Drugs	159.5	250.0	349.7	478.2	606.2	707.7	755.7	731.3	15.9%
Total	**1,482.1**	**3,403.0**	**1,438.8**	**1,548.2**	**1,693.8**	**1,813.4**	**1,881.8**	**1,878.3**	**5.5%**

Source: National Institutes of Health (NIH), US Food and Drug Administration (US FDA), World Health Organization (WHO), European Pharmaceutical Review, American Society of Gene & Cell Therapy (ASGCT), European Society of Gene and Cell Therapy (ESGCT), RNA Society (US), European Medicines Agency (EMA), Alliance for Regenerative Medicine (ARM), Annual Reports, SEC Filings, Press Releases, Investor Presentations, Corporate Presentations, Journals, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 92 UK: RNA THERAPEUTICS MARKET, BY TYPE, 2021–2028 (USD MILLION)

Type	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
mRNA Therapeutics	1,367.1	3,136.1	1,324.7	1,424.1	1,556.6	1,655.4	1,715.9	1,710.9	5.2%
RNA Interference (RNAi) Therapeutics	101.2	236.1	101.4	110.8	123.1	133.8	140.9	142.7	7.1%
Antisense Oligonucleotide (ASO) Therapeutics	13.8	30.8	12.7	13.2	14.1	14.6	14.7	14.2	2.3%
Total	**1,482.1**	**3,403.0**	**1,438.8**	**1,548.1**	**1,693.8**	**1,803.8**	**1,871.5**	**1,867.8**	**5.4%**

Source: National Institutes of Health (NIH), US Food and Drug Administration (US FDA), World Health Organization (WHO), European Pharmaceutical Review, American Society of Gene & Cell Therapy (ASGCT), European Society of Gene and Cell Therapy (ESGCT), RNA Society (US), European Medicines Agency (EMA), Alliance for Regenerative Medicine (ARM), Annual Reports, SEC Filings, Press Releases, Investor Presentations, Corporate Presentations, Journals, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 93 UK: RNA THERAPEUTICS MARKET, BY INDICATION, 2021–2028 (USD MILLION)

Indication	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Infectious Diseases	1,370.5	3,138.2	1,323.2	1,420.0	1,549.3	1,654.1	1,711.8	1,703.9	5.2%
Rare Genetic/Hereditary Diseases	89.6	211.1	91.6	101.0	113.2	124.1	131.8	134.6	8.0%
Other Indications	22.1	53.8	24.0	27.3	31.3	35.2	38.2	39.8	10.6%
Total	**1,482.1**	**3,403.0**	**1,438.8**	**1,548.2**	**1,693.8**	**1,813.4**	**1,881.8**	**1,878.3**	**5.5%**

Source: National Institutes of Health (NIH), US Food and Drug Administration (US FDA), World Health Organization (WHO), European Pharmaceutical Review, American Society of Gene & Cell Therapy (ASGCT), European Society of Gene and Cell Therapy (ESGCT), RNA Society (US), European Medicines Agency (EMA), Alliance for Regenerative Medicine (ARM), Annual Reports, SEC Filings, Press Releases, Investor Presentations, Corporate Presentations, Journals, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 94 UK: RNA THERAPEUTICS MARKET, BY END USER, 2021–2028 (USD MILLION)

End User	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Hospitals and Clinics	1,377.9	3,164.3	1,338.1	1,440.2	1,576.0	1,687.6	1,751.6	1,748.8	5.5%
Research Settings	104.3	238.7	100.6	108.0	117.8	125.8	130.1	129.5	5.2%
Total	**1,482.1**	**3,403.0**	**1,438.8**	**1,548.2**	**1,693.8**	**1,813.4**	**1,881.8**	**1,878.3**	**5.5%**

Source: World Health Organization (WHO), American Society for Clinical Laboratory Science (ASCLS), Centers for Disease Control and Prevention (CDC), National Center for Biotechnology Information (NCBI), American Association for Clinical Chemistry (AACC), American Society for Pharmacology and Experimental Therapeutics (ASPET), National Human Genome Research Institute (NHGRI), Annual Reports, SEC Filings, Investor Presentations, Company Websites, Press Releases, Interviews with Experts, and MarketsandMarkets Analysis

10.3.5 REST OF EUROPE

The Rest of Europe region comprises Italy, Spain, the Netherlands, Russia, Switzerland, Sweden, Turkey, Norway, Poland, Portugal, and Austria.

These countries host international conferences and scientific events focused on RNA therapeutics and related fields. These gatherings provide platforms for researchers, industry professionals, and experts to share knowledge, present advancements, and foster collaborations, contributing to the growth of the RNA therapeutics market.

The favorable government initiatives strengthen the healthcare infrastructure in these countries. The growth of this market in Russia is driven by the implementation of favorable government initiatives and programs to improve the healthcare infrastructure. Government initiatives such as the 2020 Healthcare Development Program and the National Project "Health" focus on improving healthcare facilities by bringing advanced technologies to its healthcare system. Russia presents a potential market for the pharmaceutical sector. Russia's Pharma 2020, a health reform plan launched in October 2009, has boosted the drug development sector in the country. The plan aims to develop clusters that will host a chain of companies with a full pharma cycle—R&D, production, professional training, distribution, logistics, and sales.

The convergence of research excellence, government support, thriving biomedical sectors, technological advancements, collaborations, translational research, education and training, industry-academia collaborations, and international scientific events propel the progress of RNA therapeutics in these regions.

TABLE 95 REST OF EUROPE: RNA THERAPEUTICS MARKET, BY PRODUCT, 2021–2028 (USD MILLION)

Product	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Vaccines	1,115.5	2,644.2	908.0	887.0	896.5	906.0	917.4	929.0	0.5%
Drugs	107.8	167.9	233.5	317.3	399.6	463.7	492.0	473.2	15.2%
Total	**1,223.3**	**2,812.0**	**1,141.5**	**1,204.3**	**1,296.1**	**1,369.7**	**1,409.5**	**1,402.2**	**4.2%**

Source: National Institutes of Health (NIH), US Food and Drug Administration (US FDA), World Health Organization (WHO), European Pharmaceutical Review, American Society of Gene & Cell Therapy (ASGCT), European Society of Gene and Cell Therapy (ESGCT), RNA Society (US), European Medicines Agency (EMA), Alliance for Regenerative Medicine (ARM), Annual Reports, SEC Filings, Press Releases, Investor Presentations, Corporate Presentations, Journals, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 96 REST OF EUROPE: RNA THERAPEUTICS MARKET, BY TYPE, 2021–2028 (USD MILLION)

Type	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
mRNA Therapeutics	1,127.8	2,589.2	1,049.7	1,106.0	1,188.8	1,247.1	1,281.4	1,272.8	3.9%
RNA Interference (RNAi) Therapeutics	83.8	196.3	81.1	87.2	95.5	102.7	107.5	108.8	6.0%
Antisense Oligonucleotide (ASO) Therapeutics	11.7	26.6	10.7	11.1	11.9	12.4	12.6	12.4	3.1%
Total	**1,223.3**	**2,812.1**	**1,141.5**	**1,204.3**	**1,296.2**	**1,362.2**	**1,401.5**	**1,394.0**	**4.1%**

Source: National Institutes of Health (NIH), US Food and Drug Administration (US FDA), World Health Organization (WHO), European Pharmaceutical Review, American Society of Gene & Cell Therapy (ASGCT), European Society of Gene and Cell Therapy (ESGCT), RNA Society (US), European Medicines Agency (EMA), Alliance for Regenerative Medicine (ARM), Annual Reports, SEC Filings, Press Releases, Investor Presentations, Corporate Presentations, Journals, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 97 REST OF EUROPE: RNA THERAPEUTICS MARKET, BY INDICATION, 2021–2028 (USD MILLION)

Indication	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Infectious Diseases	1,131.6	2,592.0	1,048.4	1,102.1	1,181.9	1,244.5	1,275.9	1,264.7	3.8%
Rare Genetic/Hereditary Diseases	73.7	175.0	73.3	79.8	88.4	96.2	101.8	104.1	7.3%
Other Indications	18.0	45.0	19.7	22.4	25.8	29.0	31.7	33.4	11.1%
Total	**1,223.3**	**2,812.0**	**1,141.5**	**1,204.3**	**1,296.1**	**1,369.7**	**1,409.5**	**1,402.2**	**4.2%**

Source: National Institutes of Health (NIH), US Food and Drug Administration (US FDA), World Health Organization (WHO), European Pharmaceutical Review, American Society of Gene & Cell Therapy (ASGCT), European Society of Gene and Cell Therapy (ESGCT), RNA Society (US), European Medicines Agency (EMA), Alliance for Regenerative Medicine (ARM), Annual Reports, SEC Filings, Press Releases, Investor Presentations, Corporate Presentations, Journals, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 98 REST OF EUROPE: RNA THERAPEUTICS MARKET, BY END USER, 2021–2028 (USD MILLION)

End User	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Hospitals and Clinics	1,137.8	2,615.9	1,062.0	1,120.6	1,206.3	1,275.0	1,312.2	1,305.6	4.2%
Research Settings	85.5	196.1	79.4	83.6	89.8	94.7	97.3	96.5	4.0%
Total	**1,223.3**	**2,812.0**	**1,141.5**	**1,204.3**	**1,296.1**	**1,369.7**	**1,409.5**	**1,402.2**	**4.2%**

Source: World Health Organization (WHO), American Society for Clinical Laboratory Science (ASCLS), Centers for Disease Control and Prevention (CDC), National Center for Biotechnology Information (NCBI), American Association for Clinical Chemistry (AACC), American Society for Pharmacology and Experimental Therapeutics (ASPET), National Human Genome Research Institute (NHGRI), Annual Reports, SEC Filings, Investor Presentations, Company Websites, Press Releases, Interviews with Experts, and MarketsandMarkets Analysis

10.4 ASIA PACIFIC

One significant growth driver for the RNA therapeutics market in the Asia Pacific region is the increasing investment in healthcare infrastructure and R&D activities. Government bodies and private entities have allocated substantial funds and launched various research programs to enhance healthcare facilities and promote scientific advancements, including RNA therapeutics technologies. These investments drive the adoption of RNA therapeutics techniques across various indications and research fields. The Asia-Pacific Economic Cooperation (APEC) Life Sciences Innovation Forum (LSIF) and its Regulatory Harmonization Steering Committee (RHSC) adopted a strategic plan (Vision 2020: A Strategic Framework: Regulatory Convergence for Medical Products by 2020). This strategic plan provided the basic proposal and rationale for achieving regional regulatory convergence of medical product approval procedures. The forum's vision is to accelerate regulatory convergence for medical products in the APAC region as much as possible by 2030 to protect people's safety, make life-saving products available, save public resources, attract investments, mitigate corruption, and improve global standing in every APAC economy.

Another driver is the rising prevalence of chronic diseases and the need for effective disease monitoring and drug development. RNA therapeutics has proven to play a crucial role in cancer treatment and has been instrumental in advancing the understanding of the disease and discovering potential therapies.

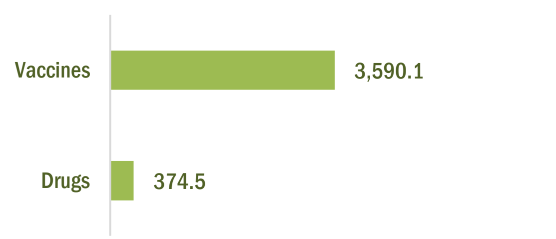
Moreover, the rapid growth of the pharmaceutical and biotechnology sectors in the region drives the demand for RNA therapeutics solutions. Pharmaceutical companies have incorporated an RNA therapeutics approach for the discovery of new RNA therapeutics.

FIGURE 24 ASIA PACIFIC: RNA THERAPEUTICS MARKET SNAPSHOT

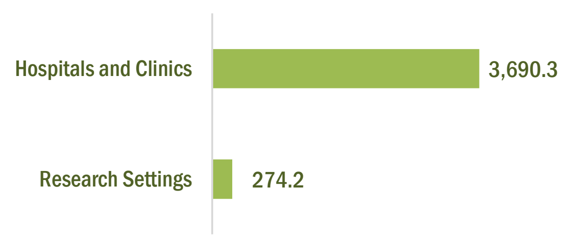
ASIA PACIFIC

2.7% CAGR (2023–2028)	USD 3,964.6 Million Market Size in 2023	China The Fastest-growing Country in the Region	33.1% Share of the Region in the Global Market in 2022

BY PRODUCT 2023 (USD MILLION)



BY END USER, 2023 (USD MILLION)



BY COUNTRY, 2023 (USD MILLION)

COUNTRY/ REGION	MARKET SIZE	CAGR (2023-2028)
Japan	2,145.3	2.6%
China	888.6	2.8%
South Korea	402.7	2.3%
Rest of Asia Pacific	528.0	3.6%

DRIVING FACTORS FOR GROWTH IN ASIA PACIFIC

- Increasing investments by larger pharmaceutical companies
- Low operating cost

Source: National Institutes of Health (NIH), US Food and Drug Administration (US FDA), World Health Organization (WHO), European Pharmaceutical Review, American Society of Gene & Cell Therapy (ASGCT), European Society of Gene and Cell Therapy (ESGCT), RNA Society (US), European Medicines Agency (EMA), Alliance for Regenerative Medicine (ARM), Annual Reports, SEC Filings, Press Releases, Investor Presentations, Corporate Presentations, Journals, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 99 ASIA PACIFIC: RNA THERAPEUTICS MARKET, BY COUNTRY, 2021–2028 (USD MILLION)

Country/Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Japan	2,478.6	5,844.5	2,145.3	2,172.5	2,266.4	2,348.1	2,408.5	2,433.9	2.6%
China	1,016.9	2,391.2	888.6	905.0	948.4	985.8	1,012.1	1,020.8	2.8%
South Korea	465.4	1,088.9	402.7	408.2	426.0	440.9	450.7	452.0	2.3%
Rest of Asia Pacific	581.1	1,350.5	528.0	548.9	584.1	612.8	629.7	631.4	3.6%
Total	**4,542.0**	**10,675.2**	**3,964.6**	**4,034.6**	**4,224.8**	**4,387.5**	**4,500.9**	**4,538.1**	**2.7%**

Source: National Institutes of Health (NIH), US Food and Drug Administration (US FDA), World Health Organization (WHO), European Pharmaceutical Review, American Society of Gene & Cell Therapy (ASGCT), European Society of Gene and Cell Therapy (ESGCT), RNA Society (US), European Medicines Agency (EMA), Alliance for Regenerative Medicine (ARM), Annual Reports, SEC Filings, Press Releases, Investor Presentations, Corporate Presentations, Journals, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 100 ASIA PACIFIC: RNA THERAPEUTICS MARKET, BY PRODUCT, 2021–2028 (USD MILLION)

Product	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Vaccines	4,370.9	10,407.3	3,590.1	3,523.0	3,576.9	3,631.5	3,694.3	3,758.2	0.9%
Drugs	171.1	267.9	374.5	511.6	648.0	756.0	806.5	779.9	15.8%
Total	**4,542.0**	**10,675.2**	**3,964.6**	**4,034.6**	**4,224.8**	**4,387.5**	**4,500.9**	**4,538.1**	**2.7%**

Source: National Institutes of Health (NIH), US Food and Drug Administration (US FDA), World Health Organization (WHO), European Pharmaceutical Review, American Society of Gene & Cell Therapy (ASGCT), European Society of Gene and Cell Therapy (ESGCT), RNA Society (US), European Medicines Agency (EMA), Alliance for Regenerative Medicine (ARM), Annual Reports, SEC Filings, Press Releases, Investor Presentations, Corporate Presentations, Journals, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 101 ASIA PACIFIC: RNA THERAPEUTICS MARKET, BY TYPE, 2021–2028 (USD MILLION)

Type	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
mRNA Therapeutics	4,185.0	9,821.6	3,642.1	3,700.9	3,869.5	3,987.6	4,083.7	4,110.6	2.4%
RNA Interference (RNAi) Therapeutics	311.9	748.9	284.0	295.0	315.2	333.9	349.2	358.8	4.8%
Antisense Oligonucleotide (ASO) Therapeutics	45.0	104.7	38.5	38.7	40.1	41.2	41.7	41.6	1.6%
Total	**4,541.9**	**10,675.2**	**3,964.6**	**4,034.6**	**4,224.8**	**4,362.7**	**4,474.6**	**4,511.0**	**2.6%**

Source: National Institutes of Health (NIH), US Food and Drug Administration (US FDA), World Health Organization (WHO), European Pharmaceutical Review, American Society of Gene & Cell Therapy (ASGCT), European Society of Gene and Cell Therapy (ESGCT), RNA Society (US), European Medicines Agency (EMA), Alliance for Regenerative Medicine (ARM), Annual Reports, SEC Filings, Press Releases, Investor Presentations, Corporate Presentations, Journals, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 102 ASIA PACIFIC: RNA THERAPEUTICS MARKET, BY INDICATION, 2021–2028 (USD MILLION)

Indication	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Infectious Diseases	4,203.2	9,836.1	3,637.0	3,685.0	3,841.6	3,971.8	4,056.2	4,071.5	2.3%
Rare Genetic/ Hereditary Diseases	272.8	666.3	256.8	270.9	293.7	315.4	334.3	347.8	6.2%
Other Indications	65.9	172.7	70.8	78.8	89.5	100.3	110.4	118.9	10.9%
Total	**4,542.0**	**10,675.2**	**3,964.6**	**4,034.6**	**4,224.8**	**4,387.5**	**4,500.9**	**4,538.1**	**2.7%**

Source: National Institutes of Health (NIH), US Food and Drug Administration (US FDA), World Health Organization (WHO), European Pharmaceutical Review, American Society of Gene & Cell Therapy (ASGCT), European Society of Gene and Cell Therapy (ESGCT), RNA Society (US), European Medicines Agency (EMA), Alliance for Regenerative Medicine (ARM), Annual Reports, SEC Filings, Press Releases, Investor Presentations, Corporate Presentations, Journals, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 103 ASIA PACIFIC: RNA THERAPEUTICS MARKET, BY END USER, 2021–2028 (USD MILLION)

End User	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Hospitals and Clinics	4,225.8	9,934.4	3,690.3	3,756.5	3,934.5	4,087.0	4,193.5	4,229.2	2.8%
Research Settings	316.1	740.7	274.2	278.2	290.4	300.6	307.3	308.9	2.4%
Total	**4,542.0**	**10,675.2**	**3,964.6**	**4,034.6**	**4,224.8**	**4,387.5**	**4,500.9**	**4,538.1**	**2.7%**

Source: World Health Organization (WHO), American Society for Clinical Laboratory Science (ASCLS), Centers for Disease Control and Prevention (CDC), National Center for Biotechnology Information (NCBI), American Association for Clinical Chemistry (AACC), American Society for Pharmacology and Experimental Therapeutics (ASPET), National Human Genome Research Institute (NHGRI), Annual Reports, SEC Filings, Investor Presentations, Company Websites, Press Releases, Interviews with Experts, and MarketsandMarkets Analysis

10.4.1 ASIA PACIFIC: RECESSION IMPACT

The Asia Pacific RNA therapeutics market has also been impacted by the recession. The Association of Southeast Asian Nations (ASEAN)-5 projected a steady decline in the economic growth of the Asia Pacific countries owing to a slower growth in their trading partners—Europe, the US, and China. The overall GDP decline in the coming year is expected to affect healthcare expenditure, spending, and wages in the industry, with reduced import and export activities.

However, in Asia, countries focus on increasing their R&D in the pharmaceutical and biotechnology sectors. South Korea has seen major progress in terms of its R&D funding and government support. According to a report by the South Korea Brand Equity Foundation (IBEF), the country's domestic pharmaceutical market was estimated at USD 42 billion in 2021, which is projected to reach USD 65 billion by 2024 and may further expand to reach USD 120–130 billion by 2030. The leading pharma companies in South Korea focus on increasing their R&D spend. For instance, Sun Pharma announced that the company intends to invest USD 600–650 million in R&D. On the other hand, Chinese pharmaceutical companies have invested nearly 10% of their sales revenue in R&D in 2021. Of all Hong Kong-listed pharma companies, 247 invested more than USD 14.99 million in R&D in 2021. China is considered to be a global manufacturing hub.

Japan's economy is expected to be more stable than the other Asian countries. However, price inflation can lead to lower consumption levels, resulting in a slight decline in Japan's economic growth in 2023.

Therefore, there will be a slight impact on the Asia Pacific pharmaceutical and biotechnology industry and associated markets, including RNA therapeutics, during the forecast period.

10.4.2 JAPAN

10.4.2.1 Increasing geriatric population and rising government initiatives for drug innovation to drive market

The growth of the RNA therapeutics market in Japan is driven by the rising demand for RNA-based therapeutics due to the increasing geriatric population in the country, growing government funding, initiatives taken by the government and private associations to improve and accelerate drug discovery processes, and the growing use of AI technology in drug discovery. According to the National Institute of Population and Social Security Research (Japan), by the end of 2035, approximately one in three people in the country will be aged 65 and over. With the rapid growth in the geriatric population, the prevalence of age-related illnesses is also on the rise, which will support the growth of the pharmaceutical industry and consequently create a higher demand for RNA therapeutics in Japan.

Over the years, Japan's RNA therapeutics market is expected to be driven by increasing government support. The Global Health Innovative Technology Fund (GHIT Fund), headquartered in Japan, is an international public-private partnership between the Japanese government, pharmaceutical and diagnostics companies, and not-for-profit organizations. It offers funds to various partnerships between Japanese and non-Japanese organizations, which include pharmaceutical companies, academic and research institutes, and product development partnerships (PDPs). In August 2023, the Japanese government granted USD 115 million in two separate grants to ARCALIS Co., Ltd., its manufacturing joint venture in Japan, to assist in the development of mRNA vaccines and therapies. The building of a factory and the acquisition of capital equipment to enable Good Manufacturing Practice (cGMP) manufacture of mRNA drug substance and mRNA drug product operations will be funded by the grants. Moreover, as of August 2022, Moderna will have a marketing license in Japan and be in charge of all import, local regulatory, development, quality assurance, and commercial activities for Spikevax (mRNA-1273).

TABLE 104 JAPAN: RNA THERAPEUTICS MARKET, BY PRODUCT, 2021–2028 (USD MILLION)

Product	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Vaccines	2,401.8	5,724.0	1,976.3	1,941.2	1,972.6	2,004.6	2,041.1	2,078.3	1.0%
Drugs	76.8	120.5	168.9	231.4	293.7	343.5	367.4	355.6	16.1%
Total	**2,478.6**	**5,844.5**	**2,145.3**	**2,172.5**	**2,266.4**	**2,348.1**	**2,408.5**	**2,433.9**	**2.6%**

Source: National Institutes of Health (NIH), US Food and Drug Administration (US FDA), World Health Organization (WHO), European Pharmaceutical Review, American Society of Gene & Cell Therapy (ASGCT), European Society of Gene and Cell Therapy (ESGCT), RNA Society (US), European Medicines Agency (EMA), Alliance for Regenerative Medicine (ARM), Annual Reports, SEC Filings, Press Releases, Investor Presentations, Corporate Presentations, Journals, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 105 JAPAN: RNA THERAPEUTICS MARKET, BY TYPE, 2021–2028 (USD MILLION)

Type	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
mRNA Therapeutics	2,284.4	5,379.1	1,971.6	1,993.9	2,077.0	2,135.8	2,187.2	2,206.8	2.3%
RNA Interference (RNAi) Therapeutics	170.0	409.1	153.2	158.1	168.1	177.5	185.4	190.8	4.5%
Antisense Oligonucleotide (ASO) Therapeutics	24.2	56.4	20.5	20.5	21.2	21.7	22.0	22.0	1.5%
Total	**2,478.6**	**5,844.6**	**2,145.3**	**2,172.5**	**2,266.3**	**2,335.0**	**2,394.6**	**2,419.6**	**2.4%**

Source: National Institutes of Health (NIH), US Food and Drug Administration (US FDA), World Health Organization (WHO), European Pharmaceutical Review, American Society of Gene & Cell Therapy (ASGCT), European Society of Gene and Cell Therapy (ESGCT), RNA Society (US), European Medicines Agency (EMA), Alliance for Regenerative Medicine (ARM), Annual Reports, SEC Filings, Press Releases, Investor Presentations, Corporate Presentations, Journals, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 106 JAPAN: RNA THERAPEUTICS MARKET, BY INDICATION, 2021–2028 (USD MILLION)

Indication	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Infectious Diseases	2,293.4	5,386.1	1,969.1	1,986.1	2,063.4	2,129.2	2,175.0	2,189.0	2.1%
Rare Genetic/Hereditary Diseases	149.1	364.3	138.4	145.0	156.2	167.0	176.6	183.8	5.8%
Other Indications	36.2	94.1	37.8	41.5	46.7	51.9	56.8	61.1	10.1%
Total	**2,478.6**	**5,844.5**	**2,145.3**	**2,172.5**	**2,266.4**	**2,348.1**	**2,408.5**	**2,433.9**	**2.6%**

Source: National Institutes of Health (NIH), US Food and Drug Administration (US FDA), World Health Organization (WHO), European Pharmaceutical Review, American Society of Gene & Cell Therapy (ASGCT), European Society of Gene and Cell Therapy (ESGCT), RNA Society (US), European Medicines Agency (EMA), Alliance for Regenerative Medicine (ARM), Annual Reports, SEC Filings, Press Releases, Investor Presentations, Corporate Presentations, Journals, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 107 JAPAN: RNA THERAPEUTICS MARKET, BY END USER, 2021–2028 (USD MILLION)

End User	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Hospitals and Clinics	2,306.1	5,438.1	1,996.2	2,021.7	2,109.1	2,185.3	2,241.6	2,265.4	2.6%
Research Settings	172.5	406.5	149.1	150.9	157.3	162.8	166.9	168.6	2.5%
Total	**2,478.6**	**5,844.5**	**2,145.3**	**2,172.5**	**2,266.4**	**2,348.1**	**2,408.5**	**2,433.9**	**2.6%**

Source: World Health Organization (WHO), American Society for Clinical Laboratory Science (ASCLS), Centers for Disease Control and Prevention (CDC), National Center for Biotechnology Information (NCBI), American Association for Clinical Chemistry (AACC), American Society for Pharmacology and Experimental Therapeutics (ASPET), National Human Genome Research Institute (NHGRI), Annual Reports, SEC Filings, Investor Presentations, Company Websites, Press Releases, Interviews with Experts, and MarketsandMarkets Analysis

10.4.3 CHINA

10.4.3.1 Low manufacturing costs and increased number of R&D activities to drive market

China is the world's fastest-growing pharmaceutical market. According to WHO ICTRP, the number of clinical trial registrations in China went from 7,010 in 2017 to 9,487 in 2021, at a CAGR of 7.9% between 2017 and 2021. Low manufacturing costs and the huge demand for medicines have attracted a large number of global players to this market.

Many global pharmaceutical companies have established their R&D centers in China. In response to increasing opportunities, major global pharmaceutical and biotechnology companies have conducted research activities to develop novel RNA therapeutics, which are as follows:

In January 2023, Orna Therapeutics and Shanghai Xianbo Biotech Co., Ltd. established a collaborative agreement to discover, develop, and commercialize oncology medications. By utilizing Simnova's development capabilities, this strategic partnership expands Orna's capacity to develop circular RNA therapeutics in China.

In July 2022, Sanofi licensed its RNA therapeutics technology to Rona Therapeutics, a biotechnology start-up headquartered in Shanghai. The deal also includes several preclinical candidates, and Rona will also obtain the rights to the small interfering RNA (siRNA) research platform of Sanofi under the terms of the agreement.

Additionally, In October 2021, Hansoh Pharmaceutical Group Co., Ltd. and OliX Pharmaceuticals, Inc. announced a licensing and collaboration agreement to discover, develop, and commercialize siRNA therapeutics in key targeted indications in Greater China, which includes mainland China, Taiwan, Hong Kong, and Macau. Under the terms of the collaboration agreement, the companies can use Hansoh's robust R&D, manufacturing, and commercialization capabilities in Greater China to address a variety of liver-based targets implicated in cardiovascular, metabolic, and other indications. They will also leverage OliX Pharmaceuticals' GalNAc-asiRNA platform.

China offers huge opportunities for drug developers to optimize the development and registration timelines for their products. The presence of a large treatment-naïve population for clinical trials, a favorable regulatory landscape, and the low cost of clinical trials are the major factors driving the clinical trials market in China.

TABLE 108 CHINA: RNA THERAPEUTICS MARKET, BY PRODUCT, 2021–2028 (USD MILLION)

Product	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Vaccines	978.6	2,331.2	804.5	789.9	802.3	814.9	829.4	844.1	1.0%
Drugs	38.2	60.0	84.1	115.1	146.1	170.9	182.7	176.7	16.0%
Total	1,016.9	2,391.2	888.6	905.0	948.4	985.8	1,012.1	1,020.8	2.8%

Source: National Institutes of Health (NIH), US Food and Drug Administration (US FDA), World Health Organization (WHO), European Pharmaceutical Review, American Society of Gene & Cell Therapy (ASGCT), European Society of Gene and Cell Therapy (ESGCT), RNA Society (US), European Medicines Agency (EMA), Alliance for Regenerative Medicine (ARM), Annual Reports, SEC Filings, Press Releases, Investor Presentations, Corporate Presentations, Journals, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 109 CHINA: RNA THERAPEUTICS MARKET, BY TYPE, 2021–2028 (USD MILLION)

Type	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
mRNA Therapeutics	936.8	2,199.8	816.3	830.2	868.8	896.1	918.6	925.0	2.5%
RNA Interference (RNAi) Therapeutics	69.9	167.8	63.7	66.2	70.8	75.1	78.6	80.8	4.9%
Antisense Oligonucleotide (ASO) Therapeutics	10.2	23.5	8.6	8.5	8.8	8.9	9.0	8.8	0.6%
Total	**1,016.9**	**2,391.1**	**888.6**	**904.9**	**948.4**	**980.1**	**1,006.2**	**1,014.6**	**2.7%**

Source: National Institutes of Health (NIH), US Food and Drug Administration (US FDA), World Health Organization (WHO), European Pharmaceutical Review, American Society of Gene & Cell Therapy (ASGCT), European Society of Gene and Cell Therapy (ESGCT), RNA Society (US), European Medicines Agency (EMA), Alliance for Regenerative Medicine (ARM), Annual Reports, SEC Filings, Press Releases, Investor Presentations, Corporate Presentations, Journals, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 110 CHINA: RNA THERAPEUTICS MARKET, BY INDICATION, 2021–2028 (USD MILLION)

Indication	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Infectious Diseases	941.1	2,203.2	815.1	826.4	862.2	892.1	911.7	915.4	2.3%
Rare Genetic/Hereditary Diseases	61.0	149.3	57.6	60.8	66.0	71.0	75.3	78.4	6.4%
Other Indications	14.7	38.7	15.9	17.7	20.2	22.7	25.0	27.0	11.1%
Total	**1,016.9**	**2,391.2**	**888.6**	**905.0**	**948.4**	**985.8**	**1,012.1**	**1,020.8**	**2.8%**

Source: National Institutes of Health (NIH), US Food and Drug Administration (US FDA), World Health Organization (WHO), European Pharmaceutical Review, American Society of Gene & Cell Therapy (ASGCT), European Society of Gene and Cell Therapy (ESGCT), RNA Society (US), European Medicines Agency (EMA), Alliance for Regenerative Medicine (ARM), Annual Reports, SEC Filings, Press Releases, Investor Presentations, Corporate Presentations, Journals, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 111 CHINA: RNA THERAPEUTICS MARKET, BY END USER, 2021–2028 (USD MILLION)

End User	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Hospitals and Clinics	946.1	2,225.4	827.2	842.7	883.4	918.4	943.1	951.6	2.8%
Research Settings	70.8	165.8	61.4	62.3	65.1	67.4	68.9	69.3	2.4%
Total	**1,016.9**	**2,391.2**	**888.6**	**905.0**	**948.4**	**985.8**	**1,012.1**	**1,020.8**	**2.8%**

Source: World Health Organization (WHO), American Society for Clinical Laboratory Science (ASCLS), Centers for Disease Control and Prevention (CDC), National Center for Biotechnology Information (NCBI), American Association for Clinical Chemistry (AACC), American Society for Pharmacology and Experimental Therapeutics (ASPET), National Human Genome Research Institute (NHGRI), Annual Reports, SEC Filings, Investor Presentations, Company Websites, Press Releases, Interviews with Experts, and MarketsandMarkets Analysis

10.4.4 SOUTH KOREA

10.4.4.1 Increasing number of government initiatives and growing R&D activities for RNA therapeutics to drive market

South Korea is fast emerging as a major destination for clinical trials. The Korean Ministry of Food and Drug Safety (MFDS) actively encourages global clinical trial participation and promotes international clinical trials. The introduction of the Clinical Trial Authorization process in 2002 also drove the number of multinational trials conducted in the country. Seoul is now one of the world's largest clinical research centers in terms of the number of trials conducted. The Korea Food & Drug Administration (KFDA) has streamlined regulatory approvals, approving trials within 30 days. According to WHO ICTRP, clinical trial registrations in South Korea went from 1,799 in 2017 to 1,940 in 2021, at a CAGR of 1.9%.

MFDS authorized conditional marketing in 2021 for Moderna's COVID-19 vaccine in South Korea. The National New Drug Development Research Project will be divided into four parts—new drug base expansion research, new drug R&D ecosystem establishment, new drug clinical development, and new drug R&D commercialization support, which will further provide opportunities for a broad category of pharmaceutical companies and organizations to explore the RNA therapeutics space and boost the market in the country. In March 2022, RNAGene Inc., a South Korean biotech company that develops mRNA-based therapeutics, announced that it had signed a collaborative research agreement with PharmAbcine Inc. Under the agreement, these companies will collaborate to develop next-generation mRNA-based antibody therapeutics by utilizing the proprietary mRNA platform from RNAGene.

TABLE 112 SOUTH KOREA: RNA THERAPEUTICS MARKET, BY PRODUCT, 2021–2028 (USD MILLION)

Product	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Vaccines	448.0	1,061.5	364.4	355.8	359.5	363.2	367.6	372.1	0.4%
Drugs	17.4	27.3	38.3	52.4	66.5	77.7	83.1	79.9	15.9%
Total	**465.43**	**1,088.87**	**402.67**	**408.21**	**425.96**	**440.87**	**450.66**	**452.01**	**2.3%**

Source: National Institutes of Health (NIH), US Food and Drug Administration (US FDA), World Health Organization (WHO), European Pharmaceutical Review, American Society of Gene & Cell Therapy (ASGCT), European Society of Gene and Cell Therapy (ESGCT), RNA Society (US), European Medicines Agency (EMA), Alliance for Regenerative Medicine (ARM), Annual Reports, SEC Filings, Press Releases, Investor Presentations, Corporate Presentations, Journals, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 113 SOUTH KOREA: RNA THERAPEUTICS MARKET, BY TYPE, 2021–2028 (USD MILLION)

Type	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
mRNA Therapeutics	428.7	1,001.3	369.7	374.2	389.8	400.2	408.3	408.8	2.0%
RNA Interference (RNAi) Therapeutics	32.0	76.6	29.0	30.0	32.0	33.9	35.3	36.2	4.5%
Antisense Oligonucleotide (ASO) Therapeutics	4.7	10.9	4.0	4.0	4.2	4.3	4.3	4.3	1.3%
Total	**465.4**	**1,088.8**	**402.7**	**408.2**	**426.0**	**438.4**	**447.9**	**449.3**	**2.2%**

Source: National Institutes of Health (NIH), US Food and Drug Administration (US FDA), World Health Organization (WHO), European Pharmaceutical Review, American Society of Gene & Cell Therapy (ASGCT), European Society of Gene and Cell Therapy (ESGCT), RNA Society (US), European Medicines Agency (EMA), Alliance for Regenerative Medicine (ARM), Annual Reports, SEC Filings, Press Releases, Investor Presentations, Corporate Presentations, Journals, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 114 SOUTH KOREA: RNA THERAPEUTICS MARKET, BY INDICATION, 2021–2028 (USD MILLION)

Indication	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Infectious Diseases	430.8	1,003.0	369.1	372.4	386.6	398.2	405.0	404.2	1.8%
Rare Genetic/Hereditary Diseases	27.9	68.1	26.2	27.7	30.0	32.2	34.0	35.3	6.1%
Other Indications	6.7	17.7	7.3	8.2	9.4	10.5	11.6	12.5	11.3%
Total	**465.43**	**1,088.87**	**402.67**	**408.21**	**425.96**	**440.87**	**450.66**	**452.01**	**2.3%**

Source: National Institutes of Health (NIH), US Food and Drug Administration (US FDA), World Health Organization (WHO), European Pharmaceutical Review, American Society of Gene & Cell Therapy (ASGCT), European Society of Gene and Cell Therapy (ESGCT), RNA Society (US), European Medicines Agency (EMA), Alliance for Regenerative Medicine (ARM), Annual Reports, SEC Filings, Press Releases, Investor Presentations, Corporate Presentations, Journals, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 115 SOUTH KOREA: RNA THERAPEUTICS MARKET, BY END USER, 2021–2028 (USD MILLION)

End User	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Hospitals and Clinics	433.0	1,013.6	375.0	380.4	397.1	411.2	420.5	422.0	2.4%
Research Settings	32.4	75.3	27.7	27.9	28.9	29.7	30.1	30.0	1.7%
Total	**465.4**	**1,088.9**	**402.7**	**408.2**	**426.0**	**440.9**	**450.7**	**452.0**	**2.3%**

Source: World Health Organization (WHO), American Society for Clinical Laboratory Science (ASCLS), Centers for Disease Control and Prevention (CDC), National Center for Biotechnology Information (NCBI), American Association for Clinical Chemistry (AACC), American Society for Pharmacology and Experimental Therapeutics (ASPET), National Human Genome Research Institute (NHGRI), Annual Reports, SEC Filings, Investor Presentations, Company Websites, Press Releases, Interviews with Experts, and MarketsandMarkets Analysis

10.4.5 REST OF ASIA PACIFIC

The Rest of Asia Pacific region includes India, Australia, Taiwan, New Zealand, Singapore, Malaysia, Thailand, Indonesia, Vietnam, and the Philippines. Singapore is an attractive market for players in the global pharmaceutical industry.

Australia is one of the most favorable locations for clinical trials due to the presence of many research institutes and the country's quality clinical infrastructure, diverse population, high volunteer rate, and fast approval process. Every year, pharmaceutical and medical device companies conduct over 1,000 new clinical trials in Australia. According to the Australian Government's Department of Health budget for 2022–23, the government aims to invest USD 6.8 billion in medical research over the next four years.

The government of New Zealand is also taking steps to advance and compete in the RNA therapeutics market globally. In March 2023, the government announced an investment of USD70 million in a new research project to establish New Zealand as a global leader in the technology used by Pfizer's COVID-19 vaccine. The RNA Development Platform, which Victoria and Auckland universities will co-host, is the main goal of the funding. The platform helps advances in manufacturing, such as developing novel RNA vaccine delivery systems or enabling smaller labs to generate RNA treatments more effectively.

Malaysia also has the advantage of a strong biomedical research base for drug discovery innovations. The Malaysian government has invested in science, research, and technology to increase the knowledge economy, encourage innovation, and support the country's best biomedical and drug development research activities.

TABLE 116 REST OF ASIA PACIFIC: RNA THERAPEUTICS MARKET, BY PRODUCT, 2021–2028 (USD MILLION)

Product	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Vaccines	542.4	1,290.5	444.8	436.1	442.5	448.9	456.3	463.8	0.8%
Drugs	38.6	60.0	83.2	112.8	141.7	163.9	173.4	167.6	15.0%
Total	**581.1**	**1,350.5**	**528.0**	**548.9**	**584.1**	**612.8**	**629.7**	**631.4**	**3.6%**

Source: National Institutes of Health (NIH), US Food and Drug Administration (US FDA), World Health Organization (WHO), European Pharmaceutical Review, American Society of Gene & Cell Therapy (ASGCT), European Society of Gene and Cell Therapy (ESGCT), RNA Society (US), European Medicines Agency (EMA), Alliance for Regenerative Medicine (ARM), Annual Reports, SEC Filings, Press Releases, Investor Presentations, Corporate Presentations, Journals, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 117 REST OF ASIA PACIFIC: RNA THERAPEUTICS MARKET, BY TYPE, 2021–2028 (USD MILLION)

Type	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
mRNA Therapeutics	535.1	1,241.3	484.4	502.7	533.9	555.4	569.6	570.0	3.3%
RNA Interference (RNAi) Therapeutics	40.0	95.3	38.2	40.6	44.2	47.4	49.8	51.0	6.0%
Antisense Oligonucleotide (ASO) Therapeutics	6.0	13.8	5.4	5.6	6.0	6.3	6.5	6.5	3.6%
Total	**581.1**	**1,350.4**	**528.0**	**548.9**	**584.1**	**609.1**	**625.9**	**627.5**	**3.5%**

Source: National Institutes of Health (NIH), US Food and Drug Administration (US FDA), World Health Organization (WHO), European Pharmaceutical Review, American Society of Gene & Cell Therapy (ASGCT), European Society of Gene and Cell Therapy (ESGCT), RNA Society (US), European Medicines Agency (EMA), Alliance for Regenerative Medicine (ARM), Annual Reports, SEC Filings, Press Releases, Investor Presentations, Corporate Presentations, Journals, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 118 REST OF ASIA PACIFIC: RNA THERAPEUTICS MARKET, BY INDICATION, 2021–2028 (USD MILLION)

Indication	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Infectious Diseases	538.0	1,243.8	483.7	500.2	529.4	552.3	564.5	562.9	3.1%
Rare Genetic/Hereditary Diseases	34.8	84.6	34.6	37.5	41.5	45.2	48.3	50.2	7.7%
Other Indications	8.3	22.2	9.8	11.3	13.3	15.2	16.9	18.3	13.4%
Total	**581.1**	**1,350.5**	**528.0**	**548.9**	**584.1**	**612.8**	**629.7**	**631.4**	**3.6%**

Source: National Institutes of Health (NIH), US Food and Drug Administration (US FDA), World Health Organization (WHO), European Pharmaceutical Review, American Society of Gene & Cell Therapy (ASGCT), European Society of Gene and Cell Therapy (ESGCT), RNA Society (US), European Medicines Agency (EMA), Alliance for Regenerative Medicine (ARM), Annual Reports, SEC Filings, Press Releases, Investor Presentations, Corporate Presentations, Journals, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 119 REST OF ASIA PACIFIC: RNA THERAPEUTICS MARKET, BY END USER, 2021–2028 (USD MILLION)

End User	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Hospitals and Clinics	540.6	1,257.4	492.0	511.8	545.0	572.1	588.3	590.3	3.7%
Research Settings	40.4	93.1	36.1	37.1	39.1	40.7	41.4	41.1	2.6%
Total	**581.1**	**1,350.5**	**528.0**	**548.9**	**584.1**	**612.8**	**629.7**	**631.4**	**3.6%**

Source: World Health Organization (WHO), American Society for Clinical Laboratory Science (ASCLS), Centers for Disease Control and Prevention (CDC), National Center for Biotechnology Information (NCBI), American Association for Clinical Chemistry (AACC), American Society for Pharmacology and Experimental Therapeutics (ASPET), National Human Genome Research Institute (NHGRI), Annual Reports, SEC Filings, Investor Presentations, Company Websites, Press Releases, Interviews with Experts, and MarketsandMarkets Analysis

10.5 REST OF THE WORLD

The rest of the World includes Latin America and the Middle East & Africa and the countries within these regions (Brazil, Saudi Arabia, Argentina, and others). The growing focus on applying RNA technologies in COVID-19 vaccine development across Brazil and Argentina has contributed to regional revenue in 2022. In September 2022, the Pan American Health Organization (PAHO) designated two biomedical facilities in Argentina and Brazil as regional focal points for the advancement and manufacture of mRNA-based COVID-19 vaccines in Latin America. The concept revolved around utilizing the current manufacturing capabilities to facilitate the transfer of vaccine technology, originally created by Moderna, Inc (US) in the US, to a region that has been severely affected by the COVID-19 pandemic and continues to face challenges in obtaining an adequate vaccine supply.

The Bio-Manguinhos Institute of Immunobiological Technology at Fiocruz, Brazil's leading biomedical laboratory, was chosen due to its established track record in vaccine production. This institute has already developed mRNA vaccine technology, as stated by PAHO. In Argentina, Sinergium Biotech, a biopharmaceutical company operating in the private sector, was designated as the chosen center to collaborate with the pharmaceutical laboratory mAbxience, both entities being part of the same group. Their partnership aims to develop and manufacture the essential components for active vaccines.

TABLE 120 REST OF THE WORLD: RNA THERAPEUTICS MARKET, BY PRODUCT, 2021–2028 (USD MILLION)

Product	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Vaccines	628.5	1,490.0	511.8	500.1	505.5	511.0	517.6	524.3	0.5%
Drugs	18.5	29.8	42.7	59.9	77.9	93.1	101.8	100.8	18.7%
Total	**647.0**	**1,519.8**	**554.5**	**560.0**	**583.4**	**604.2**	**619.4**	**625.1**	**2.4%**

Source: National Institutes of Health (NIH), US Food and Drug Administration (US FDA), World Health Organization (WHO), European Pharmaceutical Review, American Society of Gene & Cell Therapy (ASGCT), European Society of Gene and Cell Therapy (ESGCT), RNA Society (US), European Medicines Agency (EMA), Alliance for Regenerative Medicine (ARM), Annual Reports, SEC Filings, Press Releases, Investor Presentations, Corporate Presentations, Journals, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 121 REST OF THE WORLD: RNA THERAPEUTICS MARKET, BY TYPE, 2021–2028 (USD MILLION)

Type	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
mRNA Therapeutics	595.6	1,396.3	508.4	512.4	532.7	546.9	559.4	563.3	2.1%
RNA Interference (RNAi) Therapeutics	44.6	107.6	40.3	41.7	44.4	47.1	49.4	51.0	4.0%
Antisense Oligonucleotide (ASO) Therapeutics	6.7	15.9	5.8	6.0	6.3	6.6	6.8	6.9	2.7%
Total	**646.9**	**1,519.8**	**554.5**	**560.1**	**583.4**	**600.6**	**615.6**	**621.2**	**2.3%**

Source: National Institutes of Health (NIH), US Food and Drug Administration (US FDA), World Health Organization (WHO), European Pharmaceutical Review, American Society of Gene & Cell Therapy (ASGCT), European Society of Gene and Cell Therapy (ESGCT), RNA Society (US), European Medicines Agency (EMA), Alliance for Regenerative Medicine (ARM), Annual Reports, SEC Filings, Press Releases, Investor Presentations, Corporate Presentations, Journals, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 122 REST OF THE WORLD: RNA THERAPEUTICS MARKET, BY INDICATION, 2021–2028 (USD MILLION)

Indication	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Infectious Diseases	599.1	1,399.4	507.7	509.7	527.9	543.5	553.9	555.7	1.8%
Rare Genetic/Hereditary Diseases	38.6	95.3	36.5	38.5	41.8	45.2	48.2	50.5	5.7%
Other Indications	9.2	25.1	10.4	11.8	13.7	15.5	17.3	18.9	10.4%
Total	**647.0**	**1,519.8**	**554.5**	**560.0**	**583.4**	**604.2**	**619.4**	**625.1**	**2.4%**

Source: National Institutes of Health (NIH), US Food and Drug Administration (US FDA), World Health Organization (WHO), European Pharmaceutical Review, American Society of Gene & Cell Therapy (ASGCT), European Society of Gene and Cell Therapy (ESGCT), RNA Society (US), European Medicines Agency (EMA), Alliance for Regenerative Medicine (ARM), Annual Reports, SEC Filings, Press Releases, Investor Presentations, Corporate Presentations, Journals, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 123 REST OF THE WORLD: RNA THERAPEUTICS MARKET, BY END USER, 2021–2028 (USD MILLION)

End User	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Hospitals and Clinics	601.9	1,415.3	516.9	522.5	544.8	564.7	579.5	585.3	2.5%
Research Settings	45.0	104.5	37.7	37.5	38.6	39.5	40.0	39.8	1.1%
Total	**647.0**	**1,519.8**	**554.5**	**560.0**	**583.4**	**604.2**	**619.4**	**625.1**	**2.4%**

Source: World Health Organization (WHO), American Society for Clinical Laboratory Science (ASCLS), Centers for Disease Control and Prevention (CDC), National Center for Biotechnology Information (NCBI), American Association for Clinical Chemistry (AACC), American Society for Pharmacology and Experimental Therapeutics (ASPET), National Human Genome Research Institute (NHGRI), Annual Reports, SEC Filings, Investor Presentations, Company Websites, Press Releases, Interviews with Experts, and MarketsandMarkets Analysis

11 COMPETITIVE LANDSCAPE

11.1 OVERVIEW

The competitive landscape includes the analysis of the key growth strategies adopted by major players between January 2021 and August 2023 to expand their presence and increase their share in the global RNA therapeutics market. Players in this market have employed various strategies to expand their footprints and increase their market shares. The key strategies adopted by the major players in the RNA therapeutics market include product launches; business acquisitions; agreements, partnerships, and collaborations; and expansions.

Prominent players in the RNA therapeutics market are Moderna, Inc. (US), Alnylam Pharmaceuticals, Inc. (US), Novartis AG (Switzerland), Ionis Pharmaceuticals, Inc. (US), Sarepta Therapeutics, Inc. (US), Sanofi (France), Pfizer Inc. (US), Arrowhead Pharmaceuticals, Inc. (US), BioNtech SE (Germany), Orna Therapeutics (US), CRISPR Therapeutics (Switzerland), Silence Therapeutics (UK), Astellas Pharma Inc. (Japan), CureVac SE (Germany), Sirnaomics (US), Arcturus Therapeutics, Inc. (US) and Arbutus Biopharma (US).

11.2 KEY STRATEGIES ADOPTED BY MAJOR PLAYERS IN RNA THERAPEUTICS MARKET

This section includes an in-depth analysis of the major developments undertaken by the top players in this market during the study period. These developments are curated based on their overall impact on the market.

FIGURE 25 KEY PLAYER STRATEGIES IN RNA THERAPEUTICS MARKET, 2021–2023

COMPANY NAME	ORGANIC		INORGANIC	
	PRODUCT APPROVALS	REGULATORY APPROVALS	ACQUISITIONS	COLLABORATIONS AND PARTNERSHIPS
MODERNA, INC. (US)		Moderna, Inc. received authorization from Health Canada for its COVID-19 booster vaccine, mRNA-1273.214 (SpikevaxBivalent Original/Omicron). This vaccine is designed for immunization against COVID-19 in children and adolescents who are between 6 to 17 years.		Moderna, Inc., partnered with a nonprofit scientific research organization– IAVI– to apply mRNA technology for the development of RNA vaccines/therapeutics.
ALNYLAM PHARMACEUTICALS, INC. (US)		Alnylam Pharmaceuticals, Inc. received marketing authorization from the European Commission (EC) for its RNAi therapy- AMVUTTRA.		Alnylam Pharmaceuticals, Inc. entered a partnership with F. Hoffmann-La Roche Ltd. to co-develop and co-commercialize Zilebesiran. This is an Investigational RNAi medicine to treat hypertension in patients with a high risk of CVD.
NOVARTIS AG (SWITZERLAND)	Novartis AG's Leqvio RNA therapy received FDA approval expanded use as an adjunct to diet and statin therapy. The use of this therapy is approved in patients with elevated LDL-C.		Novartis AG announced the acquisition of DTx Pharma, this acquisition is targeted toward strengthening the neuroscience pipeline and RNA therapeutic capabilities of Novartis	

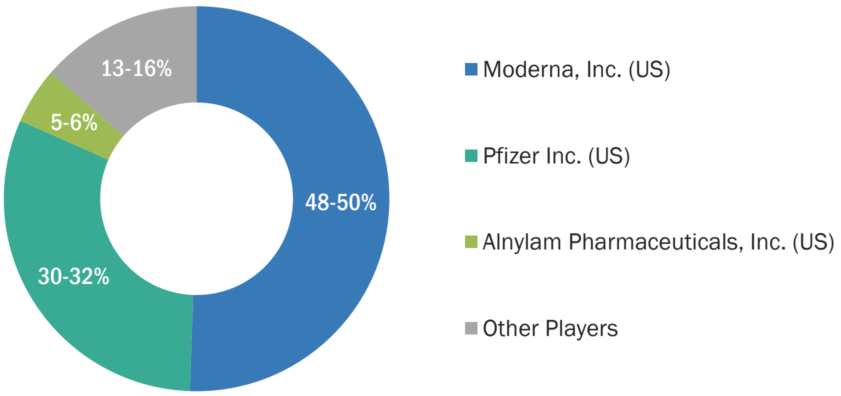
Note: Indicates events that will have a significant impact on market growth.

Source: Press Releases, Interviews with Experts, and MarketsandMarkets Analysis

11.3 MARKET SHARE ANALYSIS

The global RNA therapeutics market is a consolidated market, with Moderna, Inc. (US), Alnylam Pharmaceuticals, Inc. (US), and Pfizer Inc. (US) together accounting for ~80–85% of the global market. Novartis AG (Switzerland), Ionis Pharmaceuticals, Inc. (US), Sarepta Therapeutics, Inc. (US), and Sanofi (France) accounted for the remaining ~20–15% of the total market. Market share analysis is performed for the year 2022 and includes share analysis only for the companies that have at least one RNA therapy product (vaccines or drugs) that are approved in 2022. The remaining players have been analyzed to map competition in the RNA therapeutics market; these players include Arrowhead Pharmaceuticals, Inc. (US), BioNtech SE (Germany), Orna Therapeutics (US), CRISPR Therapeutics (Switzerland), Silence Therapeutics (UK), Astellas Pharma Inc. (Japan), CureVac SE (Germany), Sirnaomics (US), Arcturus Therapeutics Inc. (US) and Arbutus Biopharma (US).

FIGURE 26 MARKET SHARE ANALYSIS OF TOP 3 PLAYERS IN RNA THERAPEUTICS MARKET, 2022



Note: Other players include Novartis AG (Switzerland), Ionis Pharmaceuticals, Inc. (US), Sarepta Therapeutics, Inc. (US), and Sanofi (France).

Source: Annual Reports, Investor Presentations, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 124 DEGREE OF COMPETITION: RNA THERAPEUTICS MARKET

DEGREE OF COMPETITION	CONSOLIDATED
Total Market Share of the Top 3 Players	85–87%
Other Players	~13%

Note: Market shares of the above companies are calculated based on relevant segmental revenues and product sales showcased in the public domain generated by the companies.

THE DEGREE OF COMPETITION IS DEFINED AS BELOW:

Fragmented: When the top 5 players have a total market share of <25%

Competitive: When the top 5 players have a total of 25–50% market share

Consolidated: When the top 5 players have a total market share of >50%

Source: Annual Reports, Press Releases, Interviews with Experts, and MarketsandMarkets

MODERNA, INC. (US)

Moderna, Inc. is one of the leading players in the RNA therapeutics market that has gained a prominent position in the vaccine space. The company's COVID-19 booster vaccine—Spikevax—is well received globally and has shown considerable growth in sales, after its approval in 2020. Additionally, it has built a strong pipeline for other mRNA-based vaccines, with a majority in the late clinical stage. Approval and EUA for its BA.4/BA.5 Omicron-targeting bivalent COVID-19 booster vaccine across various countries has also created a strong revenue stream for the company in 2022.

The company has focused on collaborations and strategic co-developments to strengthen its vaccine manufacturing capabilities. In March 2023, it collaborated with Generation Bio Co. to co-develop genetic medicines. This strategic collaboration combines Moderna's expertise in mRNA vaccines/therapy with Generation Bio's non-viral genetic medicine platform. In February 2023, the company also collaborated with Life Edit Therapeutics Inc., an ElevateBio company with expertise in next-generation gene editing technologies, to develop in vivo mRNA gene editing therapies.

ALNYLAM PHARMACEUTICALS, INC. (US)

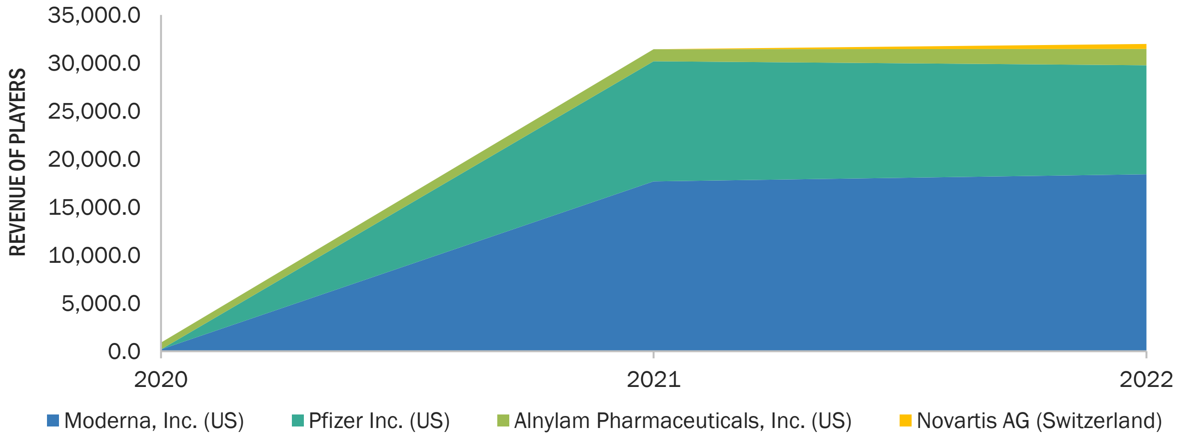
Alnylam Pharmaceuticals offers three established RNAi therapies, namely, ONPATTRO (patisiran), GIVLAARI (givosiran), and OXLUMO (lumasiran)—all these therapies have shown consistent growth in sales from 2019 to 2022, thereby, contributing to Alnylam's market position. The company has also recently added AMVUTTRA (vutrisiran) to its portfolio, which was approved in June 2022; the therapy recorded sales of approximately USD 90 million in the same year. The company has established itself as a prominent player in the RNAi therapeutics space, targeted toward ATTR treatment.

The company focuses on partnerships and collaborations to expand its reach across different geographies. It partners with biotechnology and pharmaceutical companies to strengthen their development and manufacturing capabilities. In July 2023, the company entered into a partnership with F. Hoffmann-La Roche Ltd. for the co-development and co-commercialization of Zilebesiran, which is an Investigational RNAi medicine to treat hypertension in patients with high cardiovascular risk.

11.4 REVENUE SHARE ANALYSIS OF TOP MARKET PLAYERS

The revenue analysis was calculated based on the performance of major players during the past few years and the current years in the RNA therapeutics market, which provides an estimation of the overall market size, structure, and recent technological advancements in products. Leading players such as Moderna, Inc. (US), Alnylam Pharmaceuticals, Inc. (US), and Pfizer Inc. (US) influence the RNA therapeutics market due to their vast product portfolios, recent acquisitions, and wide distribution and sales networks globally.

FIGURE 27 REVENUE SHARE ANALYSIS OF TOP 4 PLAYERS IN RNA THERAPEUTICS MARKET, 2020–2022



Notes: Segmental revenues for the top four players, along with their product sales, have been considered. The segmental share of the RNA therapeutics market may vary significantly from company to company, depending on individual company product portfolios.

Source: Press Releases and Company Websites

11.5 COMPANY EVALUATION MATRIX FOR KEY PLAYERS

The company evaluation matrix provides information on the top 17 players offering RNA therapeutics products and outlines findings on the performance of each vendor. The evaluation matrix is based on two broad categories—product offerings and business strategies. Each category carries various criteria based on which the companies have been evaluated. The evaluation criteria considered under product offerings include the breadth of offerings, products launched, the mode of delivery of products, and support. The evaluation criteria considered under business strategies include reach (based on market and geographic presence), industry coverage (based on end users catered to), viability, and inorganic growth strategies.

A comprehensive list of all vendors in this market was created through a product mapping strategy and MarketsandMarkets analysis. Based on their capabilities, innovations, breadth of product offerings, and business strategies, the vendors were shortlisted. The company matrix includes the top 17 companies that comprehensively cover the major market share. These companies are placed into four categories—Stars, Emerging Leaders, Pervasive Players, and Participants—based on their performance in each criterion.

11.5.1 STARS

Players that fall into this quadrant receive high scores for most of the evaluation criteria. They have a strong and established product portfolio, a large market share, wide application coverage, and a strong geographical footprint. These players provide mature, innovative, and reputable RNA therapeutic products. Moderna, Inc. (US), Alnylam Pharmaceuticals, Inc. (US), Novartis AG (Switzerland), Ionis Pharmaceuticals, Inc. (US), and Pfizer Inc. (US) are the key players that fall under this category.

11.5.2 EMERGING LEADERS

Emerging leaders are players who have demonstrated substantial product innovations compared to their competitors (resulting in an evolved product portfolio) and significant market presence. However, they do not have a very strong growth strategy for geographical penetration and overall business growth. Companies such as Sarepta Therapeutics, Inc. (US), Sanofi (France), and BioNtech SE (Germany) are recognized as emerging leaders in this market.

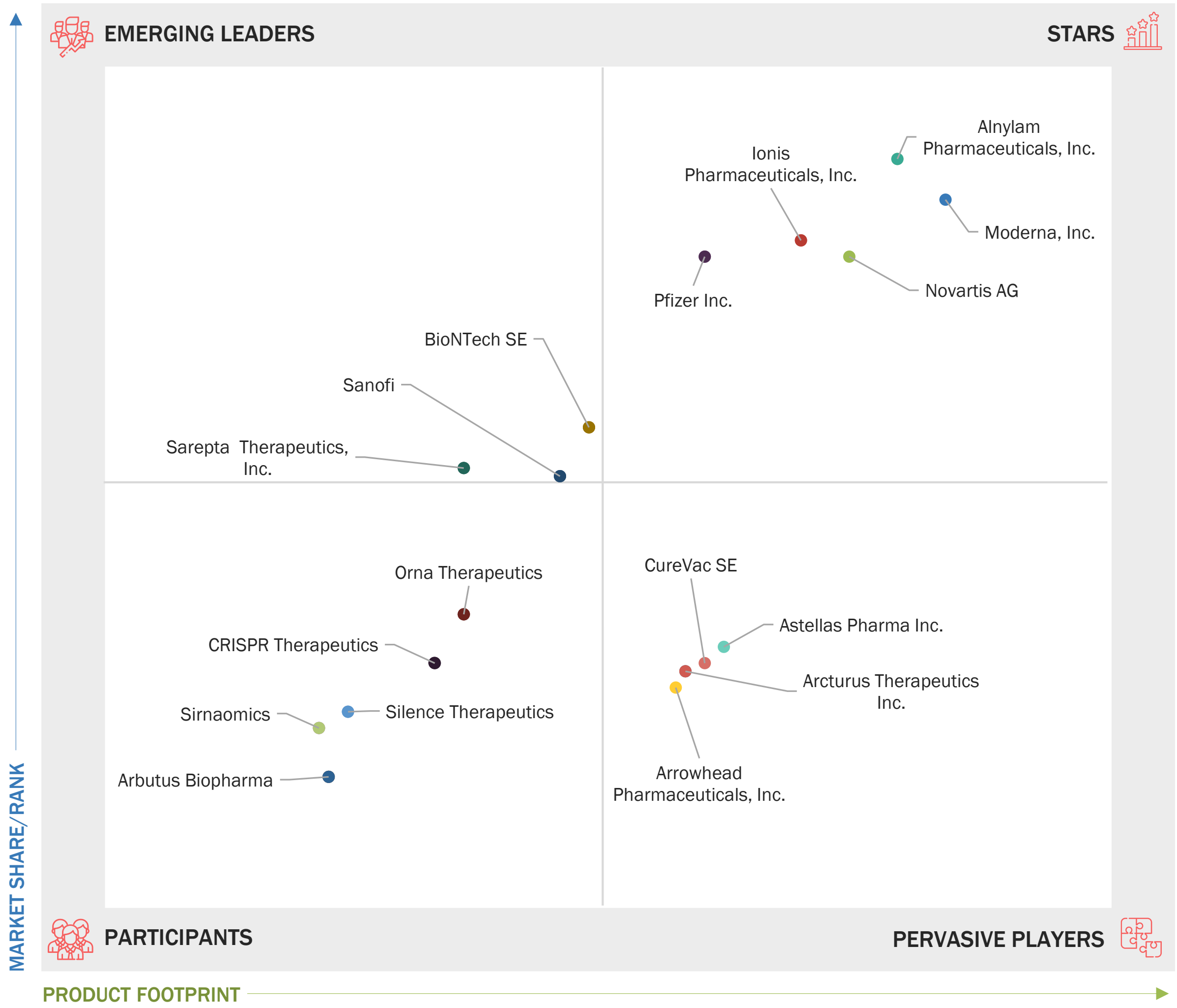
11.5.3 PERVASIVE PLAYERS

The companies falling in this quadrant are established players with very strong business strategies. However, they have a weaker product portfolio and account for lower market shares as compared to star players. Arrowhead Pharmaceuticals, Inc. (US), Astellas Pharma Inc. (Japan), CureVac SE (Germany), and Arcturus Therapeutics Inc. (US) are recognized as pervasive players in this market.

11.5.4 PARTICIPANTS

The companies that fall in this quadrant have niche product offerings, owing to which they are likely to gain a significant position in the market. They do not have strong business strategies compared to other established vendors. These companies could be new entrants in the market and may require time to gain traction in the RNA therapeutics market. Orna Therapeutics (US), CRISPR Therapeutics (Switzerland), Silence Therapeutics (UK), Sirnaomics (US), and Arbutus Biopharma (US) are the key players that fall under this category.

FIGURE 28 COMPANY EVALUATION MATRIX FOR KEY PLAYERS: RNA THERAPEUTICS MARKET, 2022



Source: MarketsandMarkets Analysis

11.6 COMPANY EVALUATION MATRIX FOR START-UPS/SMES

Vendors have been placed into four categories based on their performance in each criterion—Progressive Companies, Starting Blocks, Responsive Companies, and Dynamic Companies. The pipeline companies covered in this section include Arrowhead Pharmaceuticals, Inc. (US), BioNTech SE (Germany), Orna Therapeutics (US), CRISPR Therapeutics (Switzerland), Silence Therapeutics (UK), Astellas Pharma Inc. (Japan), CureVac SE (Germany), Sirnaomics (US), Arcturus Therapeutics Inc. (US) and Arbutus Biopharma (US).

11.6.1 PROGRESSIVE COMPANIES

Start-ups that fall under this category receive high scores for most evaluation criteria. They have an established product portfolio and a very strong market presence. Arrowhead Pharmaceuticals, Inc. (US), BioNTech SE (Germany), and Astellas Pharma Inc. (Japan) are the progressive companies in this market.

11.6.2 STARTING BLOCKS

Starting blocks are established vendors with a very strong business strategy but weaker product offerings compared to other start-ups. CRISPR Therapeutics (Switzerland) and Arbutus Biopharma (US) are the starting blocks.

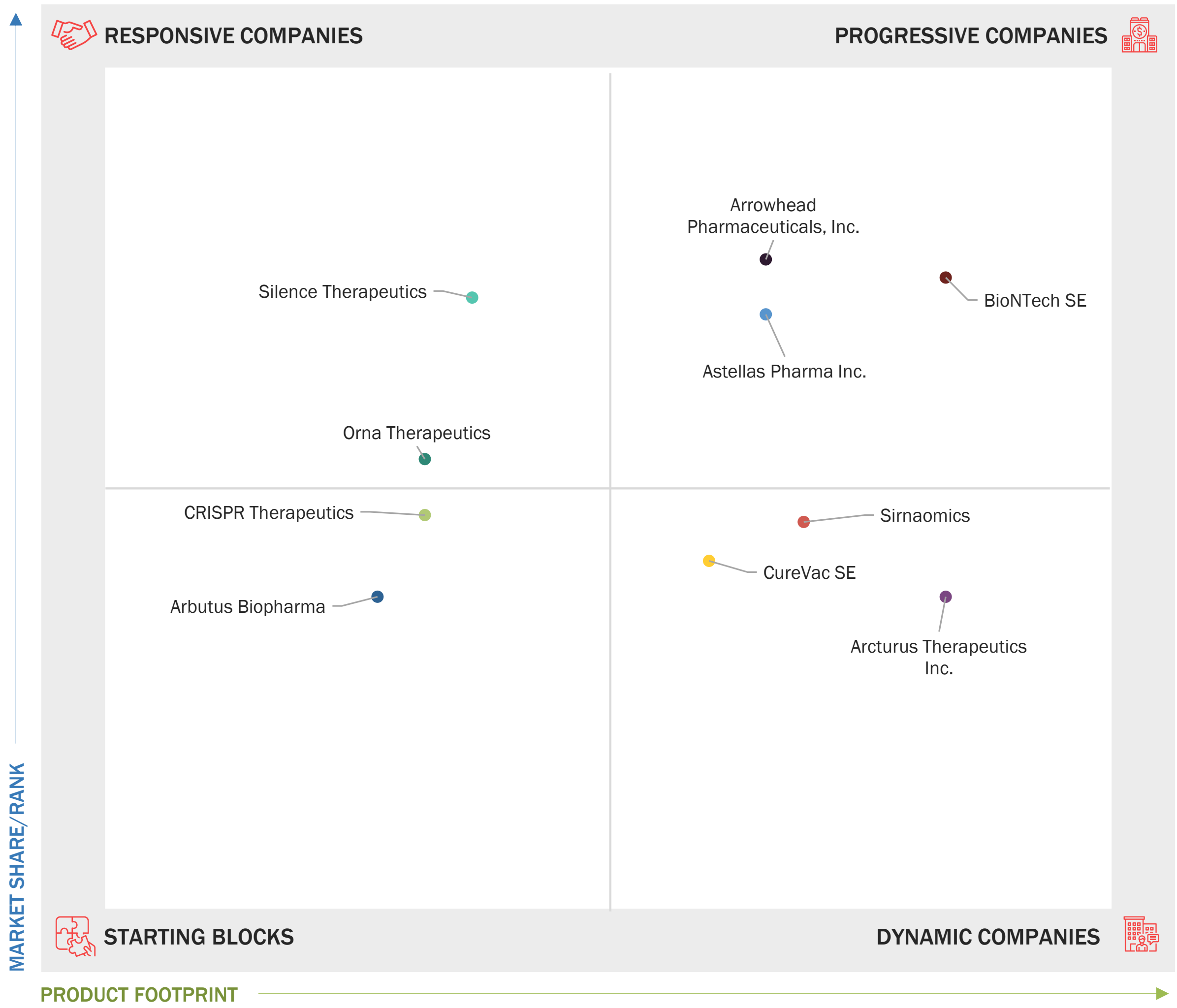
11.6.3 DYNAMIC COMPANIES

Dynamic companies are start-ups that have demonstrated significant product innovations compared to their competitors. CureVac SE (Germany), Sirnaomics (US), and Arcturus Therapeutics Inc. (US) are considered dynamic companies.

11.6.4 RESPONSIVE COMPANIES

Responsive companies are vendors with niche product offerings and are just starting to gain a position in the market. Orna Therapeutics (US) and Silence Therapeutics (UK) are responsive companies in this market.

FIGURE 29 COMPANY EVALUATION MATRIX FOR START-UPS/SMES: RNA THERAPEUTICS MARKET, 2022



11.7 COMPETITIVE BENCHMARKING

TABLE 125 DETAILED LIST OF KEY START-UPS/SMES: RNA THERAPEUTICS MARKET

COMPANY NAME	CATEGORY	OWNERSHIP STATUS	HQ LOCATION	YEAR FOUNDED	PRODUCT TYPE
Arrowhead Pharmaceuticals, Inc.	Drugs	Public	California, US	2004	RNAi Therapies
BioNTech SE	Vaccines	Public	Germany	2008	mRNA Vaccines
Orna Therapeutics	Drugs	Private	Massachusetts, US	2019	Circular RNA Therapies
CRISPR Therapeutics	Drugs	Public	Switzerland	2013	CRISPR)/Cas9-mediated RNA Therapies
Silence Therapeutics Plc.	Drugs	Private	UK	1999	siRNA Therapies
Astellas Pharma Inc.	Drugs	Public	Japan	1923	RNA Knockdown Therapy
CureVac SE	Vaccines	Public	Germany	2000	mRNA Vaccines
Sirnaomics	Drugs	Private	US	2007	siRNA Therapies
Arcturus Therapeutics Inc.	Vaccines	Public	US	2013	mRNA Vaccines
Arbutus Biopharma	Drugs	Public	US	2007	RNAi Therapies

11.8 COMPETITIVE SCENARIOS AND TRENDS

The global RNA therapeutics market is consolidated, with very few companies dominating the market. This section of the report studies the growth strategies adopted by market players between January 2021 and August 2023. Some of the key growth strategies adopted by market players include product launches; acquisitions, partnerships, collaborations; and expansions.

11.8.1 KEY PRODUCT APPROVALS

TABLE 126 KEY PRODUCT APPROVALS, JANUARY 2021–AUGUST 2023

MONTH AND YEAR	PRODUCT CATEGORY	DESCRIPTION
February 2023	mRNA vaccine	Moderna, Inc. received authorization from Health Canada for its COVID-19 booster vaccine, mRNA-1273.214 (SpikevaxBivalent Original/Omicron). This vaccine is designed for immunization against COVID-19 in children and adolescents between 6 to 17 years.
January 2023	mRNA vaccine	US FDA granted Breakthrough Therapy Designation for Moderna's investigational mRNA vaccine candidate—mRNA-1345. This vaccine was developed to prevent RSV-associated lower respiratory tract disease (RSV-LRTD) in adults aged 60 years or older.
September 2022	RNAi Therapy	Alnylam Pharmaceuticals, Inc. received marketing authorization from the European Commission for its RNAi therapy—AMVUTTRA. This treatment is developed for adult patients with stage 1 or stage 2 polyneuropathy, which is hereditary transthyretin-mediated (hATTR).

Source: Press Releases

11.8.2 KEY DEALS

TABLE 127 KEY DEALS, JANUARY 2021–AUGUST 2023

MONTH AND YEAR	DEAL TYPE	COMPANY 1	COMPANY 2	DESCRIPTION	DEAL SIZE
July 2023	Partnership	Alnylam Pharmaceuticals, Inc. (US)	F. Hoffmann-La Roche Ltd (Switzerland)	Alnylam Pharmaceuticals, Inc. entered into a partnership with F. Hoffmann-La Roche Ltd. to co-develop and co-commercialize Zilebesiran, which is an Investigational RNAi medicine to treat hypertension in patients with high cardiovascular risk.	USD 3.1 Billion
March 2023	Collaboration	Moderna, Inc. (US)	Generation Bio (US)	Moderna, Inc. collaborated with Generation Bio Co. to co-develop genetic medicines. This strategic collaboration combines Moderna's expertise in mRNA vaccines/therapy with Generation Bio's non-viral genetic medicine platform.	USD 40 million (To Generation Bio)

February 2023	Collaboration	Moderna, Inc. (US)	Life Edit Therapeutics (US)	Moderna, Inc. collaborated with Life Edit Therapeutics Inc., a company with expertise in next-generation gene editing technologies, to develop in vivo mRNA gene editing therapies.	NA
January 2022	Collaboration	Alnylam Pharmaceuticals, Inc. (US)	Novartis AG (Switzerland)	Alnylam Pharmaceuticals, Inc. collaborated with Novartis AG to discover and develop siRNA-based targeted therapy for patients with end-stage liver diseases.	NA

Source: Press Releases

11.8.3 OTHER KEY DEVELOPMENTS

TABLE 128 OTHER KEY DEVELOPMENTS, JANUARY 2021–AUGUST 2023

MONTH AND YEAR	DEAL TYPE			DESCRIPTION
March 2023	Business expansion			Moderna, Inc. entered into an agreement with the Government of the Republic of Kenya to establish a facility in the country for manufacturing mRNA vaccines/therapies. This is the company's first mRNA manufacturing facility in Africa.
October 2021	Agreement	Novartis AG (Switzerland)	BioNTech (Germany)	Novartis signed an initial agreement with BioNTech to leverage its vaccine manufacturing capabilities during the COVID-19 pandemic. Novartis utilized its sterile manufacturing facilities at its Novartis Technical Operations site in Slovenia to manufacture around 24 million doses in 2022.

Source: Press Releases

12 COMPANY PROFILES

12.1 KEY PLAYERS

12.1.1 MODERNA, INC.

12.1.1.1 Business overview

Moderna, Inc. develops and manufactures messenger RNA (mRNA)-based medicines. The company originally started off as a research-stage company to support the advent of novel mRNA vaccines; however, currently, through its established collaborations with commercial manufacturers and government authorities, Moderna has strengthened its mRNA vaccine manufacturing capabilities. The company secured the authorization and approval for mRNA vaccines against COVID-19 and distributed millions of doses of vaccines for this disease from 2021 to 2022. The company operates as a single business unit, generating its revenue from the COVID-19 vaccines based on mRNA technology.

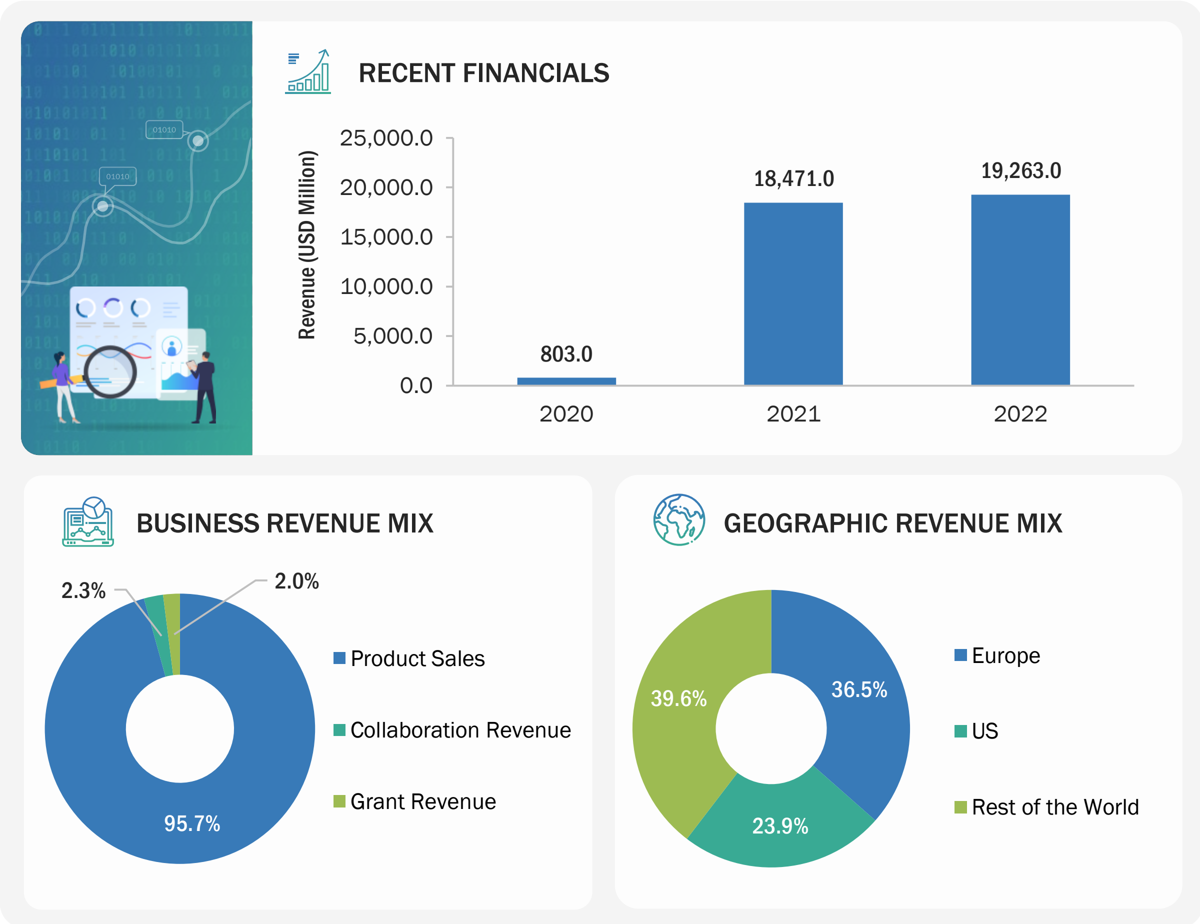
Moderna, Inc. has active commercial subsidiaries in nearly 16 countries across North America, Europe, and the Asia Pacific. Some of the subsidiaries include Brizo Ltd. (Bermuda), Moderna Australia Pty Ltd., Moderna Belgium SRL, Moderna Biopharma Canada Corporation, Moderna Biotech Ireland Limited, Moderna Biotech Manufacturing UK Ltd., Moderna Denmark ApS, Moderna Japan Co., Ltd., Moderna Malaysia Sdn. Bhd., and Moderna Switzerland GmbH.

TABLE 129 MODERNA, INC.: COMPANY OVERVIEW

Year Founded	2010
Headquarters Country	US
Headquarters State	Massachusetts
Employee Size	Approx. 3,900 (As of December 2022)
Annual Revenue	USD 19.2 Billion (As of 2022)
Ownership	Public

Source: Company Websites and Annual Reports

FIGURE 30 MODERNA, INC.: COMPANY SNAPSHOT (2022)



Note: Moderna, Inc. operates as a single business unit; hence, the business revenue mix is provided as per revenue sources, and the geographic revenue mix is reported for the product sales.

Source: Company Website, Annual Reports, and SEC Filings

12.1.1.2 Products offered

PRODUCT CATEGORY	APPLICATION	PRODUCT NAME	DESCRIPTION
mRNA vaccine	COVID-19	Spikevax	This mRNA vaccine is indicated for active immunization against SARS-CoV-2 for individuals 18 years and older.
mRNA vaccine	COVID-19	mRNA-1283	This mRNA vaccine is a Phase-3 candidate indicated for active immunization against SARS-CoV-2 for individuals 18 years and older.
mRNA vaccine	Flu Vaccine	mRNA-1010	This mRNA vaccine is a Phase-3 candidate indicated for active immunization against SARS-CoV-2 in individuals 18 years and older.
mRNA vaccine	Respiratory Syncytial Virus (RSV) Vaccine	mRNA-1345	This mRNA vaccine is a Phase-3 candidate indicated for active immunization against RSV in older adults.

mRNA therapy	Individualized Neoantigen Therapy (INT)	mRNA-4157	This mRNA therapy is a Phase-3 candidate to treat individualized neoantigen therapy (INT). This therapeutic candidate is being developed at a 50-50 global profit sharing with Merck.

Note: SPIKEVAX (COVID-19 Vaccine, mRNA) is no longer available, and the FDA has revised its recommendation for new and ongoing COVID-19 vaccinations.

Source: Company Websites and Annual Reports

12.1.1.3 Recent developments

PRODUCT AND REGULATORY APPROVALS

MONTH AND YEAR	PRODUCT CATEGORY	DESCRIPTION
February 2023	mRNA vaccine	Moderna, Inc. received authorization from Health Canada for its COVID-19 booster vaccine, mRNA-1273.214 (SpikevaxBivalent Original/Omicron). This vaccine is designed for immunization against COVID-19 in children and adolescents between 6 to 17 years.
January 2023	mRNA vaccine	US FDA granted Breakthrough Therapy Designation for Moderna’s investigational mRNA vaccine candidate—mRNA-1345. This vaccine was developed to prevent RSV-associated lower respiratory tract disease (RSV-LRTD) in adults aged 60 years or older.
December 2022	mRNA vaccine	Moderna, Inc. received emergency use authorization (EUA) for its BA.4/BA.5 Omicron-targeting bivalent COVID-19 booster vaccine, mRNA-1273.222, in children between 6 months and 5 years.
August 2022	mRNA vaccine	Medicines and Healthcare Products Regulatory Agency (MHRA) in the UK provided conditional authorization for Moderna’s Omicron-containing bivalent COVID-19 booster vaccine, mRNA-1273.214, to be used in individuals 18 years and older.

Source: Press Releases

DEALS

MONTH AND YEAR	DEAL TYPE	COMPANY 1	COMPANY 2	DESCRIPTION	DEAL SIZE
March 2023	Collaboration	Moderna, Inc. (US)	Generation Bio (US)	Moderna, Inc. collaborated with Generation Bio Co. to co-develop genetic medicines. This strategic collaboration combines Moderna's expertise in mRNA vaccines/therapy with Generation Bio's non-viral genetic medicine platform.	USD 40 million (To Generation Bio)
February 2023	Collaboration	Moderna, Inc. (US)	Life Edit Therapeutics (US)	Moderna, Inc. collaborated with Life Edit Therapeutics Inc., a company with expertise in next-generation gene editing technologies, to develop in vivo mRNA gene editing therapies.	NA

January 2023	Collaboration	Moderna, Inc. (US)	CytomX Therapeutics, Inc. (US)	Moderna, Inc. and CytomX Therapeutics, Inc. entered into a collaboration and licensing agreement to develop investigational mRNA-based conditionally activated therapies.	USD 35 million
January 2023	Acquisition	Moderna, Inc. (US)	Oriciro Genomics, Inc. (Japan)	Moderna, Inc. announced plans to acquire Oriciro Genomics, Inc., a company that develops cell-free DNA synthesis and amplification technologies.	USD 85 million
August 2022	Supply Agreement	Moderna, Inc. (US)	Government of Canada	The Government of Canada announced its plans to purchase an additional 4.5 million doses of omicron-containing bivalent COVID-19 booster vaccines that are developed by Moderna, Inc.	NA
July 2022	Supply Agreement	Moderna, Inc. (US)	Government of US	Moderna, Inc. and the US government entered into an agreement to secure 66 million doses of a Moderna COVID-19 vaccine booster candidate.	USD 1.74 billion
April 2022	Partnership	Moderna, Inc. (US)	International AIDS Vaccine Initiative (US)	Moderna, Inc., partnered with a nonprofit scientific research organization— IAVI—to apply mRNA technology for the development of RNA vaccines/therapeutics.	NA

Source: Press Releases

OTHER DEVELOPMENTS

MONTH AND YEAR	DEAL TYPE	DESCRIPTION
March 2023	Business expansion	Moderna, Inc. entered into an agreement with the Government of the Republic of Kenya to establish a facility in the country for manufacturing mRNA vaccines/therapies. This is the company's first mRNA manufacturing facility in Africa.

Source: Press Releases

12.1.1.4 MnM view

12.1.1.4.1 Key strengths

Moderna, Inc. has emerged as one of the leading players in the RNA therapeutics market. The company's COVID-19 booster vaccine, Spikevax, is well received and has shown considerable growth in sales after its approval in 2020. Additionally, it has built a strong pipeline for other mRNA-based vaccines, with a majority being in the late clinical stage. Also, the company has received approval and EUA for its BA.4/BA.5 Omicron-targeting bivalent COVID-19 booster vaccine in various countries, creating a strong revenue stream for the company in 2022.

12.1.1.4.2 Strategic choices

Moderna, Inc. has relied on collaborations and strategic co-developments to strengthen its vaccine manufacturing capabilities. In March 2023, the company collaborated with Generation Bio Co. to co-develop genetic medicines. This strategic collaboration combines Moderna's expertise in mRNA vaccines/therapy with Generation Bio's non-viral genetic medicine platform. In February 2023, Moderna, Inc. collaborated with Life Edit Therapeutics Inc., an ElevateBio company with expertise in next-generation gene editing technologies, to develop in vivo mRNA gene editing therapies.

12.1.1.4.3 Weaknesses and competitive threats

Moderna, Inc. is a leading player in the RNA-based vaccines market. However, the sudden halt on further studies regarding Spikevax may create a bottleneck for the company to maintain its revenue generation capacity and can encourage other participants to enter the market.

12.1.2 ALNYLAM PHARMACEUTICALS, INC.

12.1.2.1 Business overview

Alnylam Pharmaceuticals, Inc. develops and manufactures novel medicines based on RNA interference (RNAi technology). The company is currently developing its portfolio around RNAi therapeutics, including small interfering RNA (siRNA). The company has also built a pipeline of investigational RNAi therapeutics targeted toward four therapeutic applications, namely, cardio-metabolic diseases, genetic medicines, infectious diseases, and central nervous system (CNS) and ocular diseases. There are more than 12 RNAi therapeutics in clinical development, with multiple pipeline candidates in late-stage development. Alnylam Pharmaceuticals operates as a single business segment to develop and manufacture RNAi therapeutics.

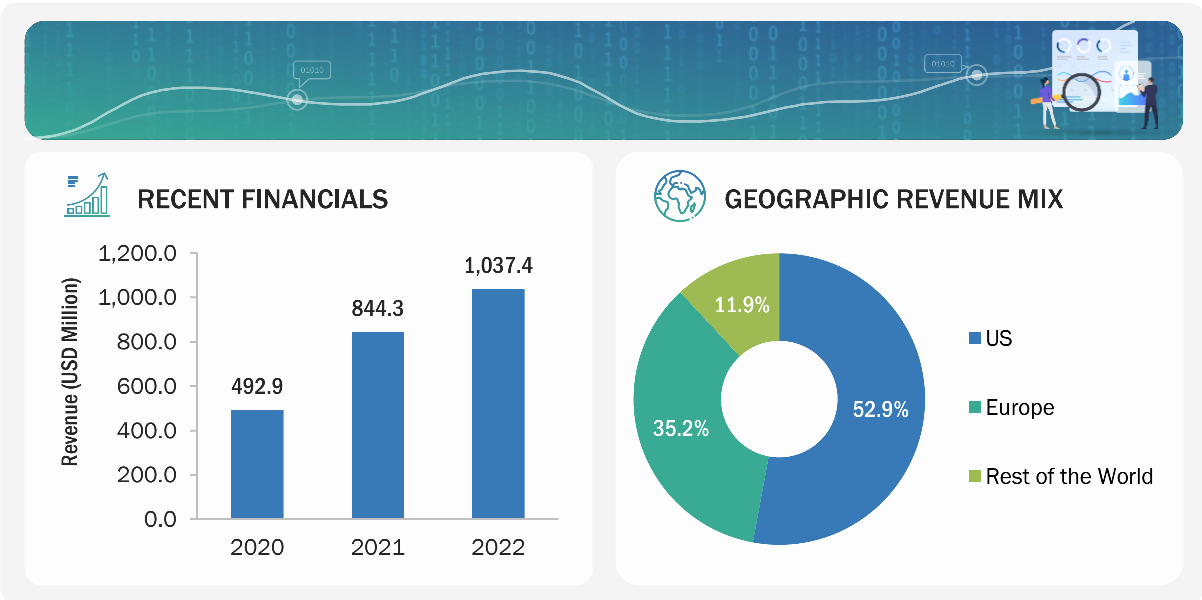
Currently, Alnylam Pharmaceuticals, Inc. controls approximately 20 wholly owned subsidiaries, some of which include Alnylam U.S., Inc., Sirna Therapeutics, Inc (US), Alnylam Austria GmbH, Alnylam Brasil Farmaceutica Ltda. (Brazil), Alnylam Switzerland GmbH, and Alnylam Taiwan Co. Ltd. The company operates across three key geographies, namely the US, Europe, and the Rest of the World.

TABLE 130 ALNYLAM PHARMACEUTICALS, INC.: COMPANY OVERVIEW

Year Founded	2002
Headquarters Country	US
Headquarters State	Massachusetts
Employee Size	Approx. 2,002 (As of December 2022)
Annual Revenue	USD 1 Billion (As of 2022)
Ownership	Public

Source: Company Websites and Annual Reports

FIGURE 31 ALNYLAM PHARMACEUTICALS, INC.: COMPANY SNAPSHOT (2022)



Note: Alnylam Pharmaceuticals, Inc. operates as a single business unit; hence, the business revenue mix of the company is not included.

Source: Company Website, Annual Reports, and SEC Filings

12.1.2.2 Products offered

PRODUCT CATEGORY	APPLICATION	PRODUCT NAME	DESCRIPTION
RNAi Therapy	hATTR Amyloidosis with Polyneuropathy	AMVUTTRA (vutrisiran) ONPATTRO (patisiran)	This RNAi-based medicine is prescribed for adults to treat hereditary transthyretin-mediated amyloidosis (hATTR amyloidosis).
RNAi Therapy	Acute Hepatic Porphyria	GIVLAARI (givosiran)	This RNAi-based medicine is prescribed for adults to treat acute hepatic porphyria (AHP).
RNAi Therapy	Primary Hyperoxaluria Type 1	OXLUMO (lumasiran)	OXLUMO (lumasiran) is a prescription medicine for the treatment of PH1 to lower oxalate in urine and blood in children and adults.

Source: Company Websites and Annual Reports

12.1.2.3 Recent developments

PRODUCT AND REGULATORY APPROVALS

MONTH AND YEAR	PRODUCT CATEGORY	DESCRIPTION
September 2022	RNAi Therapy	Alnylam Pharmaceuticals, Inc. received marketing authorization from the European Commission for its RNAi therapy—AMVUTTRA. This treatment is developed for adult patients with stage 1 or stage 2 polyneuropathy, which is hereditary transthyretin-mediated (hATTR).

Source: Press Releases

DEALS

MONTH AND YEAR	DEAL TYPE	COMPANY 1	COMPANY 2	DESCRIPTION	DEAL SIZE
July 2023	Partnership	Alnylam Pharmaceuticals, Inc. (US)	F. Hoffmann-La Roche Ltd (Switzerland)	Alnylam Pharmaceuticals, Inc. entered into a partnership with F. Hoffmann-La Roche Ltd. to co-develop and co-commercialize Zilebesiran, which is an Investigational RNAi medicine to treat hypertension in patients with high cardiovascular risk.	USD 3.1 Billion
January 2022	Collaboration	Alnylam Pharmaceuticals, Inc. (US)	Novartis AG (Switzerland)	Alnylam Pharmaceuticals, Inc. collaborated with Novartis AG to discover and develop siRNA-based targeted therapy for patients with end-stage liver diseases.	NA

Source: Press Releases

12.1.2.4 MnM view

12.1.2.4.1 Key strengths

Alnylam Pharmaceuticals offers three established RNAi therapies, namely, ONPATTRO (patisiran), GIVLAARI (givosiran), and OXLUMO (lumasiran), which have shown consistent growth in sales from 2019 to 2022 and contributed to the company's market position. It has also recently added AMVUTTRA (vutrisiran) to its portfolio, which was approved in June 2022 and recorded sales of approximately USD 90 million in the same year. The company has established itself as a prominent player in the RNAi therapeutics market for ATTR treatment.

12.1.2.4.2 Strategic choices

Alnylam Pharmaceuticals focuses on partnerships and collaborations with biotechnology and pharmaceutical companies to expand its reach across different geographies and strengthen its development and manufacturing capabilities. In July 2023, the company partnered with F. Hoffmann-La Roche Ltd. to co-develop and co-commercialize Zilebesiran, an Investigational RNAi medicine to treat hypertension in patients with high cardiovascular risk.

12.1.2.4.3 Weaknesses and competitive threats

Alnylam Pharmaceuticals faces a broad spectrum of competition from current and potential competitors, which range from key global pharmaceutical companies to small/mid-sized biotechnology companies. The competitive landscape has expanded rapidly through partnerships/collaborations among market players and several companies targeting RNAi technology, including Arrowhead Pharmaceuticals, Inc. and its collaborators Takeda Pharmaceutical, GlaxoSmithKline PLC, and Amgen Inc., among others.

12.1.3 NOVARTIS AG

12.1.3.1 Business overview

Novartis AG discovers, develops, manufactures, and markets prescription, generic pharmaceutical, and eye care products. The company operates through two global business divisions, namely, Innovative Medicines and Sandoz (Generics). In April 2022, the company announced structural changes in its operations by integrating its former Oncology and Pharmaceuticals business units into Innovative Medicines. This integration has led to two separate commercial businesses within the Innovative Medicines division, namely, Innovative Medicines US and Innovative Medicines International.

The Innovative Medicines Division caters to key treatment areas such as, namely, hematology, cardiovascular, neuroscience, immunology, solid tumor, ophthalmology, and respiratory. Novartis AG offers RNA therapeutics through its Innovative Medicines business division, which sells the products across approximately 130 countries.

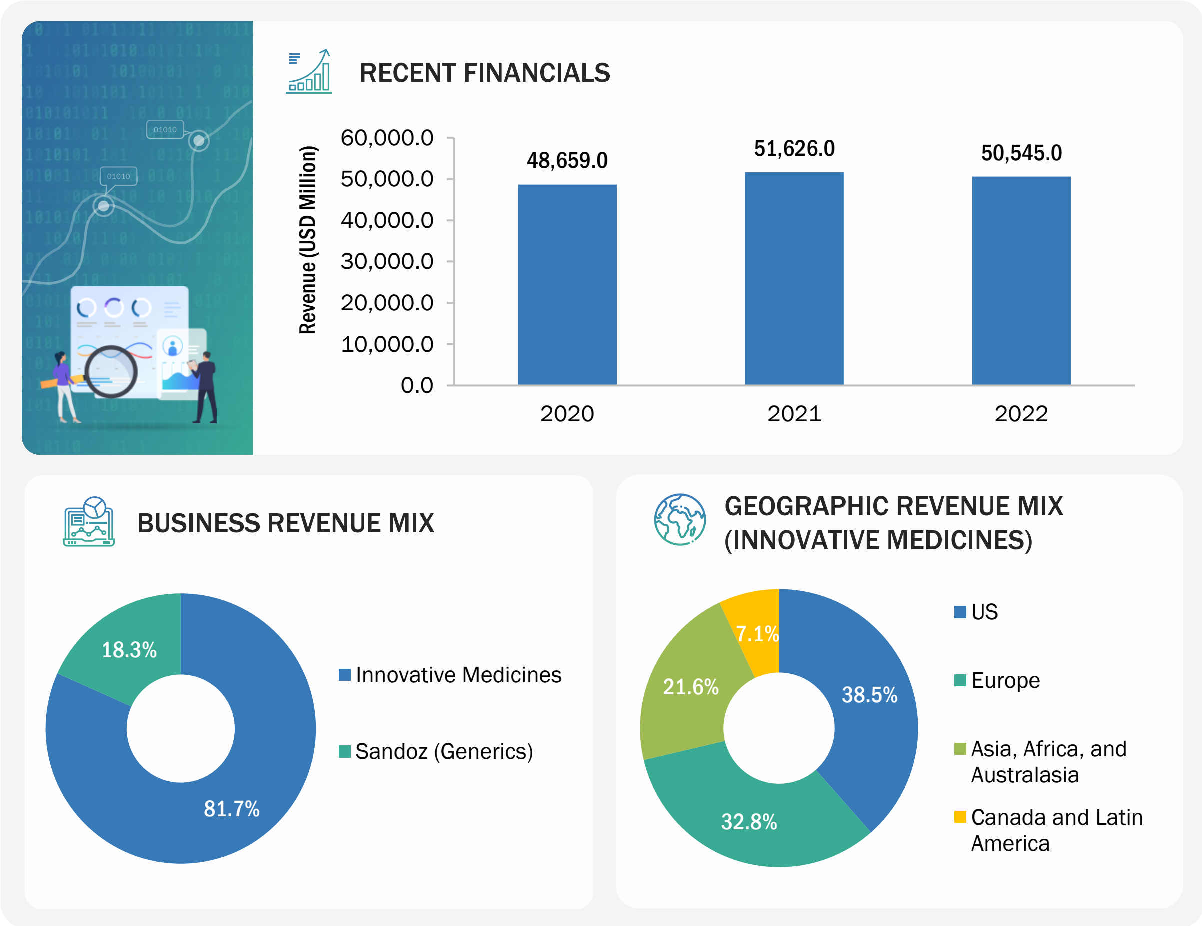
The company operates through a network of subsidiaries and offices in the US, Canada, Latin America, Europe, Asia, Africa, and Australasia. Some of its major subsidiaries include Novartis Argentina S.A., Novartis Pharma S.A. (Belgium), Novartis Biociências S.A. (Brazil), Novartis Pharmaceuticals Canada, Inc., Novartis Deutschland GmbH (Germany), Novartis India Limited, Novartis Holding Japan K.K., and Novartis Corporation (US).

TABLE 131 NOVARTIS AG: COMPANY OVERVIEW

Year Founded	1996
Headquarters Country	Switzerland
Headquarters City	Basel
Employee Size	Approx. 102,000 (As of December 2022)
Annual Revenue	USD 50.5 Billion (As of 2022)
Ownership	Public

Source: Company Websites and Annual Reports

FIGURE 32 NOVARTIS AG: COMPANY SNAPSHOT (2022)



Note: Information for the recent financials has been provided for the overall company.

Source: Annual Reports and Company Website

12.1.3.2 Products offered

PRODUCT CATEGORY	APPLICATION	PRODUCT NAME	DESCRIPTION
siRNA Therapeutics	LDL-C (bad cholesterol) reduction	Leqvio (inclisiran)	Leqvio is a small interfering RNA (siRNA) therapy to reduce low-density lipoprotein cholesterol (LDL-C). This therapy is administered every 6 months after 2 initial doses.

Source: Company Websites and Annual Reports

12.1.3.3 Recent developments

REGULATORY APPROVALS

MONTH AND YEAR	DESCRIPTION
July 2023	Novartis AG's Leqvio RNA therapy received FDA approval to increase its use as an adjunct to diet and statin therapy. The use of this therapy is approved in patients with elevated LDL-C. This patient population also includes those who have comorbidities such as hypertension and diabetes and have no cardiovascular complications.

Source: Press Releases

DEALS

MONTH AND YEAR	DEAL TYPE	COMPANY 1	COMPANY 2	DESCRIPTION	DEAL SIZE
July 2023	Acquisition	Novartis AG (Switzerland)	DTx Pharma (US)	Novartis AG announced the acquisition of DTx Pharma to strengthen its neuroscience pipeline and RNA therapeutic capabilities.	USD 1 Billion

Source: Press Releases

OTHER DEVELOPMENTS

MONTH AND YEAR	TYPE	COMPANY 1	COMPANY 2	DESCRIPTION
October 2021	Agreement	Novartis AG (Switzerland)	BioNTech (Germany)	Novartis signed an initial agreement with BioNTech to leverage its vaccine manufacturing capabilities during the COVID-19 pandemic. Novartis utilized its sterile manufacturing facilities at its Novartis Technical Operations site in Slovenia to manufacture around 24 million doses in 2022.

Source: Press Releases

12.1.3.4 MnM view

12.1.3.4.1 Key strengths

Novartis presently offers only one product in the RNA therapeutics market named Leqvio (inclisiran), which is the only small-interfering RNA that is commercialized for cholesterol reduction. The company has developed this product through its collaboration with Alnylam Pharmaceuticals, Inc., offering the company a competitive edge in the market. The product reportedly reached a revenue of around USD 500 million in 2022 globally, providing a good head start to Novartis in the RNA therapeutics market. In 2022, just eleven months after it was launched, Leqvio was covered at or near the label for 76% of patients. It has also been assigned a unique Healthcare Common Procedure Coding System Code (J-code).

12.1.3.4.2 Strategic choices

The company has focused on both organic and inorganic strategies in this market, including collaborations and acquisitions. For instance, in July 2023, Novartis AG acquired DTx Pharma to strengthen its neuroscience pipeline and RNA therapeutic capabilities. In October 2021, Novartis signed an initial agreement with BioNTech to leverage its vaccine manufacturing capabilities to address the complications of the COVID-19 pandemic.

12.1.3.4.3 Weaknesses and competitive threats

While Novartis recorded strong sales for Leqvio soon after its launch, the company faces robust competition from market players such as Ionis Pharmaceuticals, Inc. (US), Alnylam Pharmaceuticals, Inc. (US), and Moderna, Inc. (US). Moreover, ongoing collaborations with pharmaceutical companies that can explore different RNA therapeutic modalities, mRNA, and rapidly expanding development programs (for therapies/pipeline candidates) pave the way for other upcoming companies in this market.

12.1.4 IONIS PHARMACEUTICALS, INC.

12.1.4.1 Business overview

Ionis Pharmaceuticals, Inc. develops and manufactures innovative medicines for multiple diseases, particularly rare genetic disorders. The company utilizes proprietary technologies to develop RNA-targeted therapeutics. It presently markets 3 drugs, namely, SPINRAZA (gene therapy), TEGSEDI (antisense medicine), and WAYLIVRA (antisense medicine). The company operates as a single business unit, with nearly seven medicines in the Phase 3 development stage. Its present portfolio of medicines and clinical pipeline utilizes antisense technology, which targets RNA to discover novel genetic medicines. Antisense medicines bind to mRNAs and inhibit the production of disease-causing proteins.

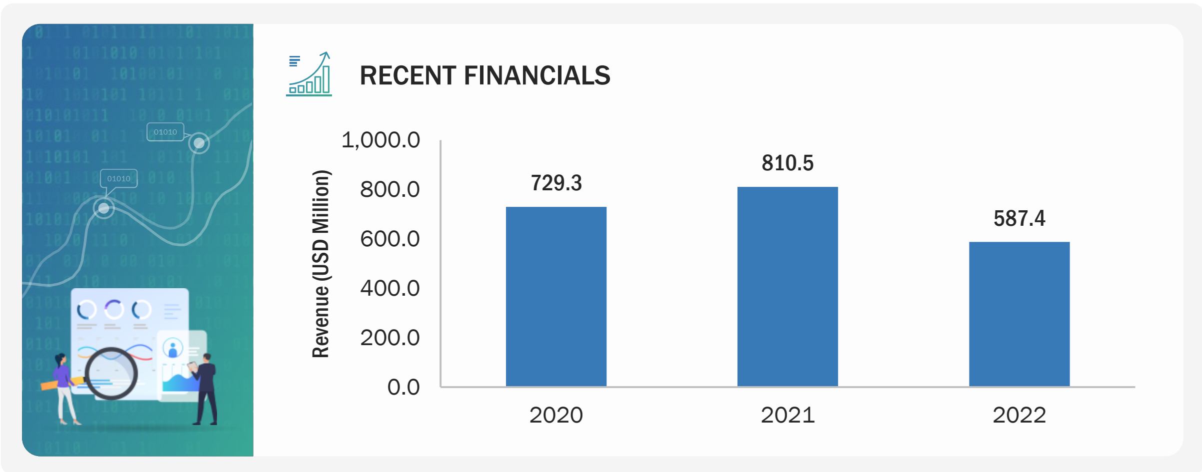
Ionis Pharmaceuticals, Inc. markets its antisense medicines across various geographies, including the US, Europe, Canada, Japan, and Brazil. Ionis Pharmaceuticals (US) and Akcea Therapeutics (US) are both registered trademarks/subsidiaries of the company.

TABLE 132 IONIS PHARMACEUTICALS, INC.: COMPANY OVERVIEW

Year Founded	1989
Headquarters Country	US
Headquarters State	California
Employee Size	Approx. 796 (As of December 2022)
Annual Revenue	USD 0.58 Billion
Ownership	Public

Source: Company Websites and Annual Reports

FIGURE 33 IONIS PHARMACEUTICALS, INC.: COMPANY SNAPSHOT (2022)



Note: Ionis Pharmaceuticals, Inc. operates as a single business unit; hence, the business and geographic revenue mix of the company is not included.

Source: Company Website, Annual Reports, and SEC Filings

12.1.4.2 Products offered

PRODUCT CATEGORY	APPLICATION	PRODUCT NAME	DESCRIPTION
Antisense RNA Therapy	Spinal muscular atrophy (SMA)	SPINRAZA (nusinersen)	SPINRAZA (nusinersen) injection is designed for intrathecal use and is a type of antisense medicine. This medicine modulates RNA splicing to increase protein production of the SMN protein, which is critical to the health and survival of nerve cells in the spinal cord that are responsible for neuro-muscular function. The SMN protein is deficient in people with SMA.
Antisense RNA Therapy	Polyneuropathy of hereditary transthyretin-mediated amyloidosis (ATTR)	TEGSEDI (inotersen)	TEGSEDI (inotersen) injection is an antisense medicine (RNA-targeted therapy) to treat ATTRv-PN in adults. ATTR is a rare genetic disorder. This medicine prevents the production of TTR protein, reducing the amount of amyloid buildup that damages organs and tissues.
Antisense RNA Therapy	Familial chylomicronaemia syndrome (FCS)	WAYLIVRA (volanesorsen)	WAYLIVRA (volanesorsen) is an antisense medicine/RNA-targeted therapy (available as a solution for injection under the skin) administered in adult patients suffering from FCS or who are at high risk for acute, potentially fatal pancreatitis.
Antisense RNA Therapy	Amyotrophic lateral sclerosis (ALS)	QALSODY (tofersen)	QALSODY (tofersen) injection is a recently approved antisense medicine to treat ALS. This binds to messenger RNA associated with the SOD1 gene and can mitigate the excess production of harmful SOD1 protein.
Antisense RNA Therapy	ATTR	Eplontersen (formerly IONIS-TTR-LRx)	This is an investigational/Phase 3 drug designed using the company's proprietary technology- Ligand-Conjugated Antisense (LICA). It is a self-administered subcutaneous injection to inhibit the production of TTR protein.
Antisense RNA Therapy	Familial chylomicronaemia syndrome (FCS)	Olezarsen (formerly IONIS-APOCIII-LRx)	This is an investigational/Phase 3 drug designed using the company's proprietary technology- Ligand-Conjugated Antisense (LICA). This drug inhibits the production of apoC-III for patients who are at risk of disease due to elevated triglyceride levels.
Antisense RNA Therapy	Hereditary angioedema (HAE)	Donidalorsen (formerly IONIS-PKK-LRx)	This is an investigational/Phase 3 drug designed using the company's proprietary technology- Ligand-Conjugated Antisense (LICA), to target the prekallikrein/PKK pathway.
Antisense RNA Therapy	Lp(a)-C (cardiovascular disease)	Pelacarsen (formerly IONIS-APO(a)-LRx)	This is an investigational/Phase 3 drug designed using the company's proprietary technology- Ligand-Conjugated Antisense (LICA) to inhibit the production of apolipoprotein(a) in the liver.
Antisense RNA Therapy	Hepatitis B (HBV)	Bepirovirsen (formerly IONIS-HBVRx)	This is an investigational/Phase 3 drug designed using the company's proprietary technology- Ligand-Conjugated Antisense (LICA) to inhibit the production of viral proteins associated with the hepatitis B virus.

Source: Company Websites and Annual Reports

12.1.4.3 Recent developments

PRODUCT AND REGULATORY APPROVALS

MONTH AND YEAR	PRODUCT CATEGORY	DESCRIPTION
April 2023	Antisense (RNA targeted) Therapy	Ionis Pharmaceuticals, Inc. announced FDA approval for its antisense therapy, namely, QALSODY (tofersen). This therapy is co-developed with Biogen, Inc. (US) injection to treat amyotrophic lateral sclerosis (ALS) in adults.
December 2022	Antisense (RNA targeted) Therapy	Ionis Pharmaceuticals, Inc. received marketing authorization from the European Medicines Agency (EMA) for its ALS treatment—Tofersen.
January 2022	Antisense (RNA targeted) Therapy	Ionis Pharmaceuticals, Inc. received orphan drug designation from the US FDA for its ALS treatment— Eplontersen.

Source: Press Releases

DEALS

MONTH AND YEAR	DEAL TYPE	COMPANY 1	COMPANY 2	DESCRIPTION	DEAL SIZE
August 2023	Collaboration	Ionis Pharmaceuticals, Inc. (US)	Novartis (US)	Ionis Pharmaceuticals, Inc. entered into a license agreement and collaboration with Novartis to develop and manufacture therapy for patients with CVD caused by lipoprotein(a).	USD 60 Million (To Ionis Pharmaceuticals, Inc.)
July 2023	Agreement	Ionis Pharmaceuticals, Inc. (US)	AstraZeneca (UK)	Ionis Pharmaceuticals, Inc. expanded its agreement with AstraZeneca to commercialize eplontersen in Latin America. This is a Phase 3/ investigational candidate for the treatment of ATTR.	USD 20 Million (To Ionis Pharmaceuticals, Inc.)
November 2022	Partnership	Ionis Pharmaceuticals, Inc. (US)	Metagenomi (US)	Ionis Pharmaceuticals, Inc. partnered with Metagenomi, through which the former can leverage Metagenomi's gene editing systems to develop RNA-targeted therapeutics.	USD 80 Million (To Metagenomi)

Source: Press Releases

OTHER DEVELOPMENTS

MONTH AND YEAR	DEVELOPMENT TYPE	DESCRIPTION
July 2023	Phase 3 enrollment	Ionis Pharmaceuticals, Inc. completed THE enrollment of CARDIO-TTRANSform for Phase 3 study in patients suffering from TTR-mediated amyloid cardiomyopathy.
October 2022	Business expansion	Ionis Pharmaceuticals, Inc. (Nasdaq: IONS) entered into an agreement with Sudberry Properties to develop and lease a new development chemistry and manufacturing site in Oceanside, California.

Source: Press Releases

12.1.5 SAREPTA THERAPEUTICS, INC.

12.1.5.1 Business overview

Sarepta Therapeutics, Inc. is a commercial-stage biopharmaceutical company that discovers and develops novel therapies such as RNA-targeted therapies and gene therapy. The company also develops therapeutic modalities against rare genetic disorders, including Duchenne muscular dystrophy, limb-girdle muscular dystrophies, and other neuromuscular and central nervous system-related disorders, through strategic partnerships and collaborations. The company applies its proprietary platforms to develop both RNA therapeutics and gene therapies and operates as a single business unit.

The company offers three commercial gene therapy products, namely, EXONDYS 51. VYONDYS 53, and AMONDYS 45. These products treat patients with Duchenne Muscular Dystrophy. It has a pipeline of approximately 8 novel therapeutic candidates (including RNA-targeted therapies, gene therapy, and gene editing-based therapies), with nearly 6 candidates in the clinical phase.

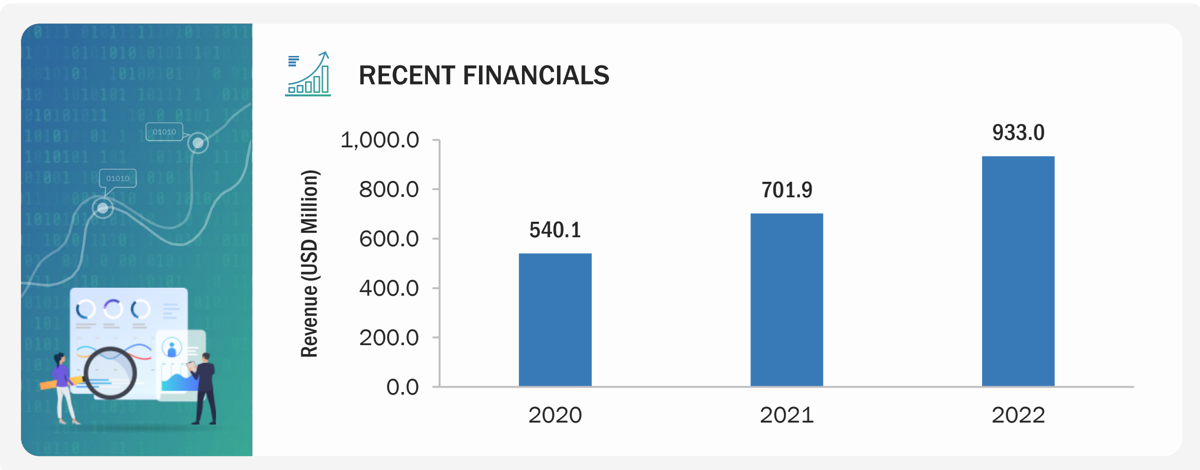
The company majorly operates its business channels in the US and in a few locations across Europe, including Ireland (Sarepta Therapeutics Ireland Limited), Switzerland (Sarepta International Holdings GmbH), and the UK (Sarepta International UK Ltd.). Some of its subsidiaries are Sarepta Securities Corp (US), ST International Holdings, Inc. (US), and Sarepta Therapeutics Three, LLC (US).

TABLE 133 SAREPTA THERAPEUTICS, INC.: COMPANY OVERVIEW

Year Founded	1980
Headquarters Country	US
Headquarters State	Massachusetts
Employee Size	Approx. 1,162 (As of December 2022)
Annual Revenue	USD 0.9 Billion (As of 2022)
Ownership	Public

Source: Company Websites and Annual Reports

FIGURE 34 SAREPTA THERAPEUTICS, INC.: COMPANY SNAPSHOT (2022)



Note: Information for the business and geographic revenue mix of the company is unavailable in the public domain. Hence, these sections are not provided here.

Source: Company Website, Annual Reports, and SEC Filings

12.1.5.2 Products offered

PRODUCT CATEGORY	APPLICATION	PRODUCT NAME	DESCRIPTION
mRNA Targeted Therapy	Duchenne muscular dystrophy	SRP-5051 (vesleteplirsen)	This therapy is a Phase 2 candidate, which is designed using the company's proprietary technology-peptide-conjugated phosphorodiamidate morpholino oligomer (PPMO). The therapy binds to exon 51 of dystrophin pre-mRNA, resulting in the exclusion of this exon during mRNA processing in patients with genetic mutations.

Source: Company Websites and Annual Reports

12.1.5.3 Recent developments

OTHER DEVELOPMENTS

MONTH AND YEAR	DEVELOPMENT TYPE	DESCRIPTION
June 2022	Product development	US FDA uplifted its clinical hold on part B of the phase 2 MOMENTUM trial (NCT04004065) of SRP-5051. The hold was originally placed in June 2022 after a patient experienced a serious adverse event (SAE) of hypomagnesemia after treatment.

Source: Press Releases

12.1.6 SANOFI

12.1.6.1 Business overview

Sanofi is a pharmaceutical company that develops vaccines and therapies to treat diseases associated with hematology, oncology, neurology, and immunology. The company has three operating segments, namely, Pharmaceuticals, Vaccines, and Consumer Healthcare. The Pharmaceuticals segment discovers and develops candidates for specialty care and general medicines. Specialty care includes therapies for atopic dermatitis (eczema), rare disorders, neurology and immunology diseases, and oncology. Sanofi offers vaccines for different indications such as influenza, booster vaccines, travel and endemic, meningitis, and Polio Pertussis Hib.

The company offers RNA drugs and vaccines through its Pharmaceuticals and Vaccines business segments, respectively. It leverages mRNA technology to develop treatment modalities, including vaccines and drugs for cancer, immune-mediated diseases, and rare diseases. In June 2021, the company launched a separate facility in the US for mRNA to accelerate the end-to-end R&D of next-generation vaccines.

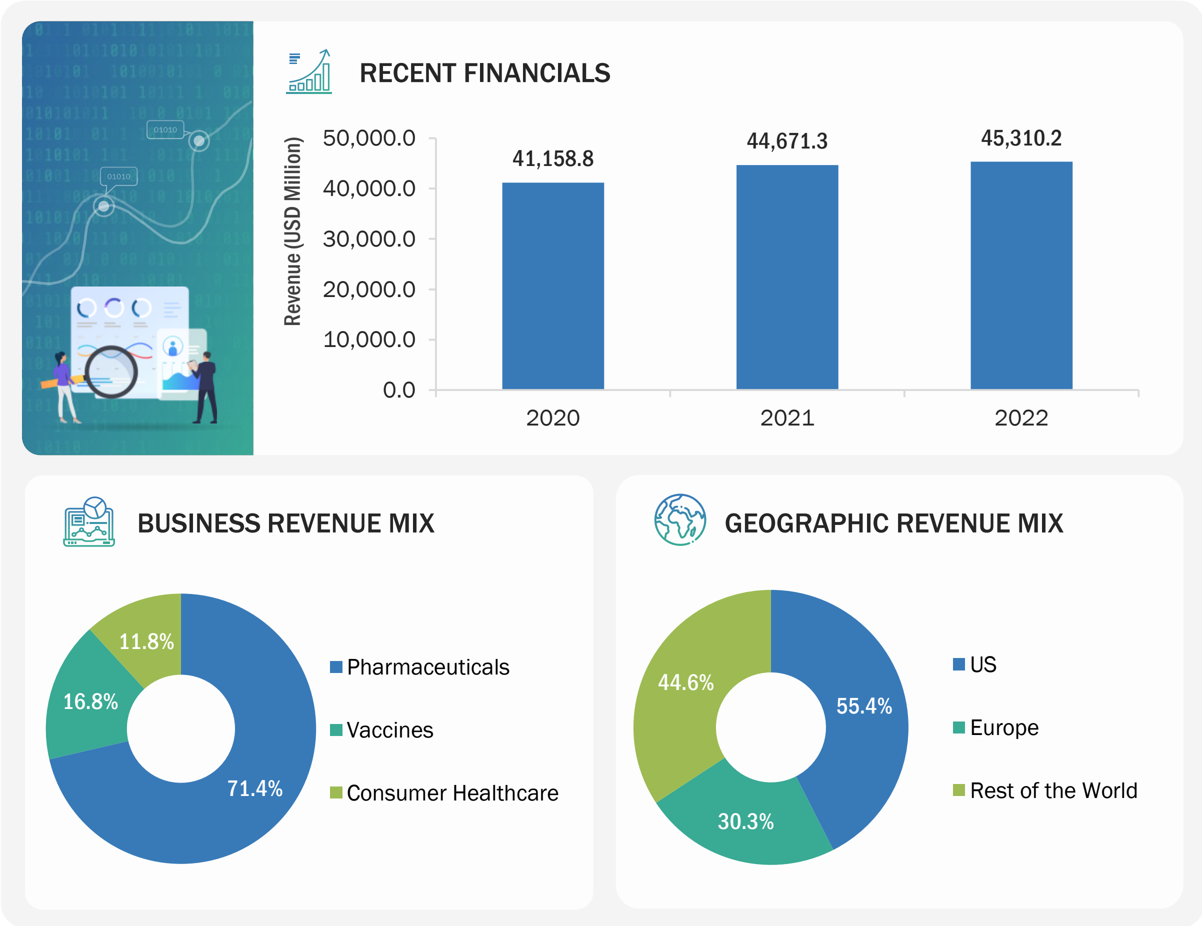
Sanofi operates across various geographies, including the US, Europe, China, Japan, Brazil, and Russia. Some of the key subsidiaries of Sanofi include Aventis Inc. (US), Genzyme Corporation (US), Genzyme Europe B.V. (Netherlands), Hoechst GmbH (Germany), Sanofi-Aventis Deutschland GmbH (Germany), Sanofi-Aventis Participations SAS (France), Sanofi-Aventis Singapore Pte Ltd. (Singapore), Sanofi Biotechnology (France), Sanofi Foreign Participations B.V. (Netherlands), Sanofi Winthrop Industrie (France), and Sanofi Pasteur Inc. (US)

TABLE 134 SANOFI: COMPANY OVERVIEW

Year Founded	1973
Headquarters Country	France
Headquarters City	Paris
Ownership	Public

Source: Company Websites and Annual Reports

FIGURE 35 SANOFI: COMPANY SNAPSHOT (2022)



Note 1: Sanofi offers RNA therapy products through its Pharmaceuticals and Vaccines business segments; hence, the business and geographic revenue mix of the company is mentioned for the overall company revenue.

Note 2: Rest of the World includes China, Japan, Brazil, and Russia.

Source: Company Website, Annual Reports, and SEC Filings

12.1.6.2 Products offered

PRODUCT CATEGORY	APPLICATION	PRODUCT NAME	DESCRIPTION
siRNA Therapeutics	Hemophilia A and B	Fitusiran	Fitusiran is designed using Alnylam Pharmaceutical Inc.'s ESC-GalNAc conjugate technology to lower antithrombin levels for maintaining hemostasis balance and preventing bleeding.

Source: Company Websites and Annual Reports

12.1.6.3 Recent developments

DEALS

MONTH AND YEAR	DEAL TYPE	COMPANY 1	COMPANY 2	DESCRIPTION	DEAL SIZE
August 2021	Acquisition	Sanofi (France)	Translate Bio (US)	Sanofi entered into a definitive agreement with Translate Bio to accelerate its manufacturing and development capabilities for mRNA therapy.	USD 3.2 Billion
November 2021	Collaboration	Sanofi (France)	Baidu, Inc. (China)	Baidu Inc. licensed its algorithm for messenger RNA (mRNA) sequence to Sanofi. This collaboration leveraged Baidu's technology to develop vaccines and therapeutic products.	NA
October 2022	Partnership	Sanofi (France)	miRecule, Inc. (US)	Sanofi partnered with miRecule, Inc. to gain control of the latter company's candidate- anti-DUX4 RNA therapy. This therapy treats facioscapulohumeral muscular dystrophy (FSHD). Sanofi plans to combine the therapy with its muscle-targeted nanobody technology to create an antibody-RNA conjugate (ARC).	USD 30 Million

Source: Press Releases

OTHER DEVELOPMENTS

MONTH AND YEAR	DEVELOPMENT TYPE	DESCRIPTION
June 2021	Business expansion	Sanofi plans to invest nearly USD 439 million each year in its mRNA Center of Excellence to accelerate the development and delivery of next-generation vaccines using mRNA technology. This center is equipped with end-to-end mRNA vaccine development modalities.

Source: Press Releases

12.1.7 PFIZER INC.

12.1.7.1 Business overview

Pfizer Inc. is a global biopharmaceutical company that discovers, develops, manufactures, sells, and distributes products in key therapeutic areas such as oncology, inflammation and immunology, and rare diseases. The company operates through two segments, namely, Biopharma and Pfizer CentreOne. It provides APIs, sterile injectable pharmaceuticals, biosimilars, and contract manufacturing services. Pfizer operates in the RNA therapeutics market through its biopharma segment. Pfizer offers RNA vaccine-COMIRNATY against COVID-19 through its Primary Care segment within the Biopharma business division.

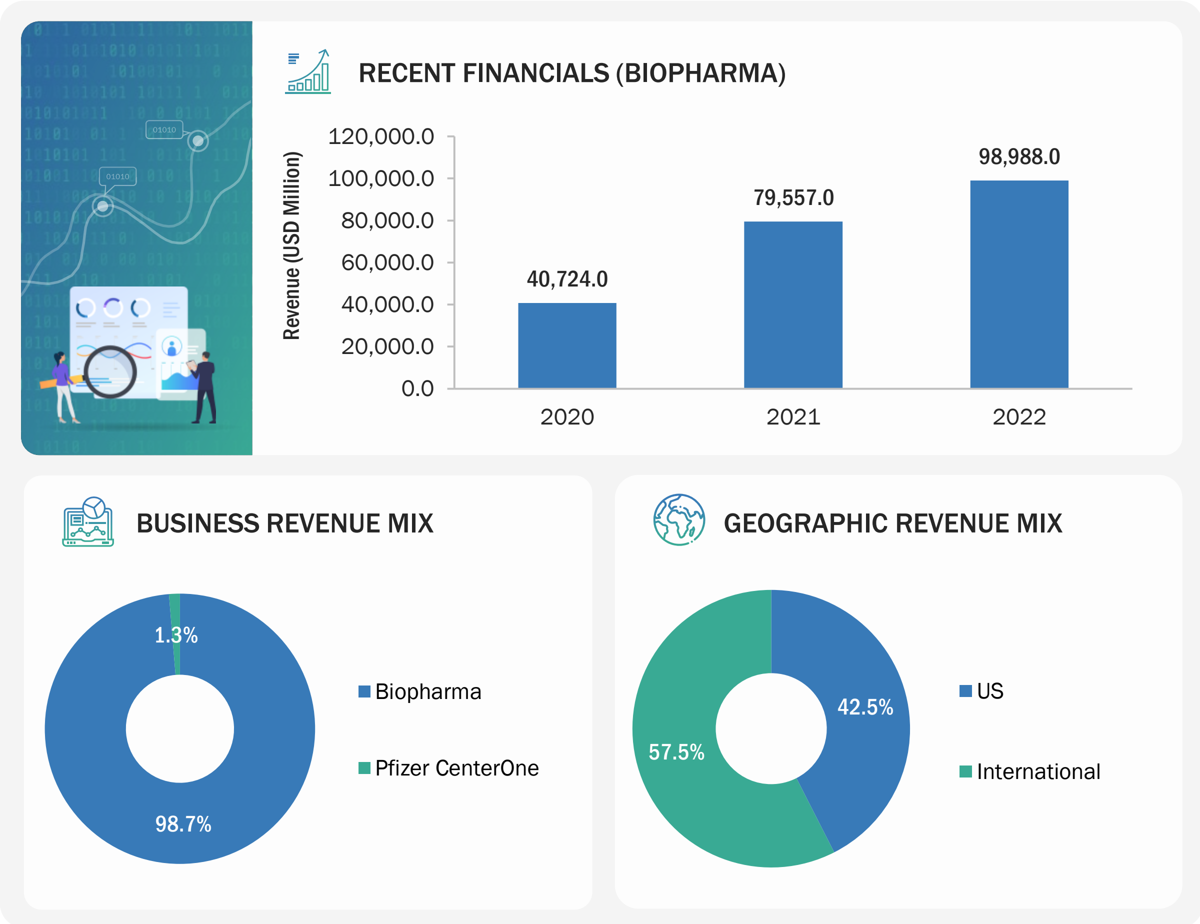
The company has a global presence across North America, Latin America, Europe, Southeast Asia, and Oceania. It operates in over 125 countries, with the US being its largest market. Outside the US, China, Japan, the UK, France, Germany, Switzerland, Sweden, Australia, and New Zealand are some of the major markets. It has major manufacturing facilities in India, China, Japan, Ireland, Italy, Belgium, Germany, Singapore, and the US.

TABLE 135 PFIZER INC.: COMPANY OVERVIEW

Year Founded	1849
Headquarters Country	US
Headquarters State	New York
Ownership	Public

Source: Company Website, SEC Filings, and Annual Reports

FIGURE 36 PFIZER INC.: COMPANY SNAPSHOT (2022)



Note: The business revenue mix and geographic revenue mix were not available for the Biopharma segment. Hence, these sections are provided for the overall company.

Source: Company Websites and Annual Reports

12.1.7.2 Products offered

PRODUCT CATEGORY	APPLICATION	PRODUCT	DESCRIPTION
mRNA Vaccine	COVID-19	Comirnaty, Omicron-based mRNA Vaccine	This is a mRNA-based vaccine, which is co-developed and co-commercialized with BioNTech against the SARS-COV-2 infection. Pfizer and BioNTech equally shared the costs of development for the Comirnaty program, which has been approved and authorized in many countries.

Source: Company Website

12.1.7.3 Recent developments

DEALS

MONTH AND YEAR	DEAL TYPE	COMPANY 1	COMPANY 2	DESCRIPTION	DEAL SIZE
December 2021	Collaboration	Pfizer Inc.	BioNTech	Pfizer, Inc. and BioNtech entered a research, development, and commercialization agreement to develop a potential first mRNA-based vaccine to prevent shingles (herpes zoster virus) based on BioNTech's proprietary mRNA technology and antigen technology.	USD 225 million
December 2021	Collaboration	Pfizer Inc.	Beam Therapeutics	Pfizer and Beam Therapeutics announced an exclusive four-year research collaboration focused on in-vivo base editing programs, which use mRNA and lipid nanoparticles for three targets for rare genetic diseases of the liver, muscle, and central nervous system.	NA

Source: Press Releases

12.2 OTHER PLAYERS

12.2.1 ARROWHEAD PHARMACEUTICALS, INC.

ARROWHEAD PHARMACEUTICALS, INC.	
Year Founded	2004
Headquarters Country	US
Headquarters State	California
Ownership	Public
Business Overview	Arrowhead Pharmaceuticals develops medicines to treat intractable diseases using the gene-silencing approach. The therapies developed by the company trigger the RNA interference mechanism to induce the knockdown of target genes. RNA interference, or RNAi, is a mechanism present in living cells that inhibits the expression of a specific gene, thereby affecting the production of a specific protein. Arrowhead's RNAi-based therapeutics leverage this natural pathway of gene silencing.
Products Offered	Targeted RNAi Molecule (TRiM) Platform, ARO-PNPLA3, HZN-457, JNJ-3989
Geographical Presence	North America and Europe

Source: Company Website

12.2.2 BIONTECH SE

BIONTECH SE	
Year Founded	2008
Headquarters Country	Germany
Headquarters City	Mainz
Ownership	Public
Business Overview	BioNTech is a next-generation immunotherapy company operational in novel medicines against cancer, infectious diseases, and other serious diseases.
Products Offered	BNT111, BNT112, BNT113, BNT116, BNT 122, BNT131, BNT141, BNT142, BNT151, BNT152, BNT153, BNT162b, BNT161, BNT163, BNT164, BNT167, BNT165
Geographical Presence	North America and Europe

Source: Company Website

12.2.3 ORNA THERAPEUTICS

ORNA THERAPEUTICS	
Year Founded	2019
Headquarters Country	US
Headquarters State	Massachusetts
Ownership	Private
Business Overview	Orna Therapeutics is a biotechnology company that designs and delivers fully engineered circular RNA (oRNA) therapeutics to treat diseases such as Duchenne muscular dystrophy (DMD) and B-cell lymphoma. Orna's proprietary platform combines novel technology to create oRNAs that drive protein expression with validated and unique delivery solutions.
Products Offered	isCAR, and Dystrophin Replacement
Geographical Presence	North America

Source: Company Website

12.2.4 CRISPR THERAPEUTICS

CRISPR THERAPEUTICS	
Year Founded	2013
Headquarters Country	Switzerland
Headquarters City	Zug
Ownership	Public
Business Overview	CRISPR Therapeutics AG is a gene-editing company that develops transformative gene-based medicines for serious diseases. The company develops its products using the Clustered Regularly Interspaced Short Palindromic Repeats (CRISPR)/Cas9 gene-editing platform, which allows for precise and directed changes to genomic DNA. The company has licensed the foundational CRISPR-Cas9 patent for human therapeutic use.
Products Offered	Guide RNA
Geographical Presence	North America and Europe

Source: Company Website

12.2.5 SILENCE THERAPEUTICS

SILENCE THERAPEUTICS	
Year Founded	1999
Headquarters Country	UK
Headquarters City	London
Ownership	Private
Business Overview	Silence Therapeutics is a biotechnology company that discovers and develops novel molecules incorporating short-interfering ribonucleic acid.
Products Offered	Zerlasiran, SLN124, SLN-COMP-1, SLN-COMP-2, SLN-HAN-1, SLN-HAN-2
Geographical Presence	Europe and North America

Source: Company Website

12.2.6 ASTELLAS PHARMA INC.

ASTELLAS PHARMA INC.	
Year Founded	1923
Headquarters Country	Japan
Headquarters City	Tokyo
Ownership	Public
Business Overview	Astellas Pharma Inc. is a global pharmaceutical company that develops, manufactures, and commercializes pharmaceutical products. The company has R&D and manufacturing bases in Japan, Europe, the US, and Asia and is present in more than 70 countries worldwide.
Products Offered	AT466 currently comprises two different gene therapy approaches, namely, exon skipping and RNA knockdown. They prevent the accumulation of the excessively long, toxic mRNA molecule associated with DM1-causing mutations in the DMPK gene.
Geographical Presence	Japan, Europe, US, and Asia

Source: Company Website

12.2.7 CUREVAC SE

CUREVAC SE	
Year Founded	2000
Headquarters Country	Germany
Headquarters City	Tübingen
Ownership	Public
Business Overview	CureVac N.V. is a clinical-stage biopharmaceutical company that develops various transformative medicines based on messenger ribonucleic acid (mRNA). It develops prophylactic vaccines, such as mRNA-based vaccine candidates CV2CoV, CV7202, and CVSQIV, and RNA-based cancer therapies, including CV8102, to treat cutaneous melanoma, adenoidcystic carcinoma and the squamous cell cancer of skin, head, and neck.
Products Offered	• FLU SV mRNA • CVSQIV • CV0501 • CV2CoV • CV7202 • CVGBM • CV8102
Geographical Presence	North America and Europe

Source: Company Website

12.2.8 SIRNAOMICS

SIRNAOMICS	
Year Founded	2007
Headquarters Country	US
Headquarters State	Maryland
Ownership	Private
Business Overview	Sirnaomics is a clinical-stage RNA therapeutics biopharmaceutical company that discovers and develops innovative drugs for indications with significant unmet medical needs in areas such as cancers, skin conditions, fibrosis-related diseases, antiviral and cardiometabolic diseases, and medical aesthetics.
Products Offered	• STP705 • STP707 • STP122G • STP125G
Geographical Presence	US and China

Source: Company Website

12.2.9 ARCTURUS THERAPEUTICS, INC.

ARCTURUS THERAPEUTICS, INC.	
Year Founded	2013
Headquarters Country	US
Headquarters State	California
Ownership	Public
Business Overview	Arcturus Therapeutics Inc. is a clinical messenger RNA medicines and vaccine company that develops infectious disease vaccines and other products for liver and rare respiratory diseases. Its technology platforms include LUNAR lipid-mediated delivery and STARR mRNA. Arcturus' pipeline includes RNA therapeutic candidates to potentially treat ornithine transcarbamylase deficiency and cystic fibrosis, along with its partnered mRNA vaccine programs for SARS-CoV-2 (COVID-19) and influenza.
Products Offered	▪ LUNAR-COV19 (ARCT-021): Phase 2 candidate, mRNA vaccine ▪ LUNAR-FLU: mRNA vaccine
Geographical Presence	US and China

Source: Company Website

12.2.10 ARBUTUS BIOPHARMA

ARBUTUS BIOPHARMA	
Year Founded	2007
Headquarters Country	US
Headquarters State	Pennsylvania
Ownership	Public
Business Overview	Arbutus Biopharma is a clinical-stage biopharmaceutical company with expertise in virology and a focus on curing a variety of conditions with unmet medical needs. The company makes use of proprietary technologies to understand and gain expertise in treating viral infections and develop, and progress novel therapeutics focused on the areas of hepatitis B virus and coronavirus.
Products Offered	▪ RNAi Therapeutic (Imdusiran) AB-729: Phase 2 candidate, RNAi therapy ▪ RNA destabilizer (oral) AB-161: Phase 1 candidate, RNA therapy
Geographical Presence	US

Source: Company Website

13 APPENDIX

Q. 1. What are your views on the growth prospects of the RNA therapeutics market? What is the current scenario in this market, and at what rate is the market expected to grow in the coming five years (2023–2028)?

Q. 2. Kindly provide the estimated market shares and growth rates of the following products in the global RNA therapeutics market. What are the key factors that will drive the fastest-growing segment in the next six years?

PRODUCT	MARKET SHARE (2022)	CAGR (2023–2028)
Vaccines		
Drugs		

Q. 3. Kindly provide the estimated market shares and growth rates of the following types in the global RNA therapeutics market. What are the key factors that will drive the fastest-growing segment in the next six years?

TYPE	MARKET SHARE (2022)	CAGR (2023–2028)
mRNA Therapeutics		
RNA Interference (RNAi) Therapeutics		
Antisense Oligonucleotide (ASO) Therapeutics		
Other Therapeutics		

Q. 4. Kindly provide the estimated market shares and growth rates of the following indications in the global RNA therapeutics market. What are the key factors that will drive the fastest-growing segment of the market in the next six years?

INDICATION	MARKET SHARE (2022)	CAGR (2023–2028)
Infectious Diseases		
Rare Genetic /Hereditary Diseases		
Other Indications		

Q. 5. Kindly provide the estimated market shares and growth rates of the following end users in the global RNA therapeutics market. What are the key factors that will drive the fastest-growing segment of the market in the next six years?

END USER	MARKET SHARE (2022)	CAGR (2023–2028)
Hospital and Clinics		
Research Settings		

Q. 6. Kindly provide the market shares and growth rates of the following regions in the global RNA therapeutics market. What are the key factors that will drive the fastest-growing segment of the market in the next six years?

REGION	MARKET SHARE (2022)	CAGR (2023–2028)
▪ North America		
• US		
• Canada		
▪ Europe		
• Germany		
• UK		
• France		
• Rest of Europe		
▪ Asia Pacific		
• China		
• Japan		
• South Korea		
• Rest of Asia Pacific		
▪ Rest of the World		

Q. 7. Kindly provide the market shares of the following key players in the RNA therapeutics market in 2022. Please validate the key players.

RANK	COMPANY NAME	MARKET SHARE (2022)
	Moderna, Inc. (US)	
	Alnylam Pharmaceuticals, Inc. (US)	
	Novartis AG (Switzerland)	
	Ionis Pharmaceuticals, Inc. (US)	
	Sarepta Therapeutics, Inc. (US)	
	Other Companies	

Q. 8. Kindly validate the drivers, restraints, opportunities, and challenges impacting the global RNA therapeutics market.

PARAMETER	DESCRIPTION
Drivers	
Restraints	
Opportunities	
Challenges	

13.1 KNOWLEDGESTORE: MARKETSANDMARKETS' SUBSCRIPTION PORTAL

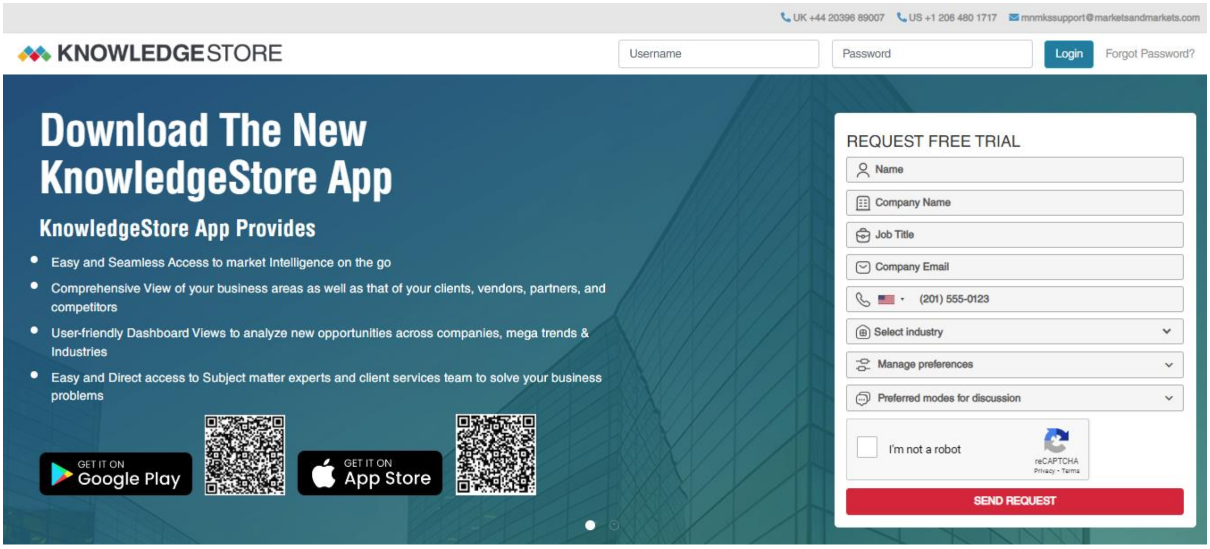
KnowledgeStore contains unique market insights of MarketsandMarkets. As a subscriber, you have access to all of the published titles and any new ones added over the course of the subscription period. You can subscribe to a specific number of reports or domains of your interest.

KNOWLEDGESTORE ENABLES YOU TO:

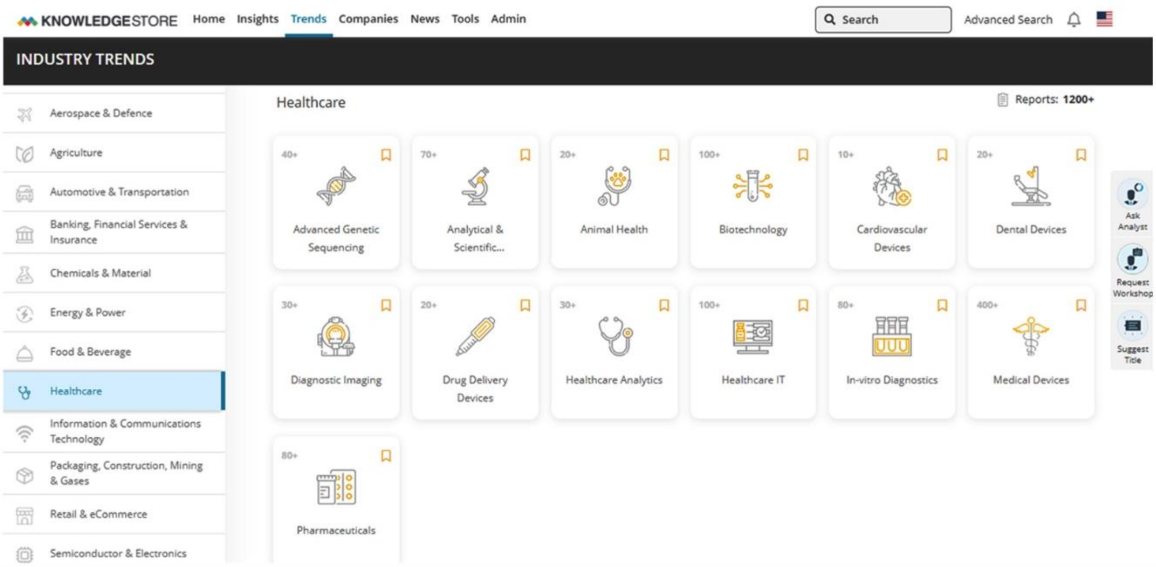
- View & download all subscribed reports from a single, online platform
- Contact our analysts to answer any questions related to our reports or request a custom research
- Share comments on specific reports with the other users of your organization
- Suggest titles/topics to our research teams that you would like us to cover in our future reports
- Identify reports published on the high-growth markets within your industry

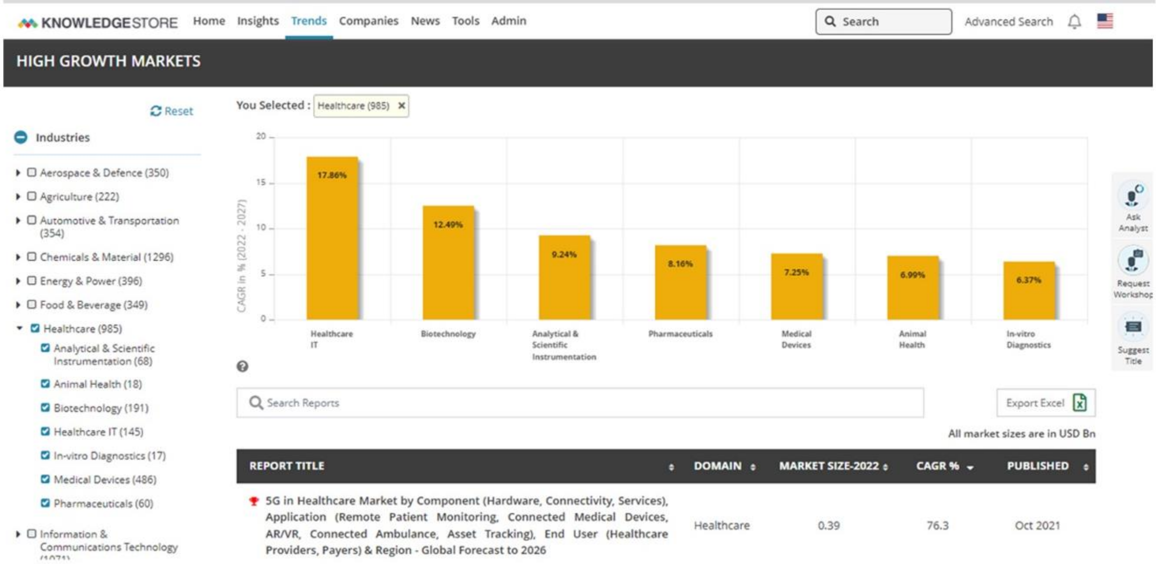
Get started now by requesting a demo and learning more about KnowledgeStore www.mnmks.com

MARKETSANDMARKETS KNOWLEDGESTORE SNAPSHOT



MARKETSANDMARKETS KNOWLEDGESTORE: HEALTHCARE SNAPSHOT





13.2 CUSTOMIZATION OPTIONS

With the given market data, MarketsandMarkets offers customizations per the company's specific needs. The following customization options are available for the report:

GEOGRAPHICAL ANALYSIS

- Further breakdown of the Rest of European market, by country
- Further breakdown of the Rest of Asia Pacific market, by country

COMPANY INFORMATION

- Detailed analysis and profiling of additional market players (up to five)

13.3 RELATED REPORTS

SR. NO.	REPORT TITLE	PUBLISHED DATE
1.	**OLIGONUCLEOTIDE SYNTHESIS MARKET- GLOBAL FORECAST TO 2027** By Product (Drug, Synthesized Oligo (Primer), Reagents), Type (Custom, Predesigned), Application (Therapeutic, (ASO, siRNA), Research (PCR, Sequencing, Diagnostics), End User (Pharma, CROs, CMOs) https://www.marketsandmarkets.com/Market-Reports/oligonucleotide-synthesis-market-200829350.html	November 2022
2.	**GENE THERAPY MARKET- GLOBAL FORECAST TO 2027** By Vectors (Non-Viral (Oligonucleotides), Viral (Retroviral, Adeno-Associated)), Indication (Cancer, Neurological, Hepatological Diseases, Duchenne Muscular Dystrophy), Delivery Method (In Vivo, Ex Vivo), Region https://www.marketsandmarkets.com/Market-Reports/gene-therapy-market-122857962.html	July 2022
3.	**VACCINES MARKET - GLOBAL FORECASTS TO 2026** By Technology (Recombinant, Toxoid, Conjugate, RNA), Type (Monovalent, Multivalent) Disease (Pneumococcal, Influenza, DTP, HPV, MMR, Covid-19), Route of Administration (IM, SC, Oral), End User (Pediatric, Adult) & Region https://www.marketsandmarkets.com/Market-Reports/vaccine-technologies-market-1155.html	January 2022

13.4 AUTHOR DETAILS

Dipak Mahajan
Associate Vice President

Dipak brings over 16 years of management consulting experience. His expertise comprises market assessment, product launch, entry strategy, opportunity assessment, pricing, market access, and commercial due diligence.

He has worked extensively on Research and Advisory projects for Pharmaceuticals, Medical Devices, and Life Sciences majors, helping them in their growth agenda and navigating through market complexities. His current focus is on the Pharmaceuticals and Biotechnology sector in MNM.

Dipak has a post-graduate degree in Business Administration – Biotechnology and an undergraduate degree in Microbiology from the University of Pune.

Ninad Nanadikar
Senior Manager

Ninad has 12+ years of Experience in Life sciences research and consulting with solutions such as market assessment, commercial due diligence, opportunity assessment, and market entry.

He has worked on growth engagements across different geographies for leading players in Pharmaceuticals, Life sciences, Medical Devices, and Consumer Healthcare. Prior to MNM, he worked with KPMG, Frost & Sullivan & SmartAnalyst.

Ninad has a post-graduate degree in Business Administration – Biotechnology from the University of Pune and an undergraduate degree in Biotechnology.

Contributors

Sampatkumar Donthula
Manager

Tanvi Sapatnekar
Senior Research Analyst